



B&G

TECH
INFO
GUIDE

NMEA 2000



Part Number Quick Guide

000-0124-69	NMEA 2000 Starter Kit (Power Cable, 2Ft cable, 15ft Cable, 2 T connectors, 2 Terminators)
000-0119-79	N2K-T-RD NMEA 2000 T Connector
000-0119-88	N2KEXT-2RD 0.6m (2ft) Extension cable
000-0127-53	N2KEXT-6RD 1.8m (6ft) Extension cable
000-0119-86	N2KEXT-15RD 4.5m (15ft) Extension cable
000-0119-83	N2KEXT-2RD 7.5m (25ft) Backbone extension cable
000-0127-45	SimNet –N2K Adapter kit Contains 44172260 SimNet Inline joiner, 24005729 Micro C-SimNet adapter, T Connector
000-0119-75	N2K-PWR-RD N2K power kit (1 Cable and T Connector)
000-0127-52	N2K Terminators 1 Female, 1 Male
24006413	N2K female to SimNet 4 m (13 ft)
24006199	SimNet to N2K (female) cable NMEA 2000® product to SimNet 0.5 m (1.6 ft)
24005729	SimNet to N2K (male) cable SimNet product to a NMEA 2000® 0.5 m (1.6 ft)

N2K Sensors

EP60- Fuel Flow



EP70- Water Speed



EP80TH- Thru-Hull Temp

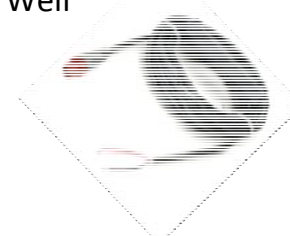


EP85- Fuel Storage



EP65- Fluid Level

Fuel, Oil, Water, Grey Water, Black Water, Live Well



EP80- Temp

Water, Inside, Outside, Engine Room Cabin, Live well, Bait well, Refrigeration, Heating



EP90- Pressure

Pitot, Atmospheric, Engine Boost, Engine Oil, Engine Water, Fuel, Transmission Oil



N2K TESTING



Practical testing equipment



T Joiner
000-0119-79



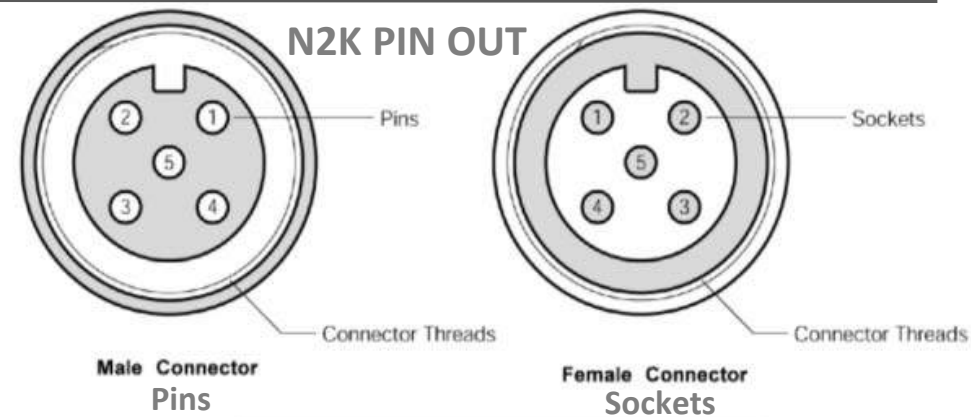
N2KEXT-2RD 0.6m (2ft) Extension cable
000-0119-88
(Cut cable to allow easy testing of exposed wires, its also helpful to fit insulated terminals for easy connection to a multimeter)

Voltage

Remove the end point termination at the furthest point from the supply and fit a T join and your modified N2K test cable, refit the terminator, measure the voltage between Red (Net-S) and Black (Net-C) 11.5VDC-14.4VDC would be an acceptable range. If the voltage is higher than this it may indicate an issue with the vessels charging system. If it is lower the power cable should be checked for poor connections, and the voltage at the battery/supply checked.

Standard rules apply with creating a SimNet/N2K network.

1. Max drop cable length: 19.5ft (6m)
2. Max connected devices: 50
3. Max current: 5A
4. Max backbone length: 490ft (150m)
5. Max total drop cable length: 200ft (60m) e.g. 10 drop cables of 19.5ft (6m)
6. N2K is designed for a 12VDC system only



- Male Connector Pins
- Pin 1: Shield
 - Pin 2: NET-S (Power Supply +ve) Red
 - Pin 3: NET-C (Power Supply -ve) Black
 - Pin 4: NET-H (CAN-H) White
 - Pin 5: NET-L (CAN-L) Blue

System resistance

With the N2K network unpowered measure fit a T join and your modified N2K test cable and measure the resistance between the blue (Net-C) and white wire (Net-L) A correctly terminated system you should read 60 Ω

A short or open circuit indicates a possible cabling fault and requires inspection of the cabling and connectors, a good method to isolate this is by disconnecting all devices and add them one at a time checking the resistance at each connection. A reading of 120 Ω will mean a single termination is fitted, a second termination should be fitted. Values of less than 60 Ω can indicate extra terminations are connected and should be removed, leaving only two terminators, one at each end of the network.

TRITON

Identifying the components



Multi configurable pages allowing Both True and apparent wind angles to be displayed as well as Data values



Split Data Boxes allow up to 9 individual areas to be displayed or the display can be configured for graphical data like wind trend lines. 8 Pages can be configured and an auto scroll function is available



Single Line data is large and very easy to see, so it can be viewed at a mast pod along with a very wide viewing angle of 170 Degrees.



Dual Gender N2K connection to allow “Daisy Chain” connections.



Pilot remote, single N2K connector, used in conjunction with Pilot configured display. Green LED= Pilot engaged Red LED= Pilot Standby



MHU, hardware based on 213 Top end and IS20 Mount. SimNet socket fitting at mast base (terminated)



DST-800 Triducer
DT-800 Depth Temp
ST-850 Speed Temp

TRITON



Triton HV Displays
 20/20 P/N:000-11088-001
 10/10 P/N:000-11087-001

The display comes fitted with a 0.5m Micro-C cable with male connector, for easy connection to N2K backbones.

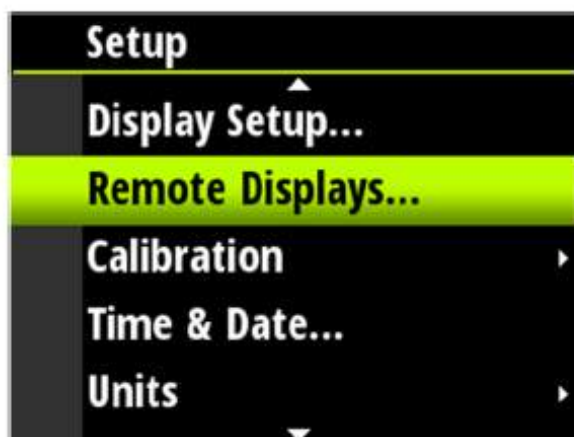


3-Way 20/20HV Mast Bracket BGH220013
 4-Way 20/20HV Mast Bracket BGH220014
 5-Way 20/20HV Mast Bracket BGH220015

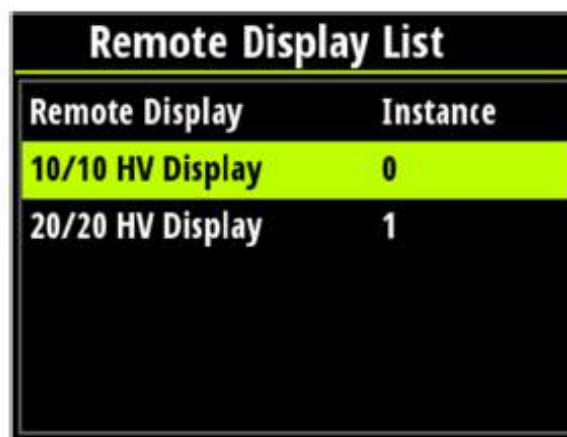
The displays are controlled via a Triton T41 display (Main menu>Setup>Remote Displays)

TIP

The HV display will flash to let you know which one you have selected



Select "Remote Displays..."



Select a display to change



Select the Data you want to display

IMPORTANT: The "Triton HV Display" is NOT the same as the "H3000 HV Display". The hardware and mechanics are the same, but the software and networking is different. They are NOT compatible.

H3000

Identifying the components



AWA 360
BGH230001



TWA 360
BGH230002



AWS 0-50kt
BGH230003



TWS 0-50kt
BGH230004



Magnified AWA
BGH230005



Heading
BGH230009



Depth 200m
BGH230007



Depth ft/fathoms
BGH230011



Boat Speed 25kt
BGH230008



Boat Speed 12.5kt
BGH230006



Rudder Angle
BGH230010



Graphical Function Display
BGH210001



Graphical Pilot Display
BGH210021



Full Function Display
BGH201001 (Non NMEA)



Full Function Display
BGH201002(NMEA)



40/40 BGH280001
30/30 BGH240001
20/20 BGH290001
10/10 BGH320001

H3000

Identifying the Components



H3000 CPU –
 Hydra BGH250001
 Hercules BGH250002
 Hercules Perf. BGH250003
 Hercules Motion BGH250004

UPGRADES:

Hydra-Hercules BGH250012
 Hydra-Herc Perf. BGH250013
 Hercules-Hercules Perf. BGH250023
 Hercules-Hercules Motion BGH250024
 Hercules Perf.-Hercules Motion BGH250034



Halcyon Gyro Processor
 BGH061001



Halcyon Gyro
 BGH060001



IMU (Inertial Measurement Unit)
 BGH260000



Gimbaled Rate
 Compass
 BGH330001



Halcyon 2000
 Compass
 486-00-009



ACP 1 Pilot Computer
 BGH171001



ACP 2 Pilot Computer
 BGH171002



Rotary Rudder
 Feedback
 RRF-ACP



Linear Rudder
 Feedback
 SEN-RUD-LF2



Hydraulic
 Pumps
 T1-12V
 T2-12V
 T2-24V
 T3-24V
 T4-24V



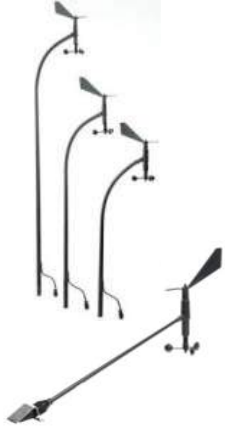
Hydraulic
 Rams
 T1-12V
 T2-12V
 T2-24V



T3-24V
 T4-24V

H3000

Identifying the Components



Vertical Masthead Units

BGH031001	VMHU 810mm
BGH031002	VMHU 1050mm
BGH031002/S	VMHU 1050mm Ocean
BGH031003	VMHU 1450mm
BGH031003/S	VMHU 1450mm Ocean
BGH031010	VMHU 1800mm
213-00-002	213 Masthead Unit



690-00-004
Clinometer



690-00-007
Barometric Pressure



224-00-066
Air Temperature



BGH040028
Digital Loadcell
Amplifier



Loadpins

BGH041001	12.7 mm (1/2")
BGH041002	16 mm (5/8")
BGH041003	19 mm (3/4")
BGH041004	23 mm (7/8")
BGH041005	25.4 mm (1")
BGH041006	28.5 mm (1 1/8")
BGH041007	32 mm (1 1/4")



202-00-064
SOV Speed



157-AA-038
SOV Depth



155-00-025
SOV Housing



SEN-SPD-HP
Standard Speed



SEN-DPT-HP
Standard Depth



SEN-DPT-A2P
NMEA Depth

H3000



Heading Source Selection

Device	Node
H3000 CPU (NMEA Input)	5
Halcyon Processor (Gyro Input)	15
Halcyon Processor (NMEA Input)	15
Halcyon 2000 Compass	16
ACP Pilot (Direct Halcyon, or GRC)	18
NMEA Input to NMEA FFD	96,97....

Red LED – This indicates FastNet communications activity and should normally be flickering quite rapidly. Slow flicker may indicate a fault with the FastNet bus interface to the CPU. If the LED is permanently off then there is no FastNet activity between the CPU and the rest of the system. In this case a full trace and inspection of the Network cabling should be carried out.

Blue – LED. This is permanently lit when there is a good supply voltage to the CPU. If the input supply is less than 10 volts the CPU will fail to operate correctly and this LED will fade / extinguish.

H3000

Setting NMEA Depth

Adjusting the terminal jumper to use NMEA Depth

Caution: *It is recommended that an approved B&G technician performs this operation.*

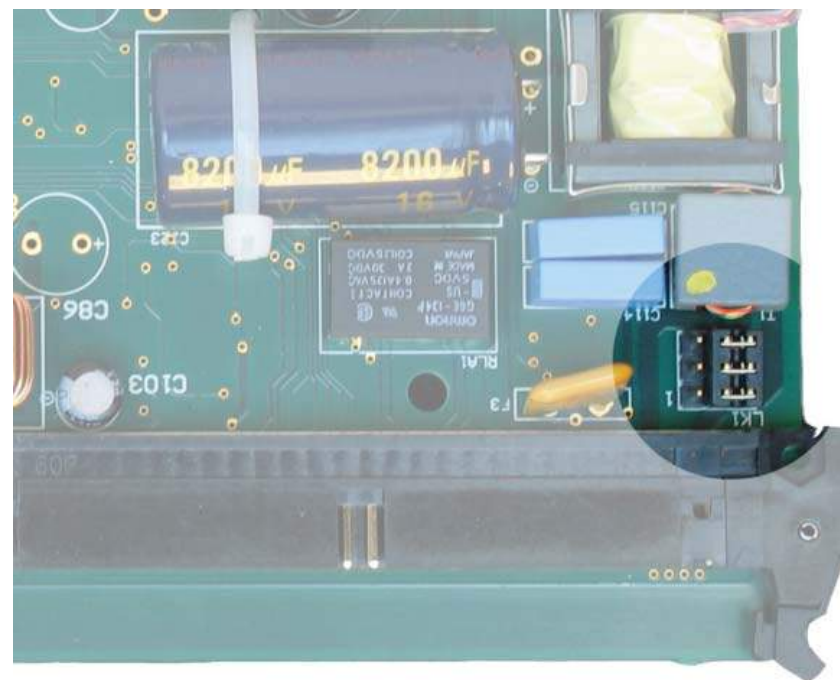
Remove terminal cover – 2 x screws

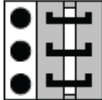
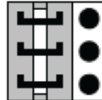
Remove top case to expose PCB – 4 x screws

Slide off terminal jumper shown below by pulling up.

Replace jumper bridging the terminals shown below by pushing down..

Note: *The jumper is set to the position shown as the standard depth setting.*

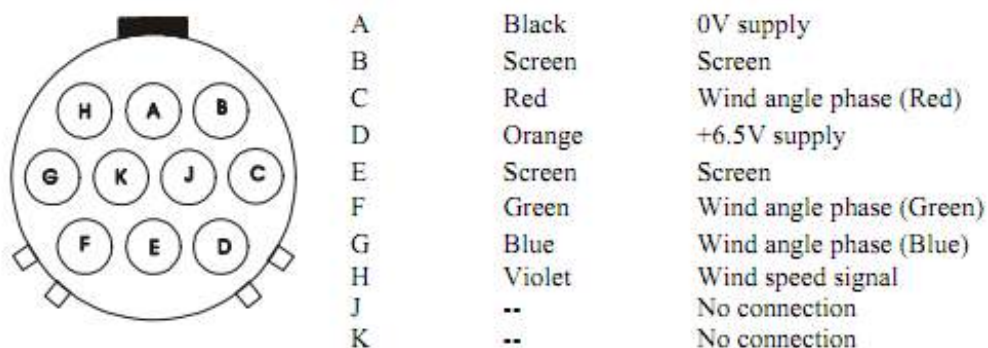


	PASSIVE DEPTH: Terminal jumper as standard bridges between centre and right pins. (Default Setting)
	NMEA DEPTH: Remove terminal jumper and place over centre and left pins

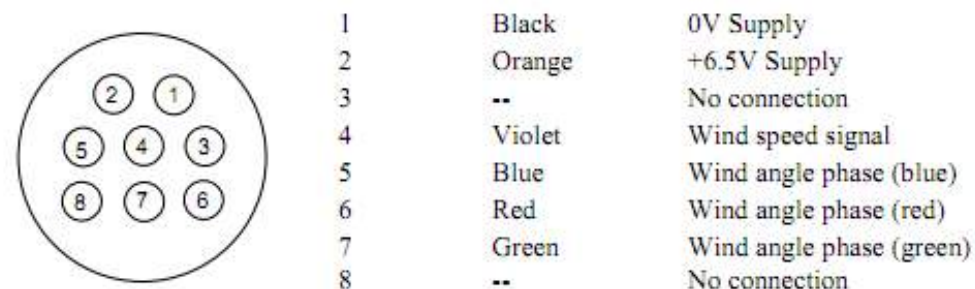
Masthead Testing

213 Masthead Tests

213 Masthead unit connector wiring



213 Vertical Masthead unit connector wiring



Wind Speed Test

1. Locate the masthead unit cable junction box at the base of the mast.
2. Disconnect the masthead unit wires in the junction box.
3. Identify which masthead unit cable goes back to the wind display or processor
4. Carefully isolate the Wind speed wire (normally Violet) and the 0V supply wire (normally Black).
5. Quickly tap the wind speed wire to the 0V wire. This will now simulate wind speed. The faster the two wires are tapped together, the faster the wind speed reading will be.

Wind angle Test

With the system switched on, and using a Digital Volt Meter, measure the wind angle phase voltages, normally Red, Green and Blue, with respect to 0V (normally Black). The measured voltages can be anywhere between 0.1V and 6.4V and will vary dependent upon the wind angle

Voltage measurements

Angle°	Red (V)	Green (V)	Blue (V)
0°	0.19	4.74	4.74
30°	0.53	3.24	5.89
60°	1.64	1.68	6.33
90°	3.25	0.47	5.93
120°	4.77	0.10	4.78
150°	5.87	0.53	3.25
180°	6.28	1.67	1.69
210°	5.87	3.25	0.53
240°	4.77	4.78	0.10
270°	3.25	5.93	0.47
300°	1.64	6.33	1.68
330°	0.53	5.89	3.24

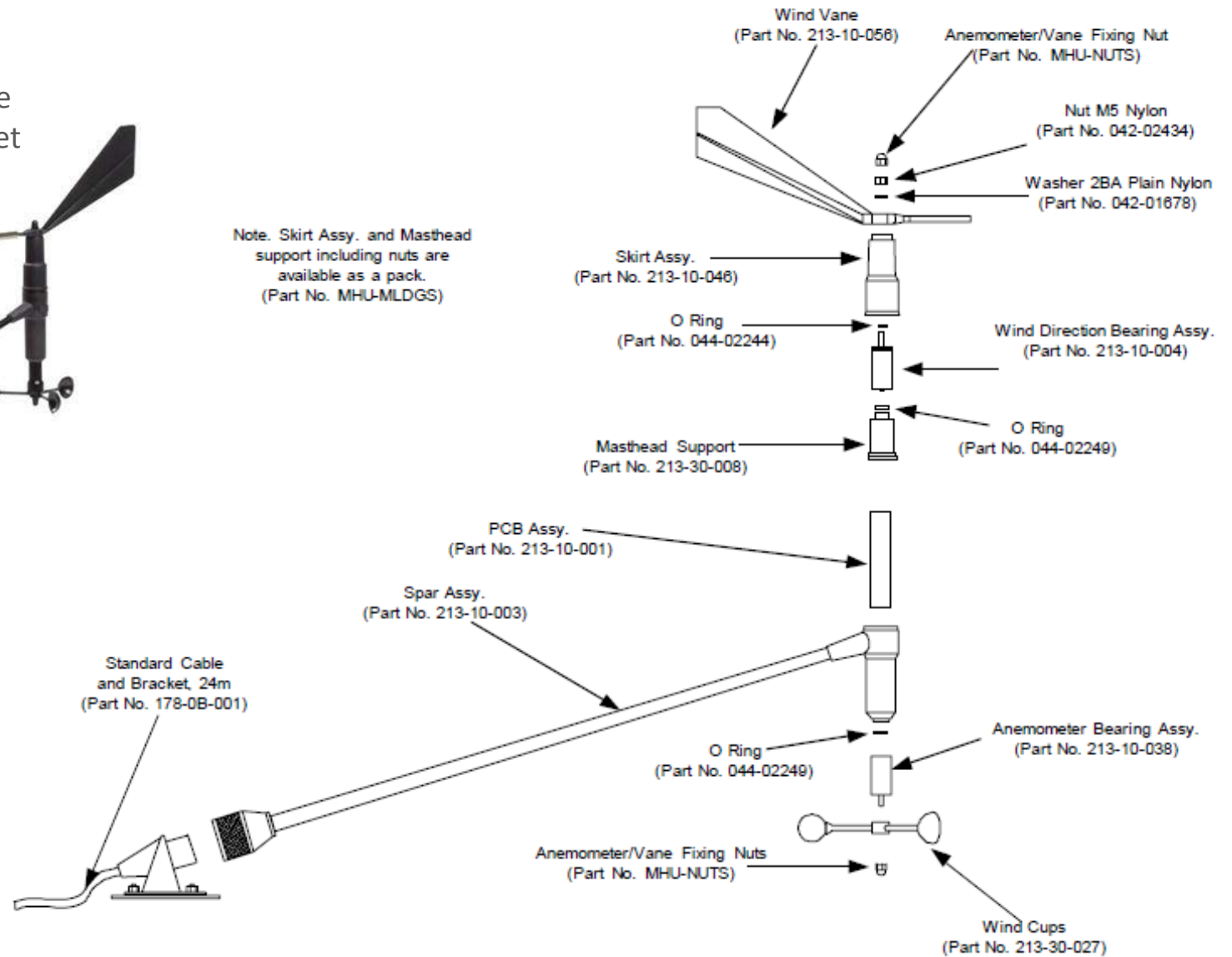
Masthead

213 Masthead Parts

508 Masthead Shares same mechanical hardware except for the spar assembly and mounting bracket



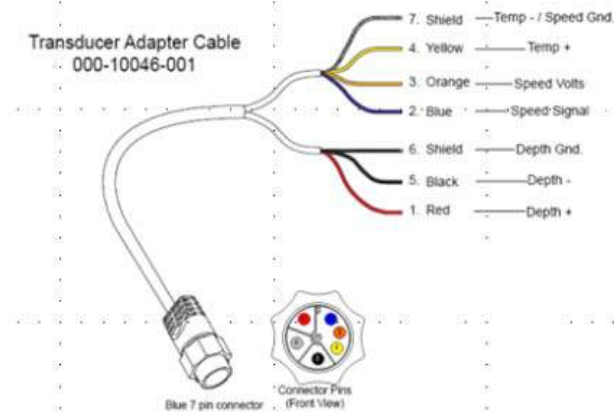
Note. Skirt Assy. and Masthead support including nuts are available as a pack.
(Part No. MHU-MLDGS)



Transducer Testing

Testing for *temp* function

With meter set to OHMS the reading should be in the 10, kohm range at 25 degrees C, (77 degrees F). If it is a Lowrance part number it will be 5 kohm range at 25 degrees C (77 degrees F). The resistance increases as the temp decreases. The sensor will read correctly in or out of water.



Testing for *speed* function

Use a 9 volt or 12 volt cordless drill battery to apply battery voltage to red and bare wires. Attach meter test leads between the green and bare wires. Turn the paddlewheel slowly by hand. The volt meter should toggle between zero volts and the input voltage with each 90 degrees of rotation.



Transducer Testing

Testing for *depth function*



Using an EDI transducer test box you can determine the resonant frequency of a transducer and confirm that it is operating properly

Attach the transducer and rotate the ***fine frequency knob*** to turn the tester on.

rotate the coarse and fine knobs until the ***res led light becomes*** the brightest.

Each frequency will typically have multiple resonant points, but the true resonant frequency of the transducer will show the lowest Impedance and brightest led.

B&G Sensors are typically 170Khz



Speed Transducer Spares

Unit Description	Hull Housing Material/Type	Sensor Body Diameter	Sensor Body Length	Cable Termination	Part Number	Impeller Spares Kit
Speed Sensor, Impeller Type	White Plastic	31mm	88mm (excluding top fittings)	Bare Wires	117-00-120	110-10-B21
Speed Sensor, Impeller Type	Bronze	31mm	88mm (excluding top fittings)	Bare Wires	117-00-060	110-10-B21
Speed Sensor, Impeller Type for Shut-Off valve Housings	Shut-Off Valve	31mm	251mm	Bare Wires	117-00-124	110-10-B21
Impeller Spares Kit (Black)	All	n/a	n/a	n/a	110-10-B21	n/a

Note: items marked in **Red** are discontinued product lines.

Unit Description	Sensor Part Number	Hull Housing Material/Type	Sensor Body Diameter	Sensor Length	Cable Termination	Paddlewheel Spares Kit
Network Paddlewheel, Low-speed (6 blades) old type	610-0A-026	White Plastic	31mm	105mm	Network Plug	202-00-129
Hydra/Focus Paddlewheel, low-speed (6-blades), old type	202-00-100	White Plastic	31mm	105mm	Bare wires	202-00-129
Network Paddlewheel, low-speed (6-blades), old type	610-0A-036	Bronze	31mm	105mm	Network Plug	202-00-128
Network Paddlewheel, high speed (4-blades), old type	610-0A-129	White Plastic	31mm	105mm	Network Plug	202-00-161
Network Paddlewheel, high speed (4-blades), old type	610-0A-130	Bronze	31mm	105mm	Network Plug	202-00-160
Hydra/Focus Paddlewheel, 6-blades, blue colour	202-00-045	Black Plastic	40mm	93mm	Bare Wires	202-00-054
Hornet 4 Paddlewheel, 6-blades red colour	202-00-036	Black Plastic or Bronze	40mm	93mm	Bare Wires	202-00-054
Shut-Off Valve Paddlewheel Sensor	202-00-064	Bronze Shut-Off Valve	31mm	232mm	Bare Wires	202-00-129
Network Paddlewheel, 4-blades	SEN-SPEED-N	Bronze or Black Plastic	31mm (insert only)	90mm	Network Plug	SEN-SPRS-SPD
Hydra Paddlewheel, 4-blades	SEN-SPEED-H	Bronze or Black Plastic	31mm (insert only)	90mm	Bare Wires	SEN-SPRS-SPD
h1000 Speed Sensor, 4-blade, 3m cable	SEN-SPEED-P	Bronze or Black Plastic	31mm (insert only)	90mm	Bare Wires	SEN-SPRS-SPD

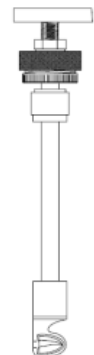
Notes:

- Items marked in **Red** are discontinued product lines.
- There is no alternative sensor for the 202-00-036 Speed Sensor
- The 202-00-045 can be replaced with SEN-SPD-HP (includes sensor, flapper insert and hull-housing)

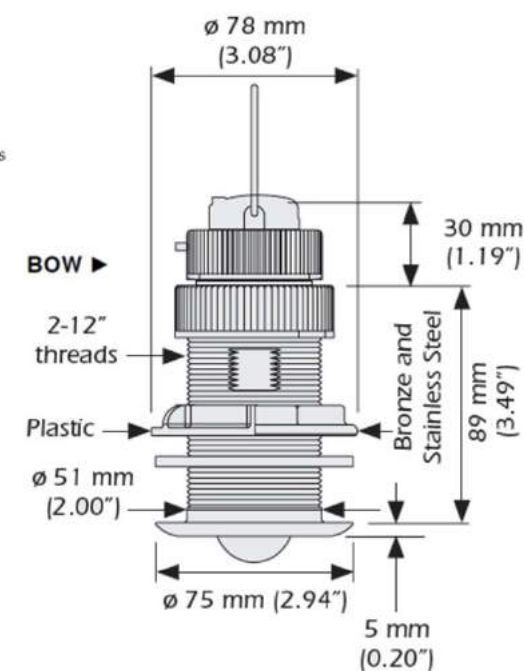
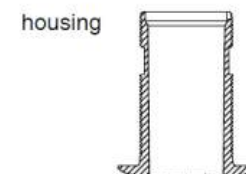
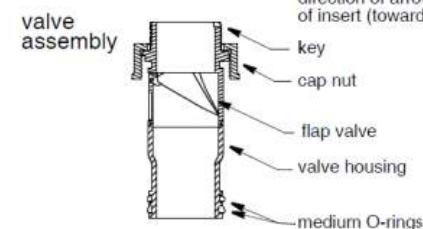
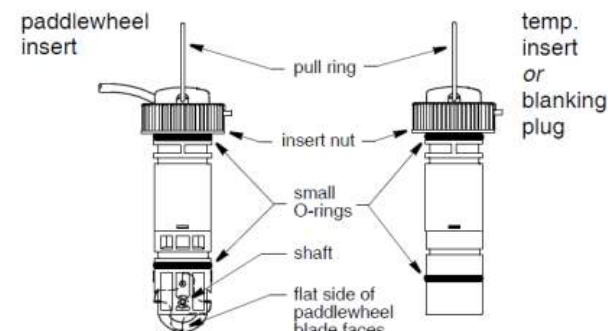
Speed Transducer Spares



Impeller Type
Speed Sensor
for use with a
Shut-Off-Valve
Part No. 117-
00-124



Six Blade
Paddlewheel Sensor
for use with a
Shut-Off-Valve Part
No. 202-00-064



Impeller Type Speed Sensor.
Part No. 117-00-120



Network Six-Blade,
Low-Speed
Paddlewheel Sensor
Part No. 610-0A-026



Network Four-Blade,
High-Speed Paddlewheel
Speed Sensor Part No.
610-0A-129

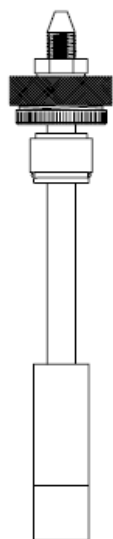
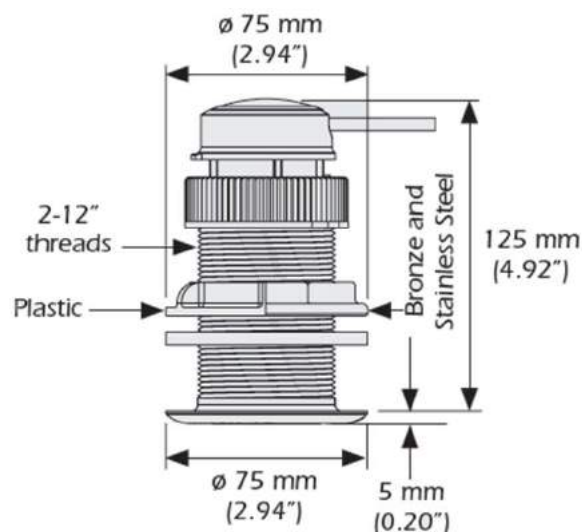


Four-Blade Paddlewheel
Sensor used on Part No.
SEN-SPEED-N, H, or P



Six-Blade Paddlewheel Speed
Sensor – Hornet 4 System Part
No. 202-00-036

Depth Transducer Spares



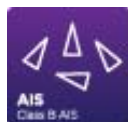
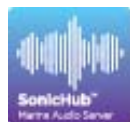
Depth Sensor for
use with a
Shut-Off-Valve
Part No. 157-AA-038

Unit Description	Part Number	Hull Housing Material/Type	Sensor Body Diameter	Sensor Length	Cable Termination	Technical
Hydra/Focus Depth Transducer	401-00-011	White Plastic	31mm	109mm	Bare Wires	Passive Coil
Network Depth Transducer	610-0A-027	White Plastic	31mm	109mm	Network Plug	Passive Coil
Hydra Transducer, shaved face	401-00-027 ^[1] (use 199-00-018)	Bronze Housing	31mm	109mm	Bare Wires	Passive Coil
Hydra/Focus Depth Transducer	199-00-018	Bronze Housing	31mm	109mm	Bare Wires	Passive Coil
Network Depth Transducer	610-0A-040	Bronze Housing	31mm	109mm	Network Plug	Passive Coil
Hecta/Hydra/Network Long Reach Depth Transducer	199-00-029	Bronze	31mm	158mm	Bare Wires	Passive Coil
Hecta Mk 6	111-00-049	Bronze	31mm	109mm	Co-ax Connector	Passive Coil
Network Depth Transducer	SEN-DEPTH-N	Black Plastic	41mm	89mm	Conexall Connector	Passive Coil
Hydra Depth Transducer	SEN-DEPTH-H	Black Plastic	41mm	89mm	Bare Wires	Passive Coil
h1000 Sail Passive Transducer 170kHz	SEN-DEPTH-P	Black Plastic or Bronze	41mm	89mm	Bare Wires	Ceramic only ^[2] Use with SDIE
h1000 Power Active Transducer 235kHz	SEN-DEPTH-A2	Black Plastic or Bronze	41mm	89mm	Bare Wires	NMEA

^[1] Items marked in **Red** are discontinued product lines.

^[2] 3m cable forms part of a tuned circuit. Do not cut cable to length.

Zeus Touch



Cables



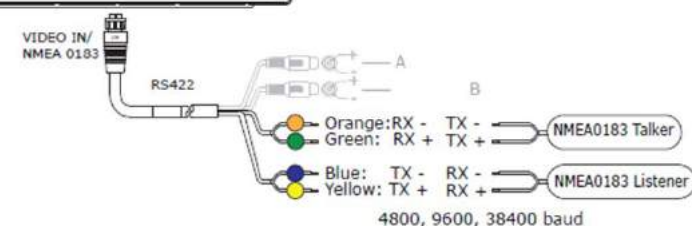
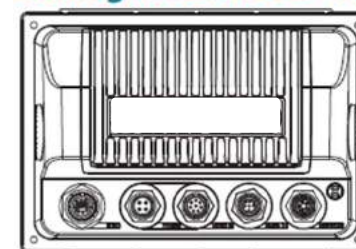
Power 000-00128-001

1. **Black** Supply (-ve)
2. **Blue** Alarm
3. **Yellow** Power Control
4. **Red** Supply (Red)

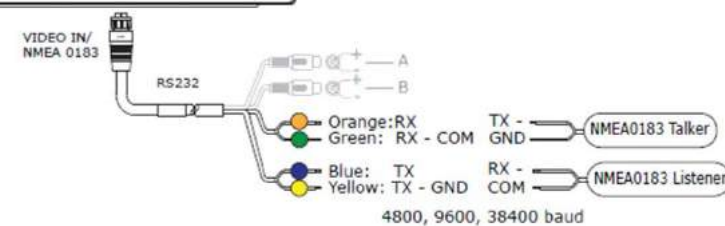
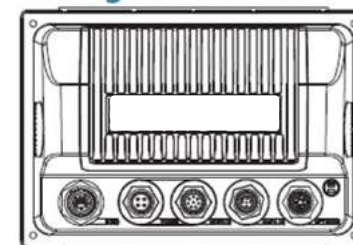
Data 000-00129-001

1. Video 1 Gnd
2. Video 1 +ve
3. Video 2 Gnd
4. Video 2 +ve
5. **RS422 TX-ve**
6. **RS422 TX+ve**
7. **RS422 RX -ve**
8. **RS422 RX +ve**

Wiring NMEA 0183 for RS422



Wiring NMEA 0183 for RS232



Helpful Tips

The Zeus Touch has a Settings page that is accessed via the Pages key (Home), or by double press of the Menu key

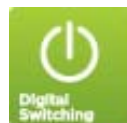
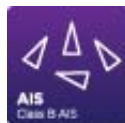
The Zeus Touch 7,8 have a broadband sounder module built in the, Zeus Touch 12 must use an external sounder source

The Zeus Touch 12 has an inbuilt 3 port ethernet switch built in.

You can access the Touch screen calibration by pressing and holding the Menu Key then powering up the unit



Zeus 1.0



Always set MMSI number, this stops false AIS alarms (Set in the VESSELS menu)

To add a scrolling Instrument Bar, access by pressing the direct access key and selecting the spanner symbol, then EDIT INSTRUMENT BAR

Advanced Settings: MENU:MENU>SYSTEM>ADVANCED...>

WAYPOINTS

Allows duplicate waypoint names

HARDWARE

Lighting controls for keypad, screen and external DI series monitors

USER INTERFACE

Options to change the way that cursor controls operate

FEATURES

This menu then allows you to turn on extra Features , for equipment that has been added like LSS-1 Structure Scan, Czone, or Sonic Hub

RADAR

Controls Radar logging function

TRAILS

Sets when logging points are dropped

H3000

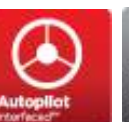
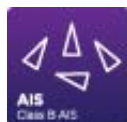
Allows Heading information to be transferred to the H3000 system

Helpful Tips

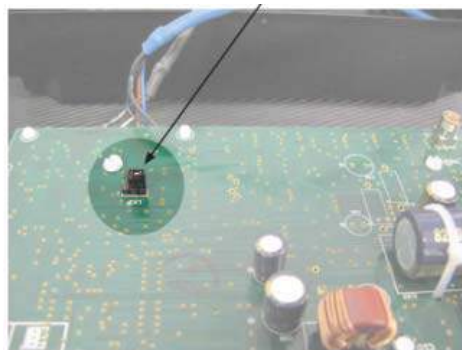
To adjust Panel sizes, access by pressing the direct access key and selecting the spanner symbol, then ADJUST PANEL SIZES

When Power Control is used, be aware that the output from the Zeus is the same as the supply voltage i.e 24VDC supply means 24VDC control voltage, important when connecting to LSS-1, or other 12VDC accessories

Zeus 1.0

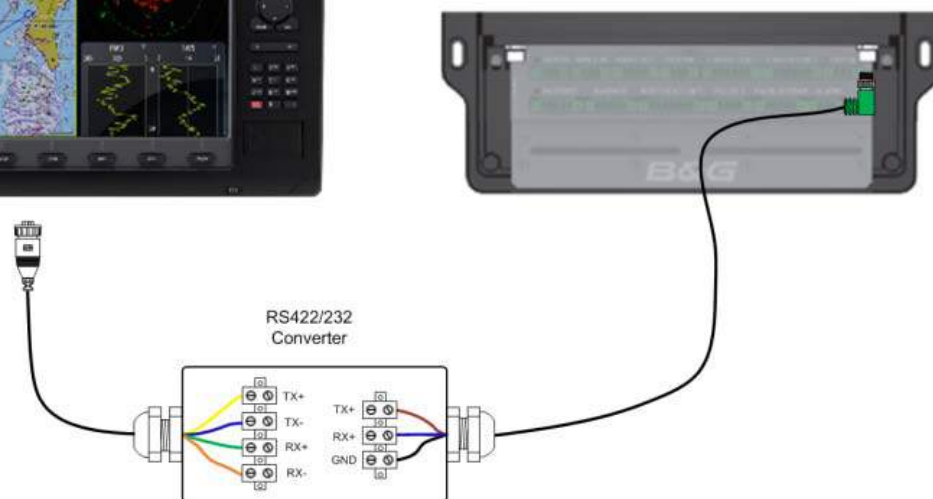
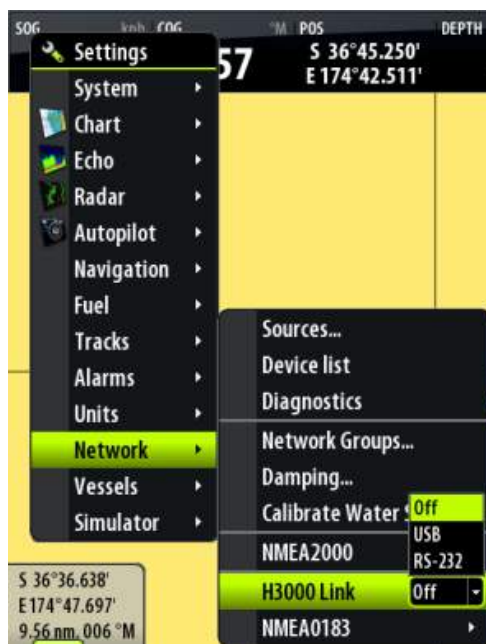


H-LINK™ Interface



	RS232: Remove terminal jumper and place over centre and left pin.
	USB: Terminal jumper as standard bridges between centre and right pin

To enable H-LINK™, you must have a connection to the H3000 via either USB (default) or RS232/NMEA Cable (Jumper on h3000 PCB must be changed) The selection will reset to “Off” if it does not detect Hlink communications. The device list will now show Hlink as a Source in the Network device list

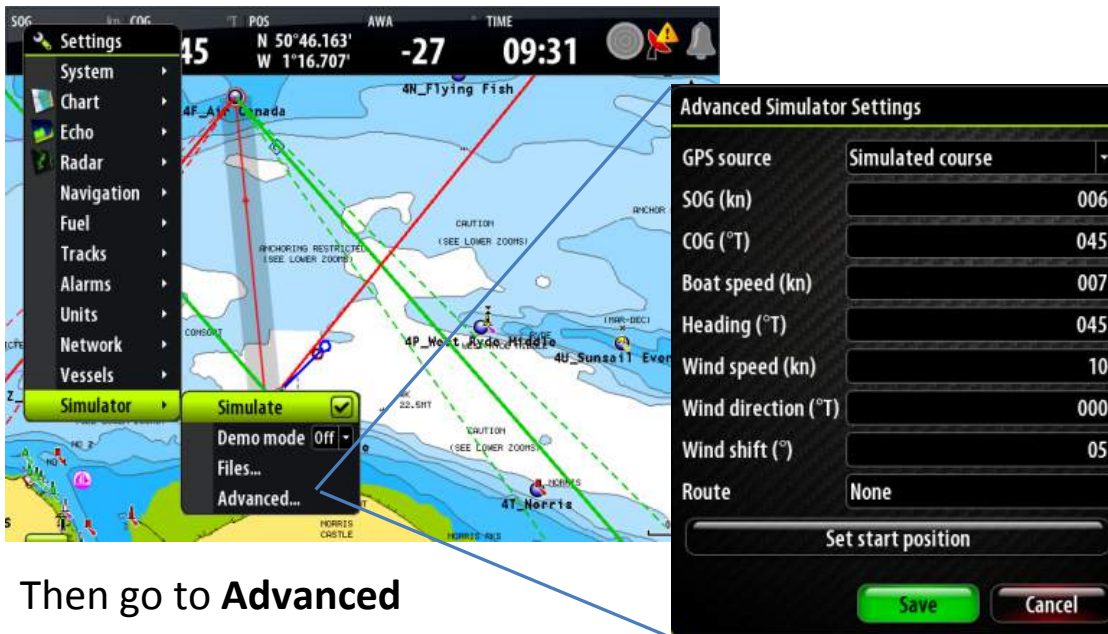


ZEUS

How to setup Zeus simulator mode

Setting up laylines:

Press **MENU:MENU** twice to get to **Settings**
Scroll down to **Simulator** and select **Simulate**



Then go to **Advanced**

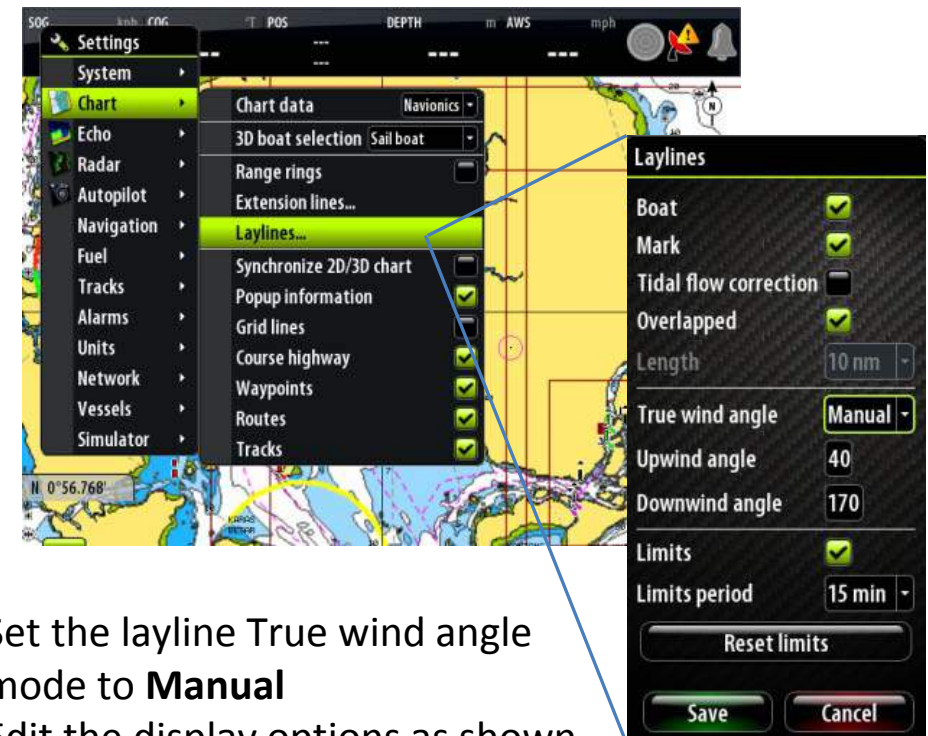
Set the GPS source to **Simulated course**

Enter your desired values for instrument data

Go to **Set start position** – this will place the boat at the cursor

Position the cursor over a waypoint or other destination and press the **GOTO** button

Press **MENU:MENU** twice to get to **Settings**
Scroll down to **Chart** and select **Laylines**



Set the layline True wind angle mode to **Manual**

Edit the display options as shown and save

ZEUS

Loading and displaying a GRIB file:

Download a wind and pressure GRIB file and save it onto a USB stick. Sources for GRIB files include:

www.grib.us

<http://www.ocens.com/wxnet.htm>

<http://www.saildocs.com/gribinfo>



Insert the USB stick in the USB port on the Zeus

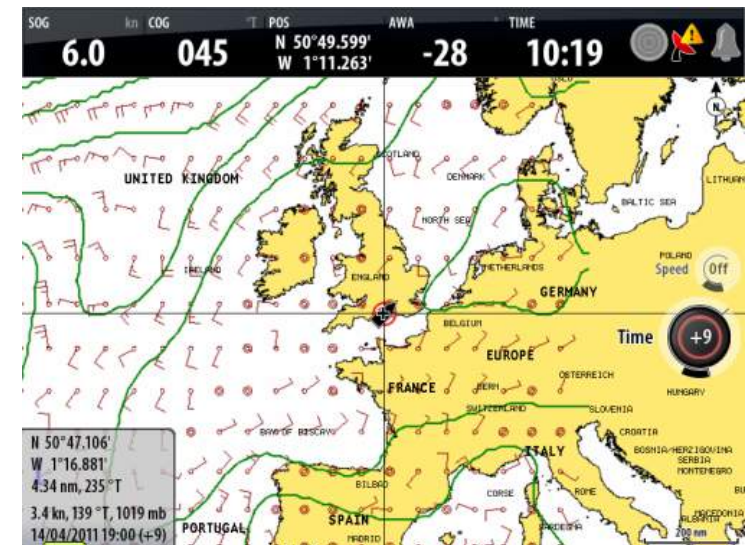
With a chart page selected press **MENU** and select the **Grib weather** overlay



Go to **Grib weather options** and select the grib file from the USB stick



The Grib file will be overlaid onto the chart. Step through the GRIB by adjusting the **Time control**. Alternatively, set the animation **Speed Control** by pressing the rotary control



ZEUS

NMEA Interface

Zeus has the facility for NMEA 0183 interfacing, the table describes what sentences are transmitted and what are received

GGA=Time, Position, Fix Data
 GLL=Position, Time of fix, Status
 GSA=GPS fix quality info
 GSV=Satellites in view info
 VTG=COG, SOG
 ZDA=Time, Date, UTC, Local offset
 RMC=Time,Date,Position,COG,SOG,Mag Var
 AAM=Waypoint Arrival Alarm info
 APB=Cross track error, Destination arrival data, Destination ID, Bearing to destination
 BOD=Bearing Origin to destination, Origin ID, Destination ID
 BWC=Bearing and Distance to waypoint, Waypoint Position
 BWR=Bearing and Distance to waypoint, Waypoint Position-Rhumb Line
 RMB=Nav Data to destination, Range, Bearing to destination
 XTE=Cross track error
 DBT=Water Depth referenced to transducer
 DPT=Water depth including offset
 MTW= Water temp
 VLW= Water Distance Travelled
 VHW=Water Speed and Heading

GPS							
Receive	GGA	GLL	GSA	GSV	VTG	ZDA	
Transmit	GGA	GLL	GSA	GSV	VTG	ZDA	
Navigation							
Receive	RMC						
Transmit	AAM	APB	BOD	BWC	BWR	RMC	RMB XTE
Echo							
Receive	DBT	DPT	MTW	VLW	VHW		
Transmit	DBT	DPT	MTW	VLW	VHW		
Compass							
Receive	HDG	HDT	HDM				
Transmit	HDG						
Wind							
Receive	MWV	MWD					
Transmit	MWV						
AIS/DSC							
Receive	DSC	DSE	VDM	AIS Sentences are not bridged to N2K			
MARPA							
Transmit	TLL	TTM	MARPA is TX only				

HDG= Heading, Deviation, Mag var
 HDT=Heading in True
 HDM=Heading in Magnetic
 MWV=Wind Speed and angle
 MWD=Wind Direction and Speed
 DSC= Info for Digital selective calling

DSE=Extra Information on DSC
 VDM= AIS vessel Data
 TLL= Target position, data for tracked data
 TTM=Target data referenced to own vessel

Zeus Engine Interface



When connecting to a engine its very important to ensure that you have the correct interface. If you want to record fuel levels then its very likely that you will require a EP85R Fuel storage device. This device can support up to 3 engines.

On multi engine installations each engine needs to be addressed on the N2K network, and if using a J2K100 the j1939 address also. It may be necessary to discuss this configuration with the engine installer.

Engine	Notes	P/N	Supply
Evinrude/ BRP		000-0120-62	Navico Agent
Honda		06328 ZZ3 760HE	Honda Agent
Yamaha	No Command Link	000-0120-37	Navico Agent
Yamaha	Command Link	MAR-GTWAY-ML-09*	Yamaha Agent
Yamaha	Command Link Plus	MAR-GTWAY-KT-00	Yamaha Agent
Suzuki		990C0-88134	Suzuki Agent
Suzuki	i Command Gauges	Micro C connector	Navico Agent
Mercury	Merc Monitor	Micro C connector	Navico Agent
Volvo	N2K Engine	3889758	Volvo Agent
Volvo	J1939 Engine	J2K100	Maretron
Caterpillar	J1939 Engine	J2K100	Maretron
Cummins/Mercruiser	J1939 Engine	J2K100	Maretron



Screen Capture: Very useful tool to allow fault recording, or for keeping history of events.



Device Monitor: Check all connected SimNet/N2K devices, view available information and configure.



Restore Defaults: Graduated resets to allow individual units and settings to be reset.



N2K Monitor: Review current state of the N2K bus for correct operation.



Custom Simulator: Allows you to run your own files that you can record in your local regions.



Power Control: Allows power control of multiple devices connected to the Zeus.



Personal Settings: Save a generic setup to allow for fast setups on Production vessels, or custom setup for an individual.

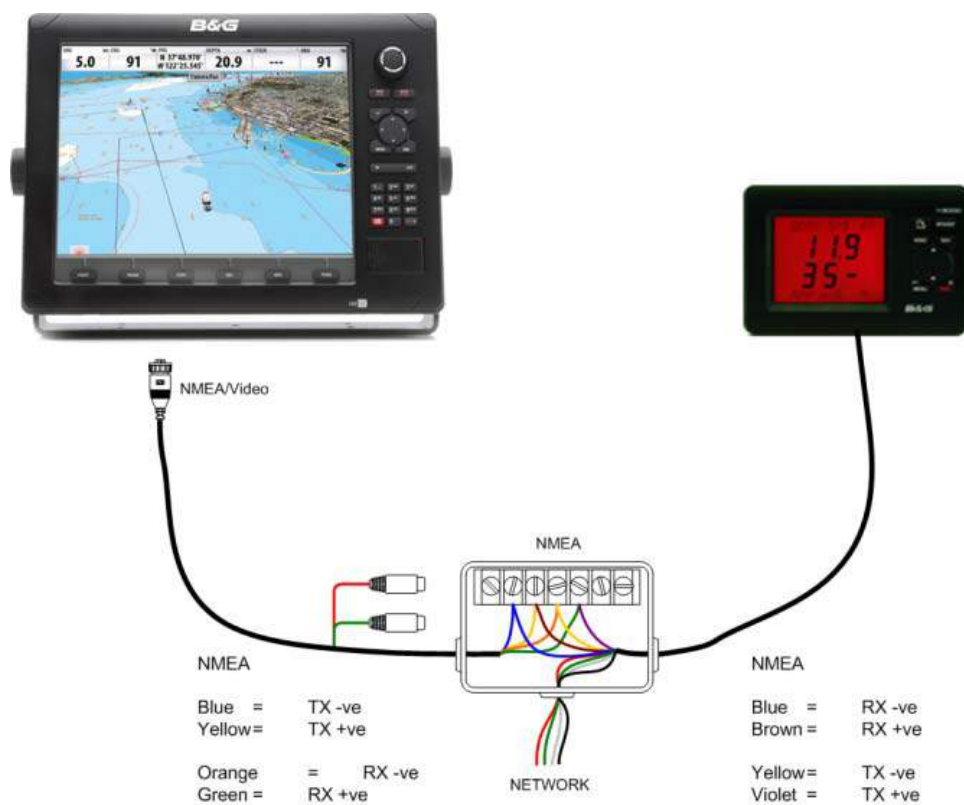


Custom Direct Access Keys: 5 Individual pages can be assigned to each key, 4 custom 1 Fixed, the setup icon allows changes to be made to page structure and Data bar info.

ZEUS

ZEUS-Instrument interfacing

ZEUS-H3000 FFD NMEA0183



ZEUS-Triton N2K standard



ZEUS-Triton Daisy Chain



Wireless



WiFi-1 Module



+

Zeus Touch Series



=



- The WIFI-1 hardware module creates a Wifi hotspot on the boat
- Simply connect to a Zeus Touch via the yellow-plug Network port
- Data from the navigation system is bridged onto Wifi via the MFD
- Install anywhere on the boat for best Wifi coverage
- View and edit WIFI-1 SSID and Passphrase from MFD



GoFree Controller App

Take control of your Zeus Touch anywhere on the boat using your iPad/Android tablet.

Zeus Touch -screen commands mirrored on iPad giving you total control*

The Zeus Touch can also select access rights to the connected devices allowing view only option.

*Autopilot/Czone control pending legal clearance.



GoFree Viewer App

View your Zeus Touch anywhere on the boat using your iPhone/Android smartphone – also available for iPad/Android tablet.

Wireless



Zeus Touch Series



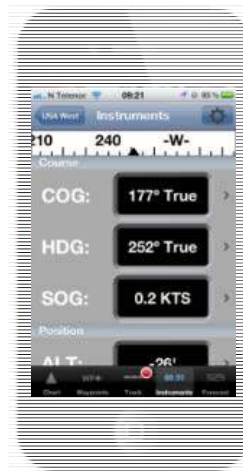
WiFi-1 Module



NMEA 0183 data is bridged from the MFD to WIFI enabling connection to any app that features 0183 TCP/IP link...including:



iRegatta



iNavx



Rosepoint Coastal Explorer

Wireless



More than 1 MFD on the network
- choose which one to mirror on the tablet



WiFi-1 Module



Only one
WIFI-1 access
point required, you can
select which MFD you want
to mirror

More than 1 iPad on the network
Only require 1 WiFi-1 Module required (you can
have 2 WiFi-1's connected to extend coverage)



Wireless setup is in the Network Settings
Menu, under the NMEA0183 Menu, as
Ethernet, it will show the IP address and
Port setting.



WTP3

The WTP3 brings new architecture with a central CPU linked to distributed data collection networks – operating at up to 1Mbit on three independent channels – the possibilities for data collection are hugely expanded, supporting recent progression in both high-end racing and super yacht systems. Data is processed in the CPU, which outputs display data, transmits serial and digital data for output via modules, communicates with Deckman via LAN and operates an on-board data logger for post-sail data analysis.

Fastnet port:

- Display connection
- Compatible sensor support
- Pilot

Terminal connection:

- System monitoring

USB port:

- Software updates
- Configuration updates
- Datalog file offload

CAN ports:

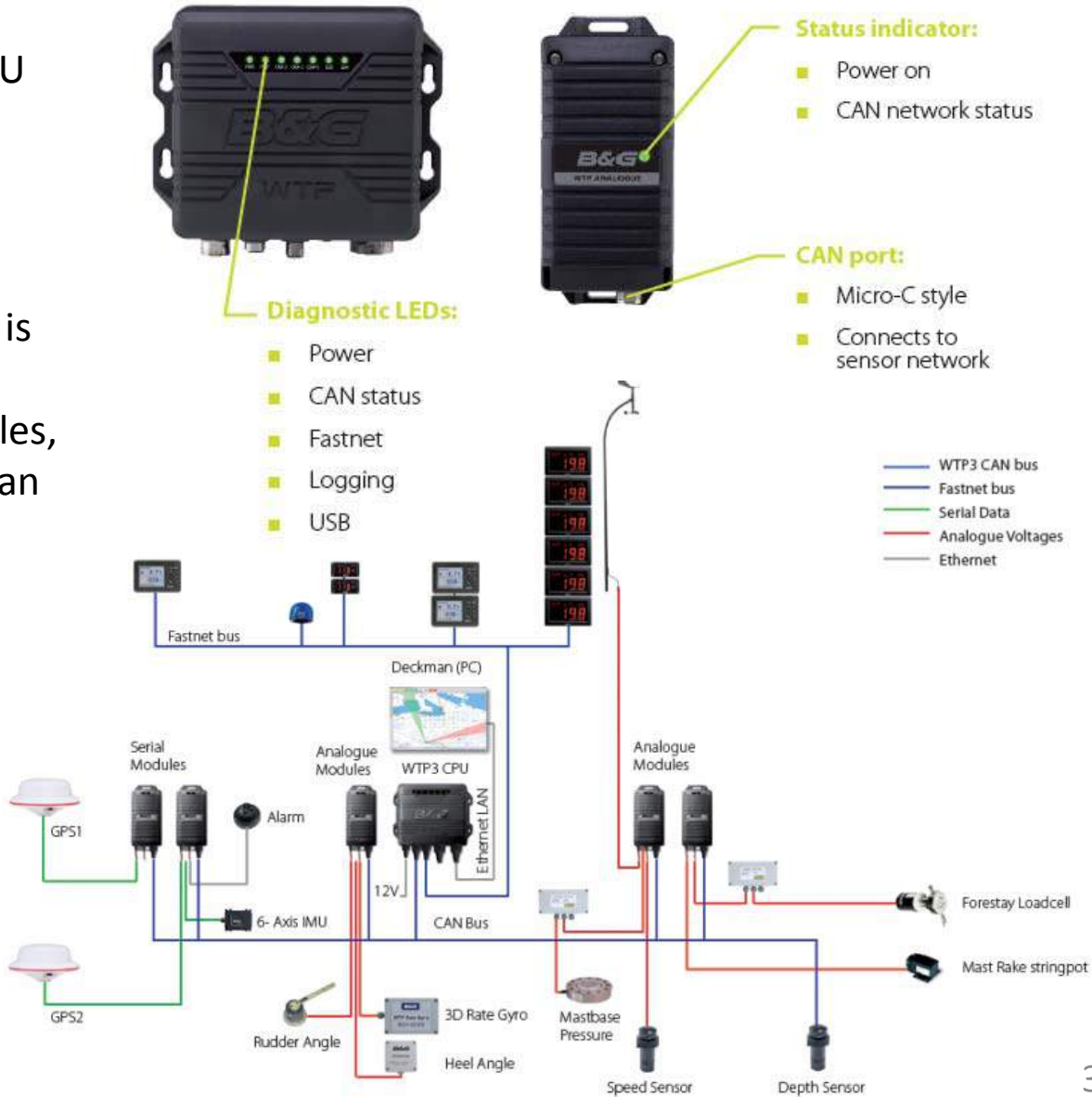
- I/O module networks
- Independently configurable
- 125 kbit > 1 Mbit speed options

CNC enclosure:

- CNC machined enclosure
- Lightweight & Robust
- IP67 protection

LAN port:

- 100Mbit Ethernet
- Deckman comms
- FTP access (file system)
- Datalog access

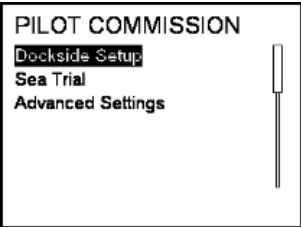


AUTOPILOTS

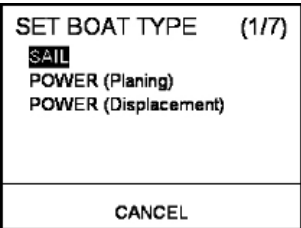
H3000



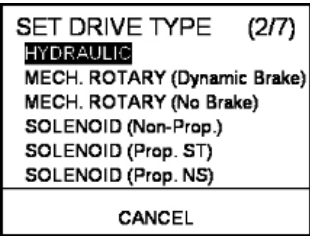
Pilot Commission-
there are 2 steps,
Dockside and Sea Trial.



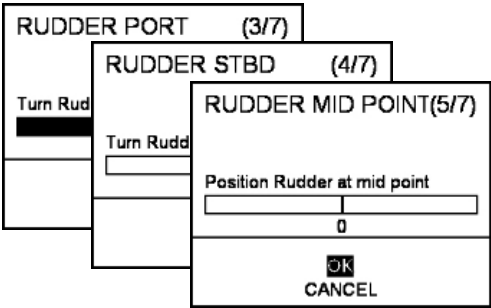
Dockside covers 7 steps-



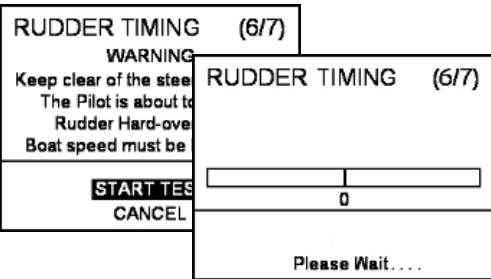
1. Boat Type



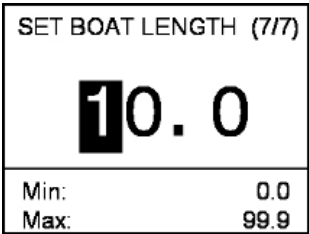
2. Drive Type



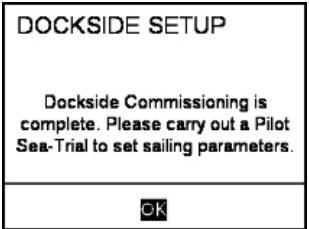
3-5. Rudder Setup



6. Rudder timing

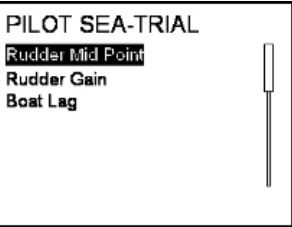


7. Boat Length

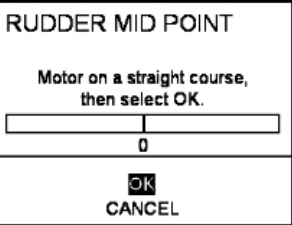


AUTOPILOTS

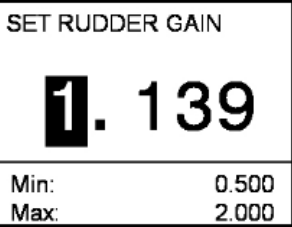
H3000



Seatrial covers 3 steps-



1. Rudder Mid-Point



2. Set Rudder Gain

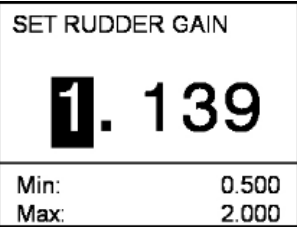
When the Pilot is part of an integrated system; boat speed data is supplied via Fastnet network from the H3000 instruments.

By monitoring boat speed and rate of turn the Pilot will automatically 'learn' the correct value

for Rudder Gain giving a rate of turn of approximately 6° per second for a sailing boat or 8° for a power boat. Steer the boat onto a suitable heading; allow time for the boat to settle on this course.

Engage the Pilot in **Compass** mode.

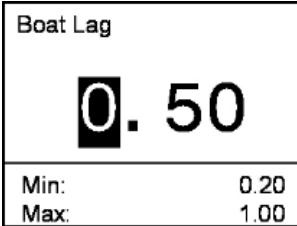
AT A SPEED NOT EXCEEDING 15 KNOTS, make at least six large course changes of at least 100° by multiple presses of the 10° course change buttons on any GPD or Handheld Controller. Observe and estimate the rate of turn. It should be approximately 6° to 8° per second. Press the OFF key to return the pilot to standby.



2. Set Rudder Gain Cont

Manual Tuning- Observe the performance of the Pilot when changing course. The rudder gain value is inversely proportional, therefore if the rate of turn is too **SLOW**, **REDUCE** the value of rudder gain and the Pilot will use more rudder. If the rate of turn is too **FAST**, **INCREASE** the value of rudder gain and the Pilot will use less rudder.

Boat Type	Factory Set Value	Typical Values
Sail	2.0	1.0 to 3.0
Power	0.8	0.3 to 1.0



3. Boat Lag

Boat Lag is the time taken for the boat to respond to changes in helm. Heavy displacement hulls require a larger value for boat lag.

Checking the boat lag:

At a speed not exceeding 15 knots, change course by 90° in either direction. Observe the Pilot steering performance. The boat should turn onto the new heading with minimal overshoot (a slight overshoot is acceptable).

If the overshoot is consistently more than 5° for course changes in both directions increase the Boat Lag value in steps of 0.1 until the overshoot is corrected. It is easier to spot overshoot than undershoot, hence if no overshoot is observed decrease the boat lag in steps of 0.1 until a small overshoot is seen. Use the smallest value of Boat Lag possible to minimise overshoot.

AUTOPILOTS

H3000 Advanced Settings

ADVANCED SETTINGS

Speed Src

Autom

Recove

Nav So

Magnet

Pilot Sp

SET SPEED SOURCE

BOAT SPEED 8.31

SOG 7.63

MANUAL SPEED 8.0

Speed Source

Select the Speed source by highlighting and pressing the Menu Key

AUTOMATIC RESPONSE

Off

Economy

Normal

Sports

Automatic Response

This option is only available if a Halcyon Gyro Stabilised Compass is fitted. The Auto Response set-up enables the Pilot to automatically alter the response level. There are four options available:

Off: The Pilot will always remain in the response mode selected

Economy: The Pilot will need to sense large environmental changes before increasing the response setting.

Normal: The Pilot will respond to moderate environmental changes state before increasing the response setting.

Sport: The Pilot will be most sensitive to changing conditions and will automatically increase its response rate to counter environmental changes.

The Automatic Response mode will never reduce the response setting below the manually set value. When the conditions have improved the Pilot will automatically return to the manual response setting.

RECOVERY MODE

Off

Narrow

Medium

Wide

Recovery Mode

This function is only available when a Halcyon Gyro Stabilised Compass is connected to the system. Recovery Mode allows the user to set the sensitivity to course errors. Pilot to react to unexpected events, for example sudden wave or wind shifts. This

function allows the Pilot to instantaneously increase the steering response to its maximum setting (PERF 4), and make a rapid recovery. The Recovery Mode will automatically switch off after 15 seconds or when the heading error has been corrected. The Pilot will then resume the previous response setting and continue normal operation.

There are four options available.

Off: The Recovery Mode function is switched off.

Narrow: The Pilot is most sensitive to sudden course changes corrected.

Medium: The Pilot is configured to the medium value when correcting sudden course changes.

Wide: The Pilot is least sensitive to sudden course changes.

Nav Source

Nav Source allows the selection of a NMEA input on the system to use as the primary source of navigation data for the Pilot Steer to Waypoint mode. In most systems, where there is only one source of NMEA navigation data, this can be left on the default setting of zero which will automatically prioritise the navigation data from that source.

If you have more than one source of data it is necessary to enter the node address of the source you wish to use, normally this value will be for either the CPU (node 5) or a NMEA FFD (node 96, 97...).

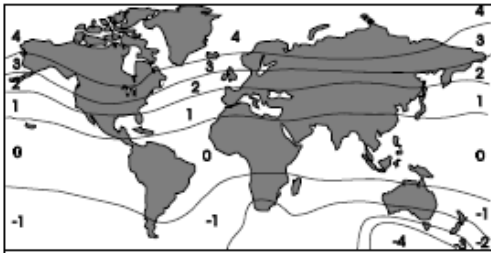
AUTOPILOTS

Set DIP Compensation (°)
0
Min: -4
Max: 4

Magnetic Dip Zone

All magnetic compasses are affected by “northerly turning errors” in the northern hemisphere or “southerly turning errors” in the southern hemisphere, which increase with boat speed and magnetic DIP angle in higher latitudes.

These can cause heading instability at boat speeds greater than 20 knots when steering with a Pilot. By entering the dip value indicated on the compensation chart, the Pilot will be able to correct for these errors and improve the heading stability.



PILOT SPEED CAL (hz)
6.25
Min: 1.50
Max: 9.99

Pilot Speed Cal

The ACP computer unit can take a direct speed input from a paddle wheel with a hall effect output. Normally the Pilot uses boat speed supplied via the Fastnet network from the instrument system, this facility is only used when the installation does not include a compatible instrument system. The Hertz/Knot value is entered into the system to ensure the Pilot steering response is controlled with reference to an accurate boat speed. The default Hertz/Knot value is 3.80; this is the default setting for B&G speed sensors. **Note** The speed cal reading is inversely proportional; i.e. to increase the boat speed, decrease the Hertz/Knot value.

Nav Source

Nav Source allows the selection of a NMEA input on the system to use as the primary source of navigation data for the Pilot Steer to Waypoint mode. In most systems, where there is only one source of NMEA navigation data, this can be left on the default setting of zero which automatically prioritise's the navigation data from that source. If you have more than one source of data it is necessary to enter the node address of the source to use, normally this value will be for either the CPU/Zeus (node 5) or a NMEA FFD (node 96, 97...).

Joystick Type

The joystick allows direct control of the rudder via the Pilot computer for quick and responsive steering. Joystick steering is engaged and disengaged with the separate Red Joystick Button. The lever can only be moved to port or starboard

Normal Steering: The rudder moves in the direction of the joystick movement, when the joystick returns to the central position the rudder movement stops. The greater the movement of the joystick, the faster the response of the rudder.

Proportional Steering: The position of the rudder follows the position of the joystick. When the joystick returns to the central position the rudder returns to its initial position.

Rudder Max Angle

This sets the maximum angle of the rudder between 25° and 45°. The default angle is 40°. The angle should be set to match the physical value from centre line to rudder end stop on the boat.

Watch Alarm

Watch Alarm Lock removes the ability to disable the watch alarm via the normal Alarms menu. The Watch Alarm lock must be unlocked before alarms can be disabled

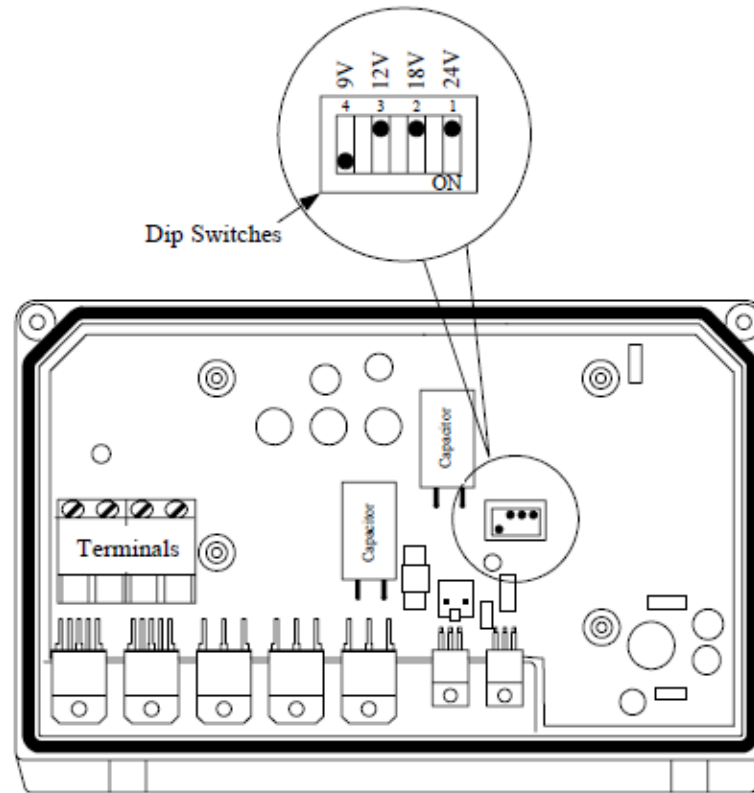
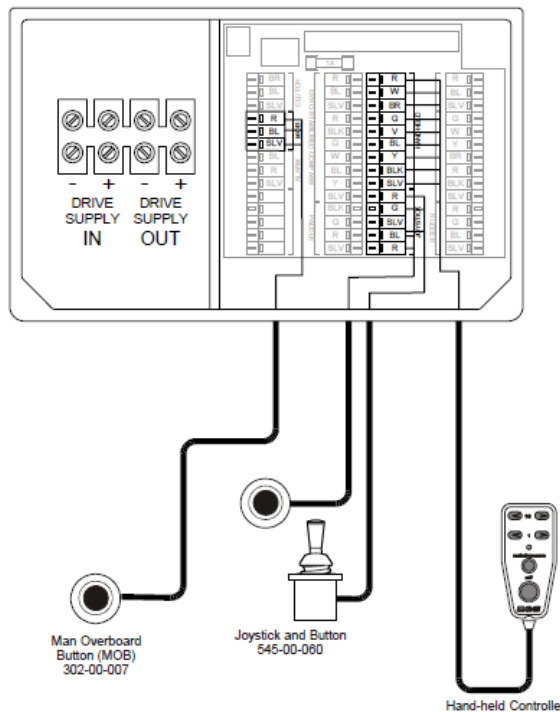
AUTOPILOTS

H3000



Dip Switch Clutch Voltage Drive Size / Type

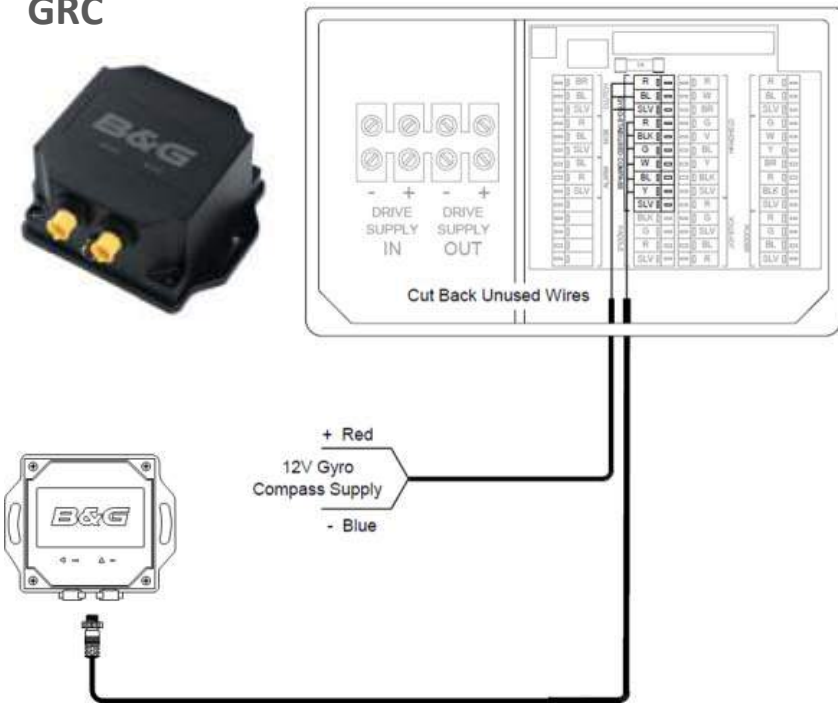
- 1 24V 24V Rotary
- 2 18V Size 3, 24V Ram
- 3 12V 12V Rotary
- 4 9V Size 1 or 2 12V Rams



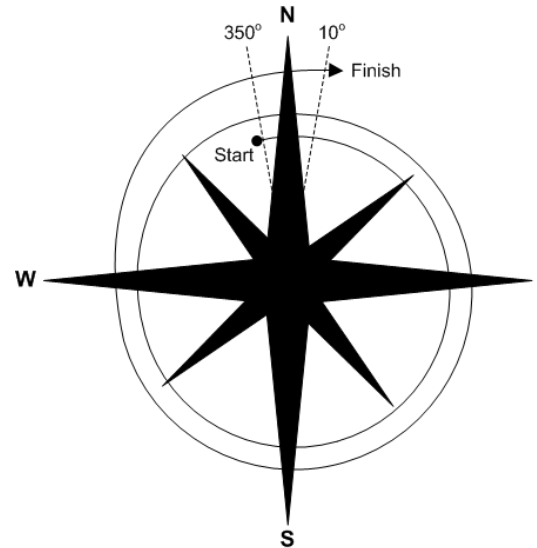
Device	Code
Pilot not Commissioned	100
Pilot Compass Failure	101
Rudder Sensor Out of Range	102
Rudder Drive not Responding	103
No (or Low) Boat Speed	104
System Compass Failure	105
No NMEA Data	106
<NOT USED>	107
Poor Quality NMEA Data	108
No (or Low) Wind Speed	109
No Optimum Wind Data	110
Current Trip: Drive or Clutch	111
Network Communication Error	112
Memory Card Changed - Reset End Stops	113
Memory Card Error	114
Drive Power Failure - Check Supply	115
Network Communication Error - cannot detect Pilot Computer	NO PILOT

AUTOPILOTS

GRC



When Connecting a GRC direct to a H3000 ACP it will require an external power source to be connected (See diagram)



The compass swing for the GRC is an automatic process, you do not need to use a display to complete the swing. Make two 360° turns to starboard within 5 mins after powering on the GRC. Try to have your rate of turn between 2-3 sec/Degree (180 secs/3min for the swing) you must ensure that you pass through north 3 times, as soon as north is passed for the 3rd time the compass swing will complete



If the GRC is connected via a N2K network with a Zeus that is connected to a H3000 processor with Hlink, you can allow the Zeus to transmit Heading data to the H3000. To select this the heading source should be set to 5.

AUTOPILOTS



Zeus Integrated, Triton

Zeus- Ideal for installations where a dedicated control is not required

Triton- For use in Triton instrument system or in conjunction with Zeus

Four Remote Options

WR20 –Full Remote operation.

JS10 – NFU Remote operation.

R3000X-NFU Remote operation

S35-NFU Remote operation

Compasses

FC40 - Fluxgate Compass (Black Case)
Not suitable for MARPA.

RC42 - Rate Compass (white case) also has NMEA 0183 output.

GRC- Gimbaled Rate Compass
Single Axis gyro stabilised sensor that also contains Heel and Trim sensors

Computers

AC12 – Low Power

AC42- High Power

Three Rudder Feedbacks

RF25 - SimNet , Data available with AC unit turned off.

RF300 - Freq

LF3000 - Linear Freq (Uses a Converter LFI3000)

Multiple Drive options

RPU - Reversing Pumps

HLD - Hydraulic Linear Drives

DD15 - Direct Drive

SD10 - Sail Drive

Setup Menus

Rudder Limits - Always change the default value, even if you return it to the same number. The wider the limit you set the less perceived load is calculated on the pump, this is because for the same output the rudder moves a greater amount.

- i.e. 30 Degree Rudder value = 79%
- 45 Degree Rudder value = 53%
- 60 Degree Rudder value = 40%

Rudder - Determines the drive direction and speed required, it also detects if Solenoids are connected by checking if increased voltage increases rudder movement speed.

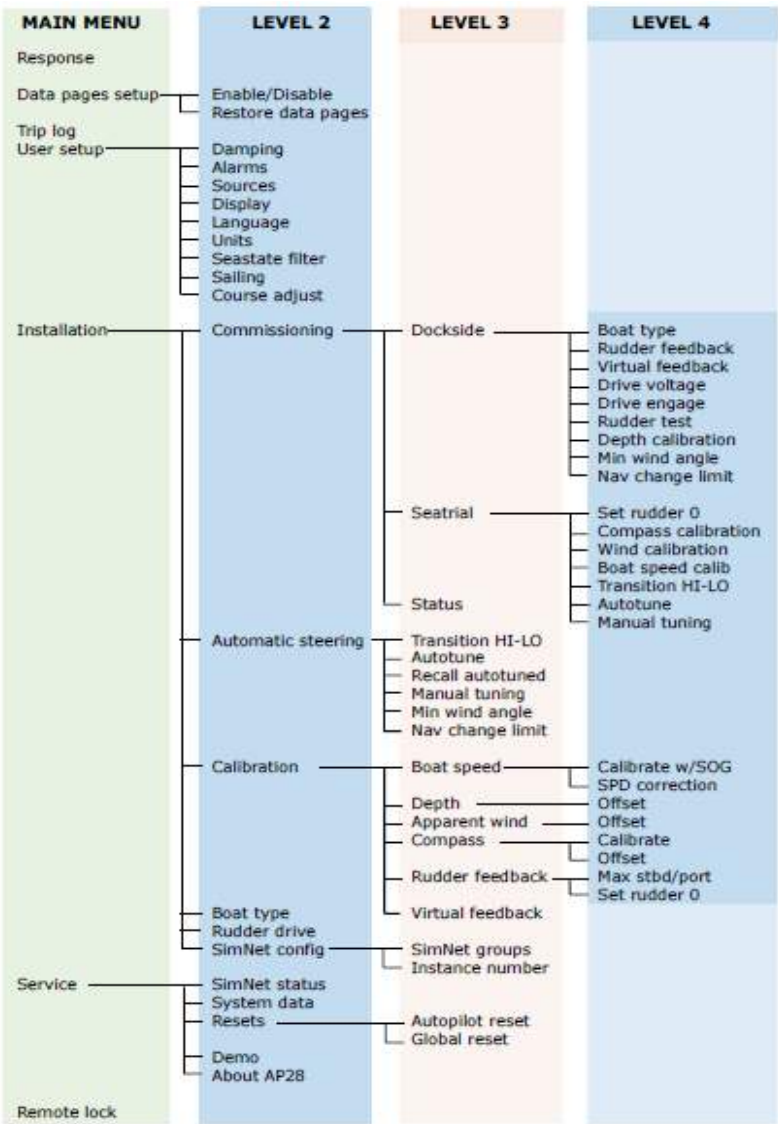
High Voltage - You can use this setting to force Solenoid operation, or set a starting voltage for the drive output. Normally the computer works out this start point by looking for the voltage required to see a rudder movement.

Vessel Transition - Determines when the Autopilot computer applies Hi or Lo gain settings (there is a 1 knot damping to ensure no hunting occurs right at the transition speed).

Steering Parameters – Gain - The default values change with what boat type you select i.e. Sailing, Displacement, Outboard. Counter Rudder - Typically 2x the gain value as a reference.

Compass Calibration - Stored in the compass, so a autopilot reset will not cause you to have to recalibrate the compass. Max recommended Variation value of 30%

Nav Change- Maximum allowable course change that takes place automatically, if the pilot needs to make a turn larger than this value an alarm will sound requiring the operator to approve the course change.



Seatrial

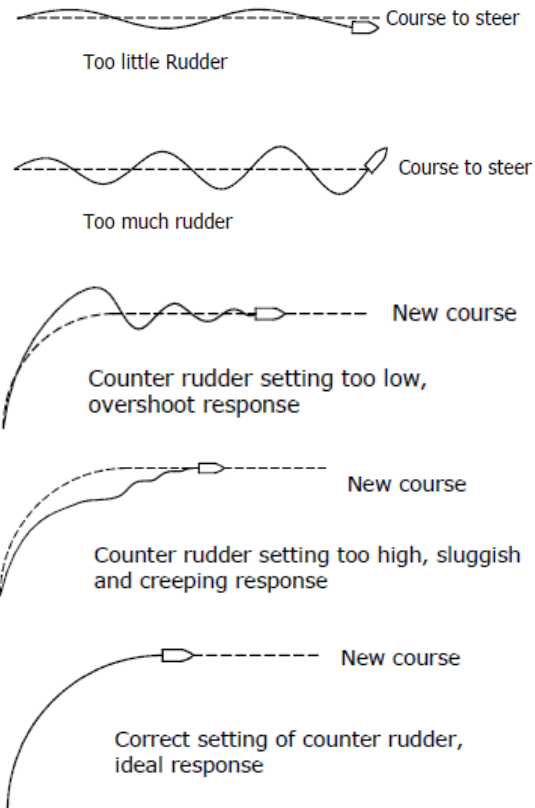
Recommended speed during Automatic tuning should not exceed 10 knots. It should be performed in calm or moderate sea conditions, preferably when steering East or West.

Recall auto tuned - To recall the parameter values that were achieved during the Auto tune procedure, if you have made adjustments and vessel response has got worse, this is an easy way to return to a start point.

Rudder Gain - Too little Rudder and the autopilot fails to keep a steady course. Too much Rudder gives unstable steering and reduces speed.

Counter Rudder

Is the parameter that counteracts the effect of the boat's turn rate and inertia. The best way of checking the value of the Counter Rudder setting is when making turns. The figures illustrate the effects of various Counter Rudder settings.



Maintenance Considerations

The Control units & computer - will under normal use require little maintenance – just keep them dry and clean.

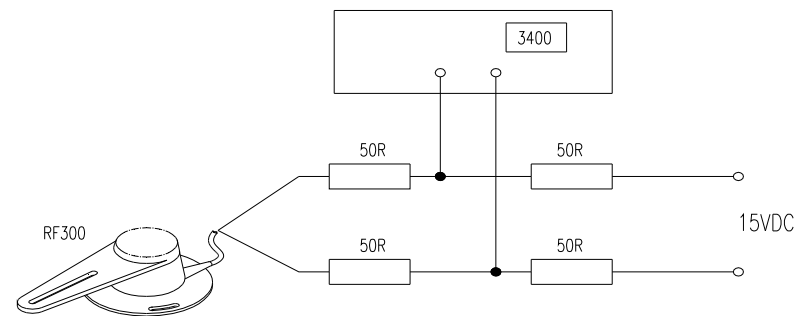
Rudder Feedback - make a visual inspection at 2-3 month intervals and at the start of each season. Apply some grease at the RF25/300 ball joints when required.

Compass - if it is exposed to the weather, make a visual inspection at 2-3 months intervals, and at the start of each season.

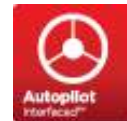
Drive unit - Refer to the manual for maintenance instructions – pay particular attention to the oil reservoir and the motor brushes.

How to check a RF300 outside an autopilot system:

Connect the suspected RF300 to a 15V power supply through a resistor network and oscilloscope or a frequency counter as shown. A working RF will give 3400Hz (square wave) in centre position, changing $\pm 20\text{Hz}$ per degree of movement.



Compass



FC40 - Fluxgate Compass (Black Case)
Not suitable for MARPA.



RC42 - Rate Compass (white case) also has NMEA 0183 output.



GRC - Gimballed Rate Compass has NMEA 0183 output, and Trim and Heel outputs.

COMPASS CALIBRATION AND OFFSET ADJUSTMENT

There are two ways of calibrating the compass depending on type of equipment to be connected.

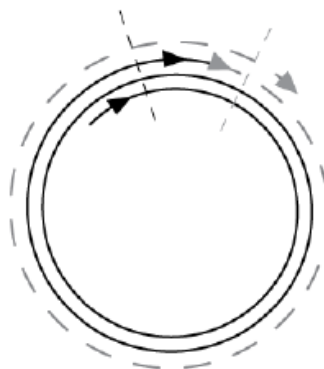
If connected to SimNet, the calibration and offset is carried out using the autopilot or instrument calibration menu. Refer to the autopilot or instrument manual.

Calibration data is stored in the compass and is presented on the autopilot and instrument display.

If connected for NMEA0183 output only, use the following procedure:

1. Make 2 consecutive turns of 360° within 5 minutes after power is applied
 - The calibration procedure will now start automatically

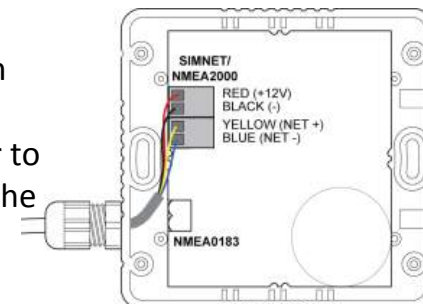
2. Continue with a third 360° turn to complete the calibration



SimNet/NMEA2000 output

This is default connection from factory.

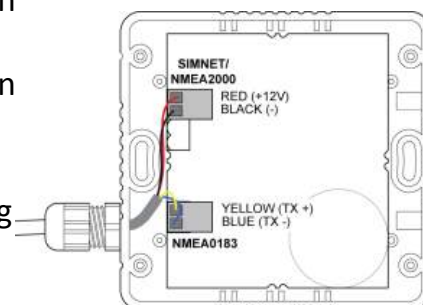
Connect the SimNet connector to an available SimNet socket in the SimNet backbone.



NMEA0183 output only

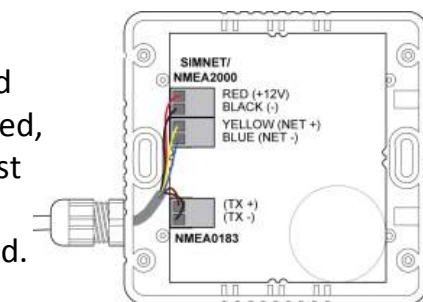
When used as stand-alone with only NMEA0183 output, the cable is connected as shown on the illustration.

Prepare the other end of the cable for hard wiring by cutting off the SimNet connector and stripping the cable.



SimNet/NMEA2000 and NMEA0183 output

If both SimNet/NMEA2000 and NMEA0183 outputs are required, an additional 2-wire cable must be used (not included). Connect the cable as illustrated.



System Resets

Zeus



MENU > MENU > SYSTEM >
RESTORE DEFAULTS

AP



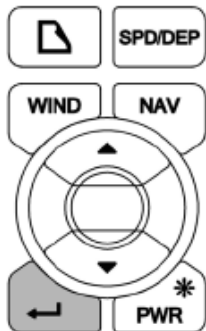
MENU > SERVICE > AUTOPILOT
RESET

Triton



MENU > SETUP > SYSTEM>
RESET TO FACTORY

H3000



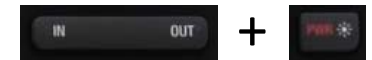
Press and Hold Menu/Enter key, then press Pwr
The SYSTEM DIAGNOSTICS menu is then displayed on the GFD. Now
use the cursor keys to highlight the appropriate Diagnostic
function and press Enter to select.

“System Reset Options”

This option provides the facility to Reset a specific, individual unit
on the system or a group of units as appropriate and returns all the
settings held on that unit, back to factory defaults

Hard Resets

Zeus
Zeus Touch



Zoom In+out key
then power on

Triton



Page+Enter key then
power on

Radar



Broadband Radar



BR24 software version **2.9.273***
Broadband 3G software version
2.9.273*

RI10 AA010189 SimNet connection
for heading, Ethernet for data

RI11 AA010204 Serial connection for
heading and data

BR24 Fault Codes

- Fault Code Meaning
- 0x00010001 Zero bearing fault
- 0x00010002 Bearing pulse fault (parameter 1 =sensor error. Parameter 2 = quadrature sensor error)
- 0x00010003 Motor not turning
- 0x00010004 No communication with antenna
- 0x00010006 Modulation problems
- 0x00010007 Trigger problems

Broadband 3G has a High speed option at 36 Rpm selectable from the Radar Menu, this activates when the radar range is reduced below 2 NM

No Radar

Indicates a Ethernet Communication Failure,
Check Cables and Power Supply

No Scanner

Indicates an internal failure of the BR24

Radar Status Menu, quick overview of system
radar configuration, and fault codes
MENU>MENU>RADAR>INSTALLATION>Radar
Status

The screenshot shows the 'Radar Status' menu with two expandable sections: 'Scanner info' and 'Scanner features'. Arrows point from these sections to detailed views on the right.

Scanner info	
Type	BR24
Software version	2.9.273
Serial number	0907A10108

Scanner features	
Target boost	yes
STC curve	yes
MARPA	yes
Local IR	yes
Fast scan	yes
Sidelobe suppression	yes
Multiradar support	yes

* Check www.bandg.com for the latest version

Radar



Broadband Radar



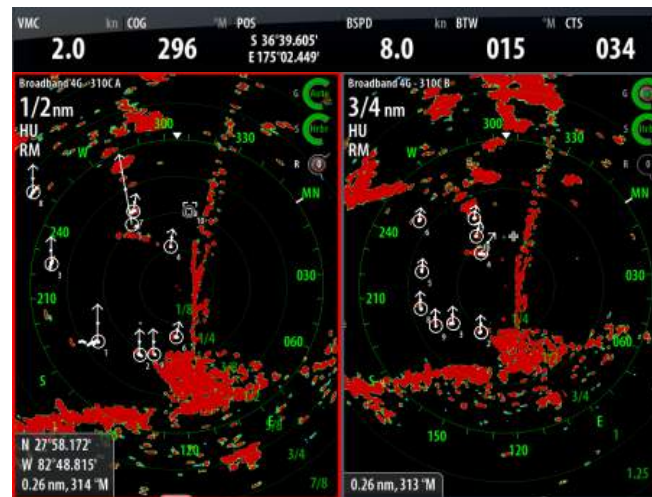
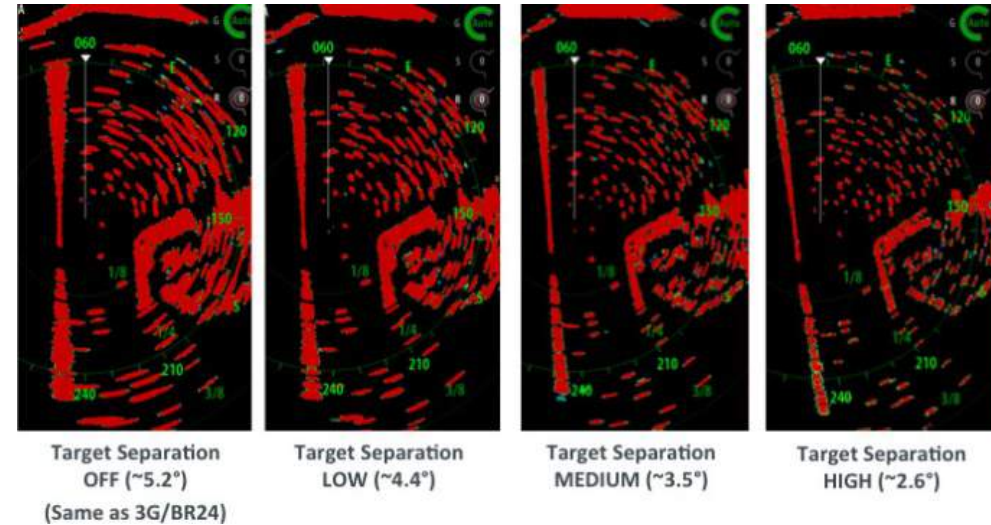
4G software version **4.1.42***

MARPA Target Tracking

Track up to 10 targets as standard or up to 20 in Dual Range mode with independent control.

Speed Modes

24/36/48rpm this adjust based on range and connected product



* Check www.Simrad-Yachting.com for the latest version

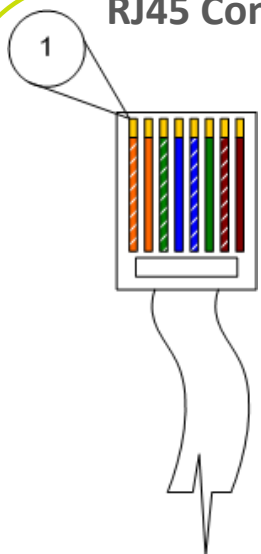
Radar



Broadband Radar

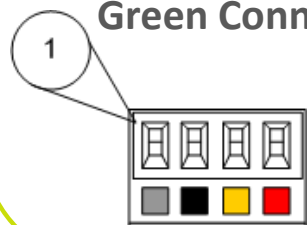


RJ45 Connector

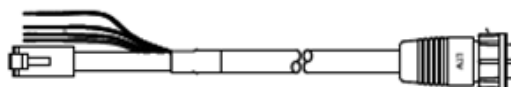


Pin 1 Orange/White
Pin 2 Orange
Pin 3 Green/White
Pin 4 Blue
Pin 5 Blue/White
Pin 6 Green
Pin 7 Brown/White
Pin 8 Brown

Green Connector



Pin 1 Screen
Pin 2 Black
Pin 3 Yellow
Pin 4 Red



Radar plug



Radar Antenna Cables

Part Number	Length
AA010211	10m
AA010212	20m
AA010213	30m

Radar Plug Pin Number	Wire Colour	Radar Interface Pin Number
1	Black	Green Connector Pin 2
2	Red	Green Connector Pin 4
3	Yellow	Green Connector Pin 3
4	Screen	Green Connector Pin 1
5	N/A	N/A
6	Blue	RJ45 Pin 4
7	Blue/White	RJ45 Pin5
8	Brown/White	RJ45 Pin7
9	Brown	RJ45 Pin 8
10	Green/white	RJ45 Pin3
11	N/A	N/A
12	Orange/White	RJ45 Pin 1
13	Green	RJ45 Pin 6
14	Orange	RJ45 Pin 2

Radar



HD Pulse Radar



DX64 4 kW Radome
 TX06 6 kW Open Array
 TX10 10 kW Open Array
 TX25 25 kW Open Array
 AA010014 6 kW Processor
 AA010015 10 kW Processor
 AA010016 25 kW Processor

AA010070 NMEA Heading sensor cable (MARPA)

AT10HD SimNet Heading input device

No Radar

Indicates a Ethernet Communication Failure,
 Check Cables and Power Supply

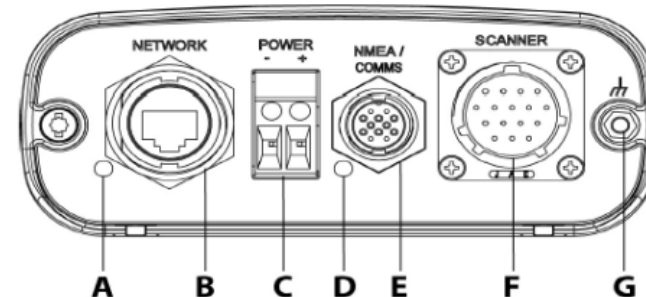
No Scanner

Indicates a failure between the radar
 processor and the scanner, Check cable and
 connections

MARPA Mini Automatic Radar Plotting Aid

available on NSE/NSO/NSS, and GB40 systems. Must have the
 following to be enabled:

- Heading sensor (RC42)
- GPS Fix



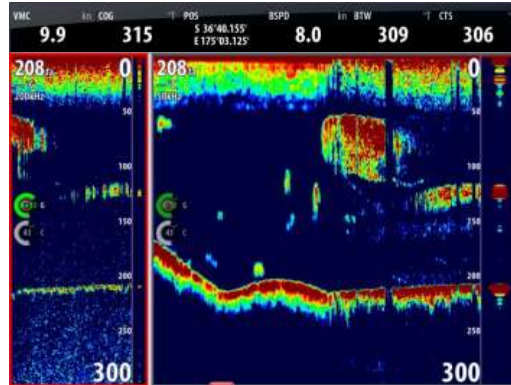
Key	Item Function	Connects to..
A	LED Ethernet	Flashing indicates communication
B	Network Connector	Use adapter P/N 000-10438-001 to convert to 5 pin yellow Ethernet connector
C	Power Terminals	2Kw 12VDC 4KW 12-24VDC 6KW 12-24VDC 10KW 24VDC 25 KW 24VDC
D	LED Power	Solid on, powered on
E	NMEA/Serial Coms	For high speed heading input required for MARPA
F	Scanner Communications	Scanner
G	Ground	Vessel Ground

* Check www.Simrad-Yachting.com for the latest version

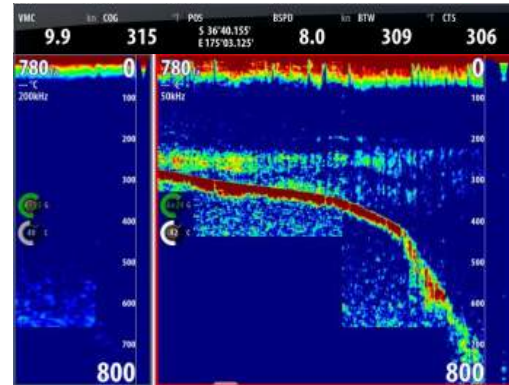
Sounder



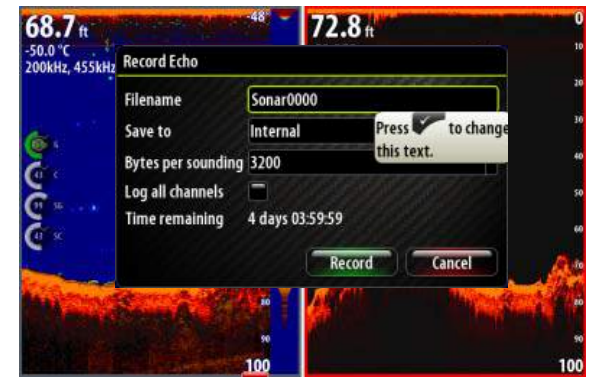
Broadband Sounder BSM-1 000-0132-05



208 Fathoms (380 m, 1248 Ft)
Using BSM-1 on 200 kHz



780 Fathoms (1426 m, 4680 Ft)
Using BSM-1 on 50 kHz



Echo Logging on Zeus

BSM-1 LED Guide

- Power LED** - Green all ok
- Power LED** - Red, System starting
- Power LED** - Red Flashing, Internal error
- Network LED** - Green, Network data
- Transducer LED** - Green, All ok
- Transducer LED** - Green Flashing
Searching for Bottom

Transducer Options

Part Number	Description
000-0136-02	Airmar P319 Low profile th depth/temp
000-0136-03	Airmar P79 In-hull depth only
000-0136-04	Airmar B60 LP-BTH Depth/Temp w/12 degree tilt
000-0136-05	Airmar B744V Thru hull DST w/high fairing blk
000-0136-06	Airmar B164 1kW D/T w/12 degree tilt
000-0136-00	Airmar B258 BTH D/T w/High speed fairing kit
000-0106-82	Airmar B260 BTH w/ High speed fairing kit
000-0106-91	Airmar M260 depth only w/In hull tank kit
000-00021-001	Airmar B60 LP-BTH Depth/Temp 50/200 kHz w/20 degree tilt
000-00022-001	Transducer Adapter Cable (LTW 6 pin to 7 pin Blue)

Transducer Connector



Pin	Color	Description
1	Red	Depth +ve
2	Blue	Speed Signal
3	Orange	Speed Pwr
4	Yellow	Temp +
5	Black	Depth -ve
6	Depth	Shield
7	Shield	Temp/Speed -ve

Sounder

Transducers

Select Transducer

Airmar P319 (5k)

Airmar P79

Airmar P66 (5k)

Airmar B45 (5k)

Airmar B60 (5k)

Airmar B164 (5k)

Airmar B258 (5k)

Airmar B260 (5k)

Airmar M260 (5k)

Airmar B744 (5k)

Generic 50/200kHz (10k)

Airmar P319 (10k)

Airmar P66 (10k)

Airmar B45 (10k)

Airmar B60 (10k)

Airmar B164 (10k)

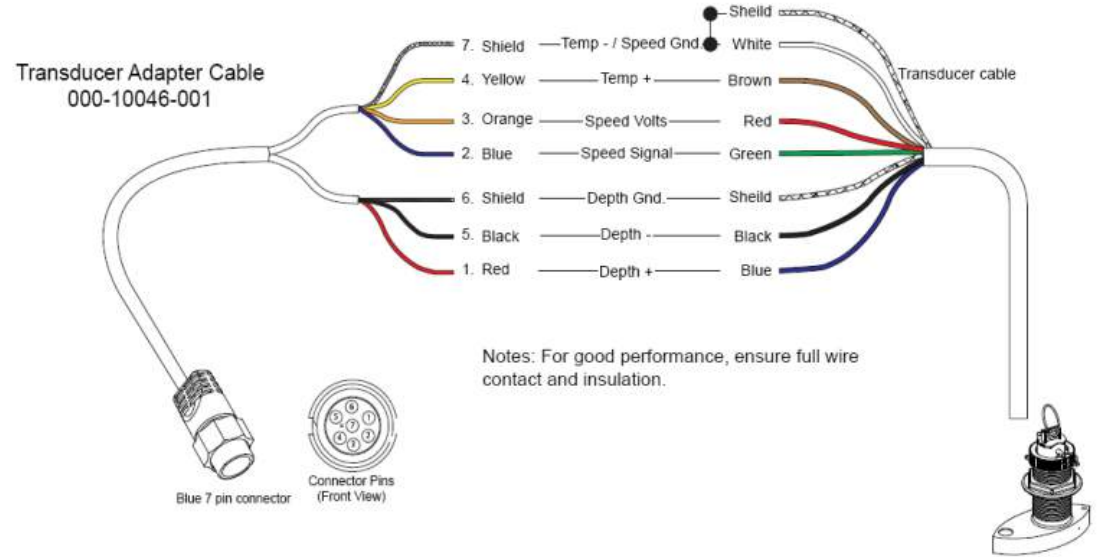
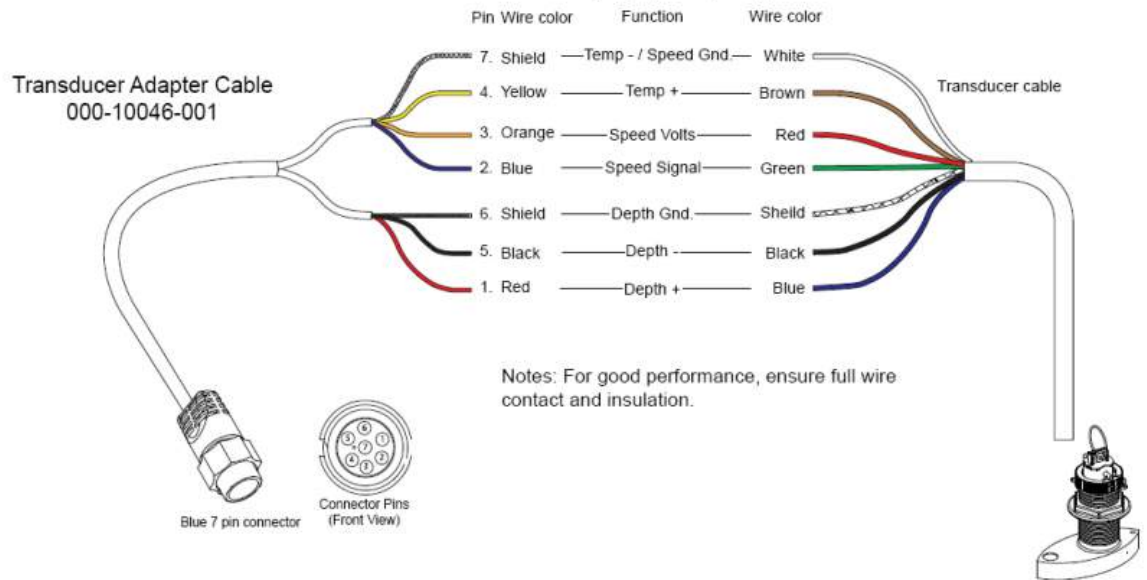
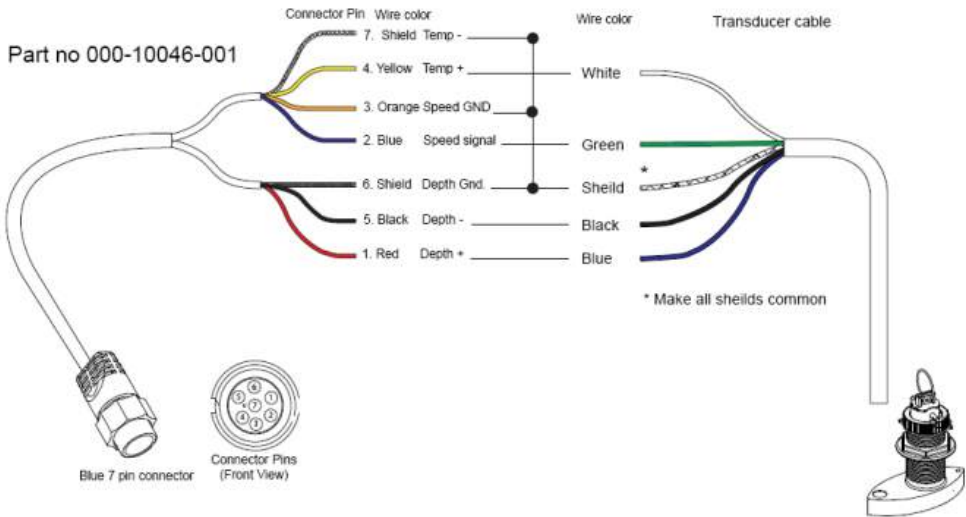
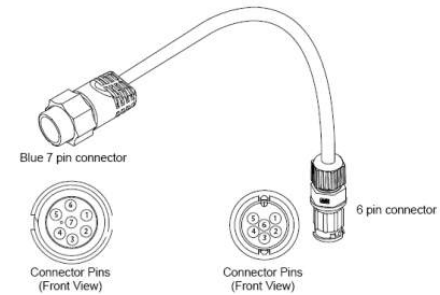
Airmar B258 (10k)

Airmar B260 Simulating

Transducer Type Selection is accessed through:
MENU>MENU>ECHO>INSTALLATION>
TRANSDUCER TYPE

You can select the different type of transducer
and thermistor resistance values

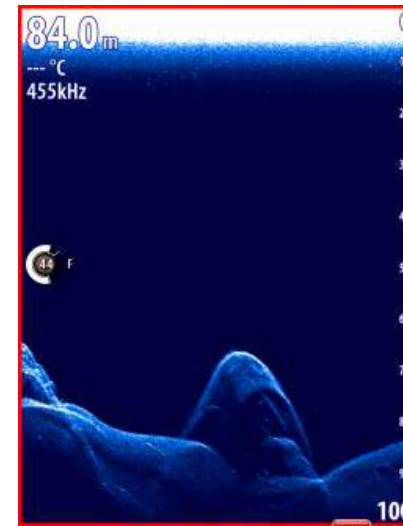
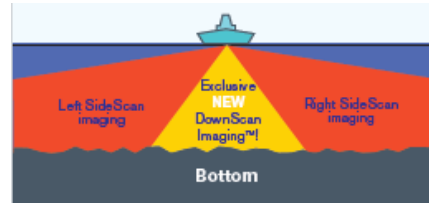
LTW 6 Pin to 7 Pin Adapter cable
Part No. 000-00022-001



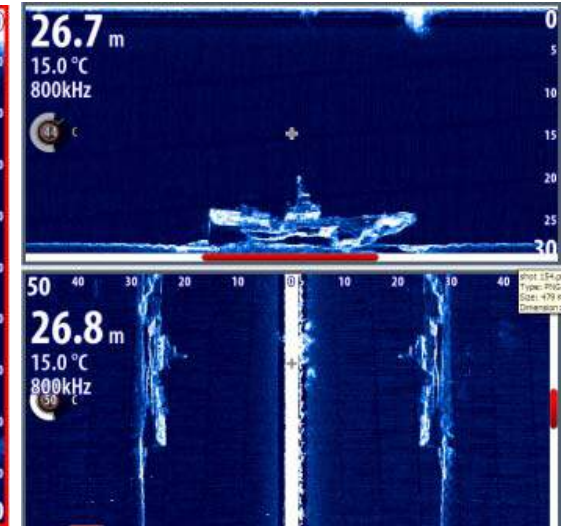
StructureScanHD



StructureScan

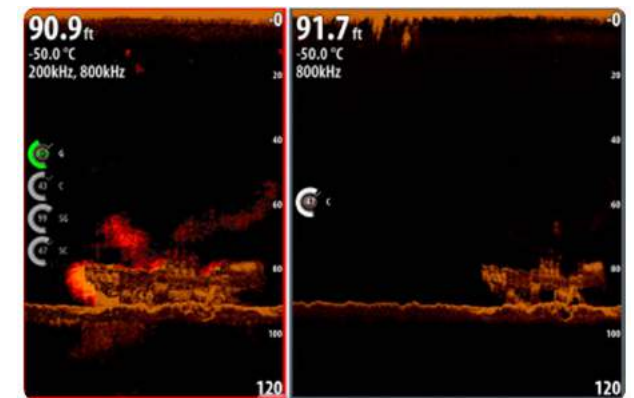
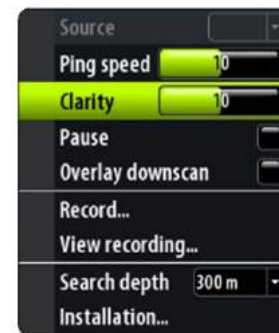


84 meters (275ft) Using LSS-1
on 455Khz



Wreck In Hauraki Gulf NZ
on 800 kHz

Structure ScanHD has improved over LSS-1 by decreasing the beamwidth of the transducer to 3.2 degrees from 6 degrees (455Khz) this allows higher resolution images to be displayed, also range has been improved with hardware modifications to the Structure scan module and the ability to use the 455KHz with higher resolution.



Structure Scan Overlay with BSM-1 on Zues

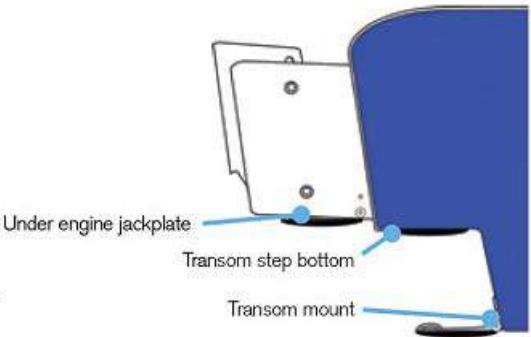
StructureScanHD



StructureScan
LSS-2 000-10801-001

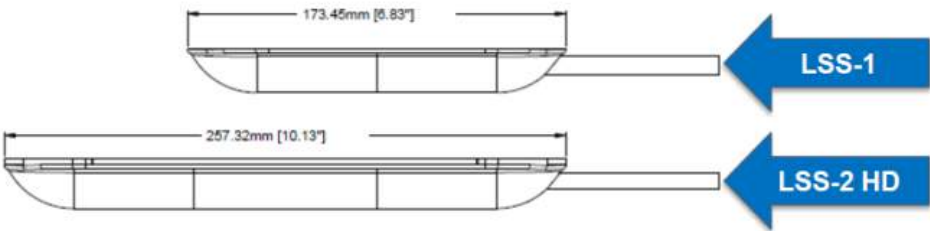


- 000-10800-001 StructureScan **HD** With Transom-Mount Transduce
- 000-10801-001 StructureScan **HD** Without Transducer
- 000-10802-001 StructureScan **HD** Skimmer Transducer
- 000-10946-001 StructureScan HD Bronze THRU-HULL Single
- 000-10947-001 StructureScan HD Bronze THRU-HULL Pair
- 000-10948-001 StructureScan HD Bronze THRU-HULL Port
- 000-10949-001 StructureScan HD Bronze THRU-HULL Starboard



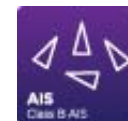
StructureScan LED Guide

- Transducer** LED Off Transducer not connected or cable/ transducer is broken
- Transducer** LED Green/ stays on Transducer detected/locked on to bottom
- Transducer** LED Green/flashing Transducer detected/not locked on to bottom
- Power** LED Off No power or not switched on
- Power** LED Red/stays on Product in booting process
- Power** LED Green/stays on Product has power/operating
- Power** LED Red/Green/slow flashing Product in factory mode; Ethernet communication working
- Power** LED Red/Green fast flashing Product in factory mode; no Ethernet communication (cable unplugged or bad cable)
- Ethernet LED Off No Ethernet communication (cable unplugged, bad cable or remote unit not powered)
- Ethernet** LED Yellow/Flashing-solid Ethernet communication is working (link/active light)



Compatibility Chart		
LSS-1 StructureScan/LSS-2 StructureScanHD		
	HD Txdr	LSS-1 Txdr
LSS-1 StructureScan	YES	YES
LSS-2 StructureScanHD	YES	With Adapter

AIS



NAIS-400

SMA Connector
GPS Antenna

SO239 Connector
VHF Antenna
or NSPL-400



Power /NMEA
Connector

Power
supply
12-24VDC
3A Fuse

Mini- B USB Connector
N2K Network

Micro C Connector
N2K Network

*Amber Flashing =waiting for GPS Fix

PWR LED Status (GREEN)

The AIS transceiver is powered up, has a position fix and has transmitted at least one vessel information report.

BIIT LED Status (RED)

The AIS transceiver has detected a system error. The likely causes of this are detailed in the troubleshooting guide in chapter 6. Diagnostic messages displayed in proAIS2 may also help troubleshoot the cause of the error.

TX LED Status* (AMBER)

The AIS transceiver is in 'transmit timeout' mode. This can be for a number of reasons:- the unit has only recently been powered on and is obtaining a position fix prior to transmitting its first vessel information report. (This can take several minutes). - position fix has been lost. The AIS transceiver will attempt to regain position fix for 30 minutes before entering an error state. - the AIS radio channels are exceptionally busy, so there is currently no available timeslot for transmission. - the unit has been in silent mode and after deactivating silent mode this amber indicator will illuminate until the first AIS message has been sent. - the AIS transceiver has been commanded by the local authority (via an AIS base station) to cease transmissions.

RX LED Status (BLUE)

When silent mode is activated, using the optional silent mode switch, the blue and amber LED are illuminated to show that the transmitter is disabled.

Power/NMEA Connector Pinout

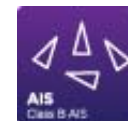
Wire colour	Pin no.	Description	Function
Red	8	Power in +	Power supply connections 12 V to a24 V DC
Black	9	Power in -	
Light green	12	Switch input-	External switch connection for silent mode
Orange	10	Switch input+	
Brown	1	NMEA 0183 port 1 TX+ (Transmit +)	High speed NMEA 0183 - Port 1: (38,400 baud) intended for connection to chartplotters
Blue	2	NMEA 0183 port 1 TX- (Transmit -)	
White	3	NMEA 0183 port 1 RX+ (Receive +)	
Green	4	NMEA 0183 port 1 RX- (Receive -)	Low speed NMEA 0183 - Port 2: (4,800 baud) intended for connection to other NMEA 0183 compatible devices
Purple	11	NMEA 0183 port 2 TX+ (Transmit +)	
Pink	7	NMEA 0183 port 2 TX- (Transmit -)	
Grey	6	NMEA 0183 port 2 RX+ (Receive +)	
Yellow	5	NMEA 0183 port 2 RX- (Receive -)	

Colour-coding of wires in the accessory cable

Part numbers

000-10980-001	NAIS-400 Class B AIS W/GPS ant
000-10981-001	NAIS-400 Class B AIS (No GPS ant)
000-10982-001	NSPL-400 VHF Antenna Splitter
000-10400-001	GPS 1330-N antenna

AIS



NSPL-400

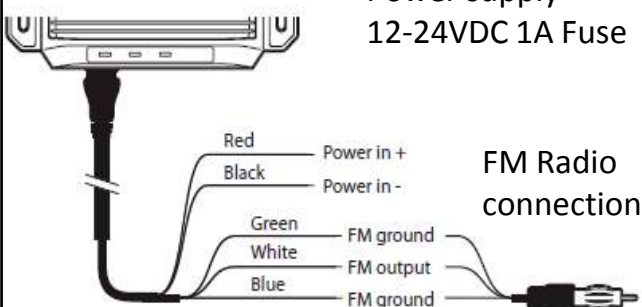
BNC Connector
NAIS-400

SO239 Connector
VHF Radio

SO239 Connector
VHF Antenna



Power /FM Radio/
Sonic Hub
Connector
Power supply
12-24VDC 1A Fuse



PWR LED Status (GREEN)

Indicates power is connected

VHF (YELLOW)

Indicates VHF Transmission

AIS (YELLOW)

Indicates AIS Transmission

Note-TX AIS and TX VHF may both illuminate with some brands of VHF radio when transmitting in high power (25W) this is normal behaviour and does not indicate a fault

Masthead SS Antenna



30 metres Cable



Whip Antenna



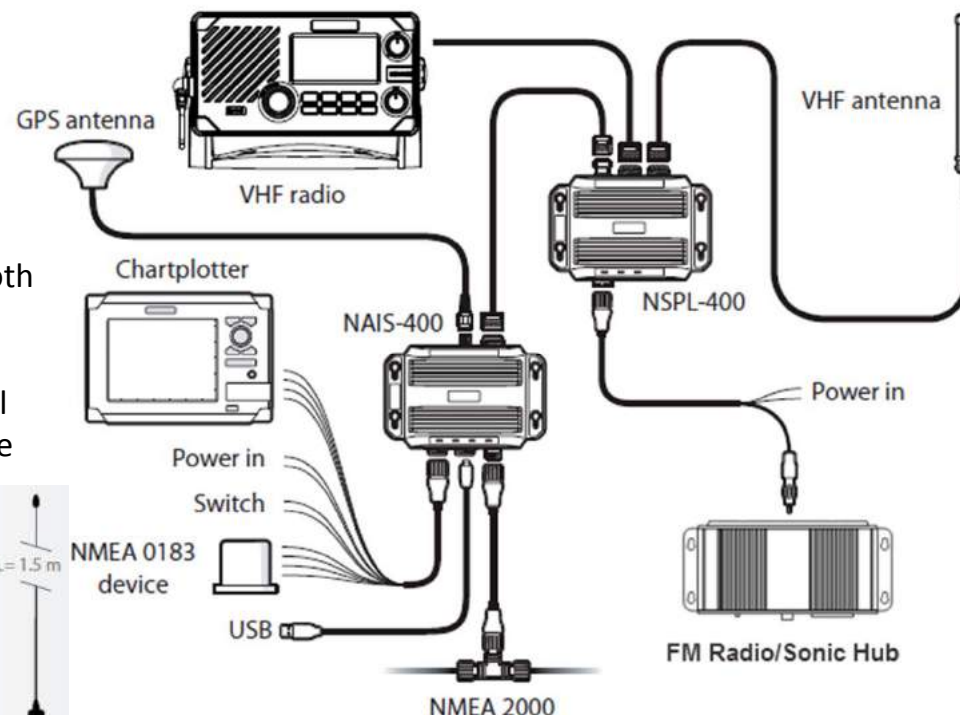
Antenna mount Stainless Steel
Stainless Steel 4-way adjustable
antenna mount.



Plastic Whip Antenna



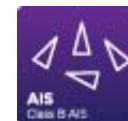
Antenna Mount Heavy Duty Nylon
White plastic 4-way adjustable
antenna mount.



VHF Antennas

AA000352	1.1 m stainless masthead whip antenna
AA000223	Nylon quick-fit antenna mount
AA000224	Stainless steel quick-fit antenna mount
AA000355	2.4 m (8-ft) fiberglass quick fit VHF antenna
AA000358	1 m (3.3-ft) fiberglass quick fit VHF antenna

AIS

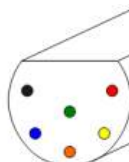


AI50



Vessel Configuration Information	
MMSI	260619550
Vessel Type	Pleasure Craft
Vessel Dimensions & GPS Reference	Length - 030m + 005m Beam - 03m + 03m
Call Sign	ABC123
Vessel Name	SHAMROCK
▲▼ to Select, ►/◄ MENU to Adjust, ◀ to Exit	

Black= -ve 12VDC
Red= +ve 12VDC
Blue = NMEA OUT -ve
Orange = NMEA OUT +ve
Green = Not Used
Yellow= Not Used



AI50 will only operate as a receiver, until you have fully completed the ship configuration. Once the vessel MMSI is entered and confirmed it will become locked and shown as greyed out, indicating that it can be viewed, but not edited.

NMEA 0183 Data Connections

Out +ve Orange (NMEA0183-HS is specified at 38.4kbaud and
Out -ve Blue is not compatible with conventional
NMEA0183, which operates at the lower
speed of 4800 baud)

AI50 has 2 SimNet ports

NAIS-300 Has a single SimNet port

NAIS-300 Uses PC software to configure for the first time, and to allow for vessel monitoring also.

NAIS300 and AI50 must have their own GPS Antenna you can not use a SimNet GPS Source

AI50 and NAIS-300 can have MMSI reset by an authorised Simrad Distributor

AI50 GPS Antenna P/N 003-7679-00

NAIS-300

PWR LED Status (GREEN)

After the initial five seconds the LED will do two short flashes and a longer pause before repeating itself. This will continue until a GPS position fix is acquired. Once a fix has been obtained, the LED will stay on continuously.

If, after a GPS fix, the transmitter has been disabled, (Silent Mode), the LED will flash at a rate of one second on and one second off.



BIIT LED Status (RED)

- 1 flash - transponder not respond
- 2 flashes - transmitter fault
- 3 flashes - receiver fault
- 4 flashes - VHF antenna VSWR fault
- 5 flashes - no GPS sensor
- 6 flashes - noise threshold exceed
- 7 flashes - voltage supply fault

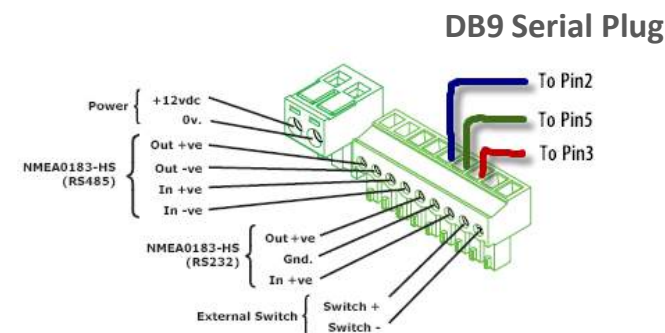
TX LED Status (YELLOW)

After the initial five seconds the LED will then go out. It will light for one second every time the unit transmits its AIS data. The LED will also light continuously if a transmitter time-out fault occurs.

RX LED Status (BLUE)

After the initial five seconds the LED will flash every time an AIS message is received.

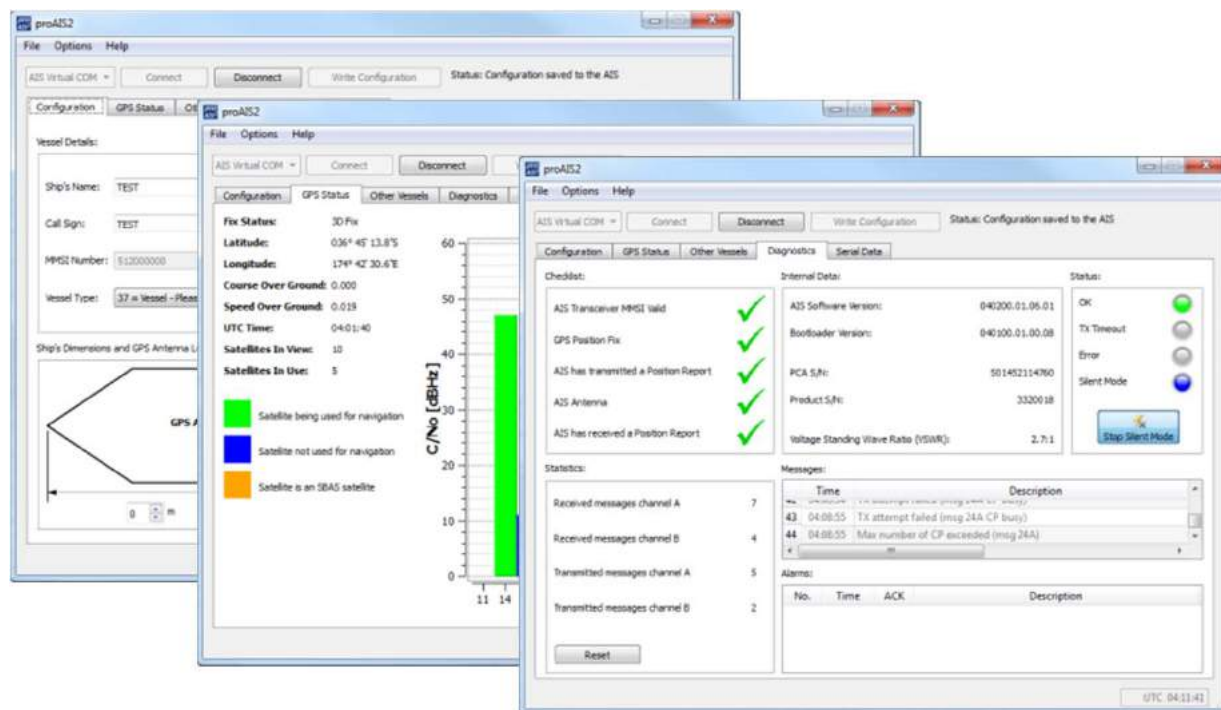
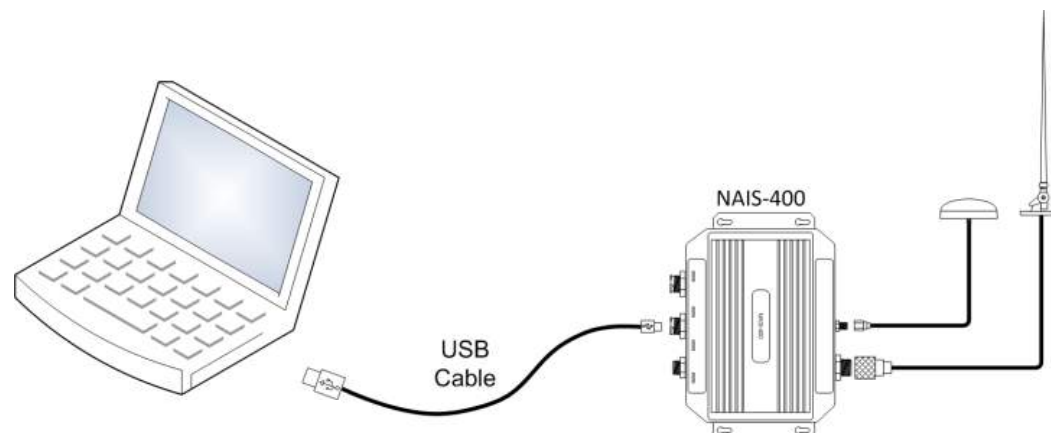
NAIS-300 GPS Antenna P/N 000-10400-001



AIS Software ProAIS2



The ProAIS2 software allows programming of the NAIS unit. MMSI, Radio Call Sign and Ship's Name are entered, so that the correct information is transmitted. If this information is not entered, transmission is inhibited and the NAIS will only receive transmissions from other vessels.



The NAIS-400 uses a USB connection to a PC and does not require power to be connected for programming. The software is provided on CD with the NAIS-400. Follow the onscreen instructions to load and operate the software.

The Diagnostics and Serial Data pages are useful tools for checking the operation and status of the AIS. The MMSI is reset using a separate software program called MMSI Reset, Contact Navico Tech Support if you require it

AIS



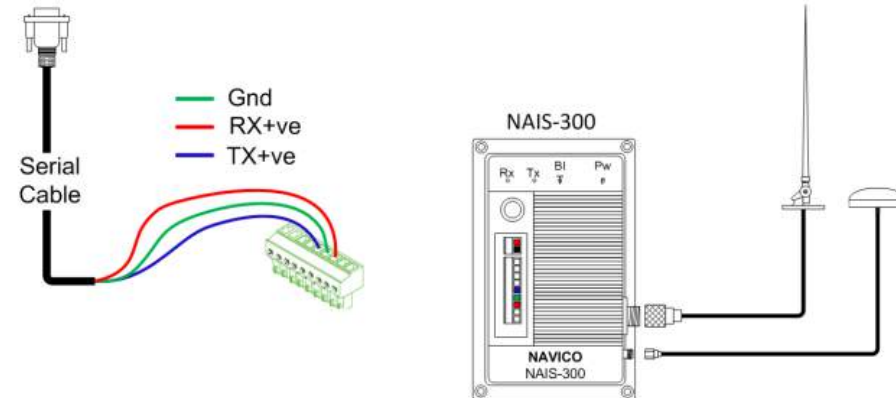
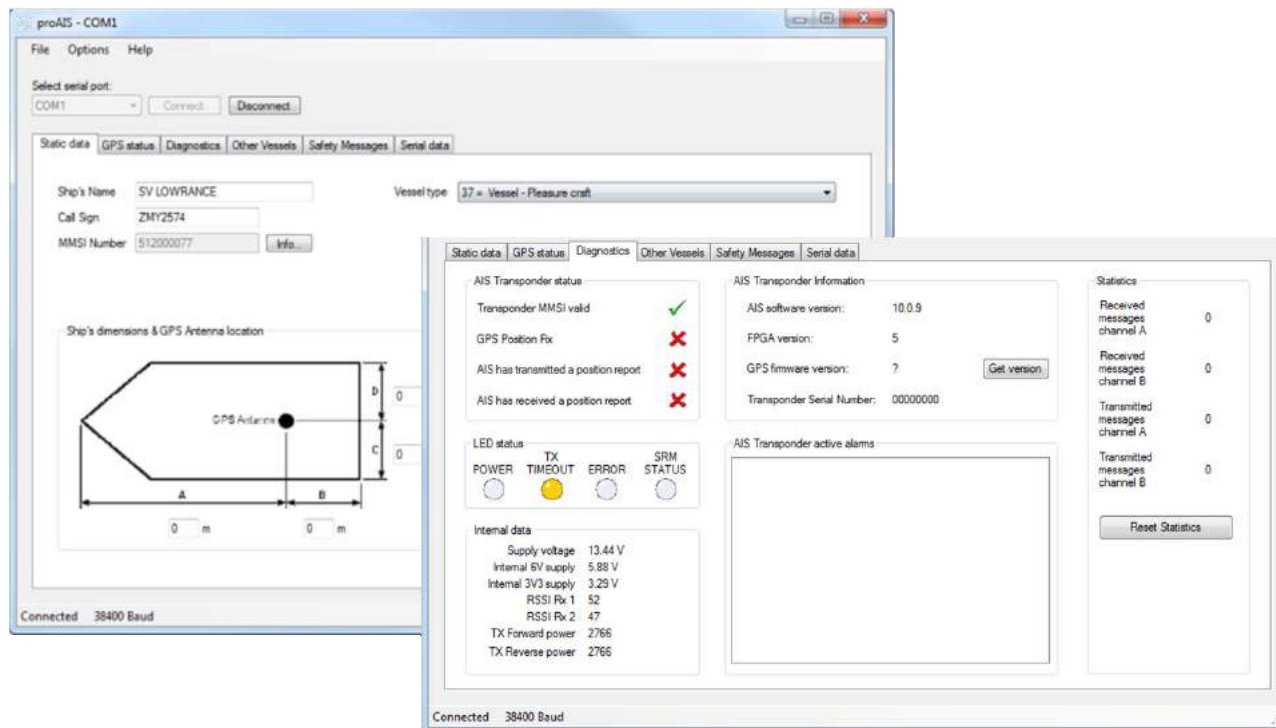
AIS Software ProAIS



The ProAIS software allows programming of the NAIS unit. MMSI, Radio Call Sign and Ship's Name are entered, so that the correct information is transmitted. If this information is not entered, transmission is inhibited and the NAIS will only receive transmissions from other vessels.



The NAIS-300 is connected via a serial cable (Included NAIS-300) to a computer for programming. The software is provided on CD. Follow the onscreen instructions to load and operate the software. The Diagnostics and Serial Data pages are useful tools for checking the operation and status of the AIS



To reset the MMSI, specific serial command must be sent to the NAIS-300, contact Navico Tech support if you require this

Class B AIS should transmit Message 18

- 30s if $SOG > 2 \text{Knots}$
- 3min If $SOG \leq 2 \text{Knots}$.

Class B AIS should transmit Message 24

Every 6 min in addition to and independent to Message 18
Message 24B should be transmitted within 1 min of 24A

Static

- Identification (MMSI)
- Name of ship
- Type of ship
- Vendor ID (optional)
- Call sign
- Dimensions of ship and reference for position.

The default value for type of ship should be 37 (pleasure craft).

Dynamic

- Ship's position with accuracy indication and integrity status
- Time (UTC seconds)
- Course over ground (COG)
- Speed over ground (SOG)
- True heading (optional).

Message 18: Standard Class B equipment position report

Parameter	Description
Message ID	Identifier for Message 18; always 18
Repeat indicator	Used by the repeater to indicate how many times a message has been repeated. Should be 0 for "CS" transmissions
User ID	MMSI number
Spare	Not used. Should be set to zero. Reserved for future use
SOG	Speed over ground in 1/10 knot Max 102.2Knts Will display this if above.
Position accuracy	1 = high ($\leq 10 \text{ m}$) 0 = low ($> 10 \text{ m}$) 0 = default
Longitude	Longitude in 1/10 000 min
Latitude	Latitude in 1/10 000 min
COG	Course over ground in 1/10 Degree
True heading	Degrees (0-359) (511 indicates not available = default)
Time stamp	UTC second when the report was generated by the EPFS (0-59)

Message 24: Static data report

Message 24 Part A and Part B should be used by Class B “CS” shipborne mobile equipment. The message consists of two parts. Message 24B should be transmitted within 1 min following Message 24A.

Simrad Class B AIS will also transmit and receive Message 14 Safety Related Messages

Parameter	Description
Message ID	Identifier for Message 24; always 24
Repeat indicator	Used by the repeater to indicate how many times a message has been repeated. 0 = default; 3 = do not repeat any more
User ID	MMSI number
Part number	Identifier for the message part number; always 1 for Part B
Type of ship and cargo type	0 = not available or no ship = default 1-99 = as defined in § 3.3.2 Not applicable to SAR aircraft
Vendor ID	Unique identification of the Unit by a number as defined by the manufacturer (option; “@@@@@@@” = not available = default)
Call sign	Call sign of the MMSI-registered vessel. 7 X 6 bit ASCII characters, “@@@@@@@” = not available = default
Dimension of ship/reference for position. Or, for unregistered daughter vessels, use the MMSI of the mother ship	Dimensions of ship in metres and reference point for reported position (see Fig. 41 and § 3.3.3). Or, for an unregistered daughter vessel, use the MMSI of the associated mother ship in this data field. For SAR aircraft, the use of this field may be decided by the responsible administration. If used it should indicate the maximum dimensions of the craft. As default should A = B = C = D be set to “0”

Parameter	Description
Message ID	Identifier for Message 24; always 24
Repeat indicator	Used by the repeater to indicate how many times a message has been repeated. 0 = default;
User ID	MMSI number
Part number	Identifier for the message part number; always 0 for Part A
Name	Name of the MMSI-registered vessel. Maximum 20 characters 6-bit ASCII, @@@@@@@@@@@@@@@@@@@@ = not available = default. For SAR aircraft, it should be set to “SAR AIRCRAFT NNNNNNN” where NNNNNNN equals the aircraft registration number

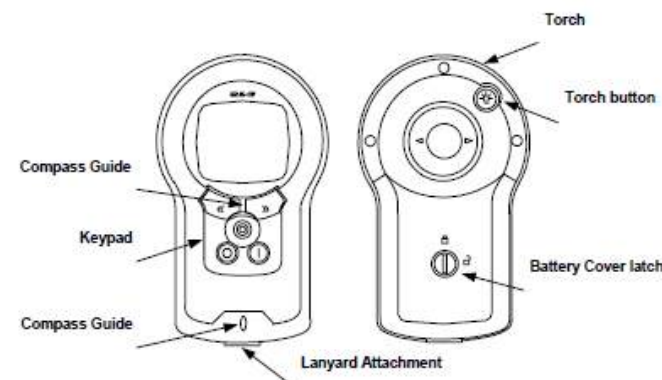
Wireless Remote Control

H3000 Only

Remote Vision



- 100 x 128 pixel, transfective LCD display
- Rugged ABS enclosure, sealed to IP68 standards
- Backlit electroluminescent display, with white LED for keypad
- Automatic backlight illumination
- 20° fully gimballed fluxgate compass
- Powered by two AA 1.5V batteries, standard or rechargeable
- Wireless link to B&G RemoteVision Wireless Port
- High brightness white LED torch
- Up to 60 hours continuous usage battery life



The RemoteVision is a wireless link to your H3000 system, as well as a Pilot control, all in the palm of your hand. View data from your system, including Boat Speed, Wind, Heading or Depth information and check your Heading and Bearing to Waypoint, then alter course from anywhere onboard up to 50m from the wireless port

Discharged batteries can cause the compass calibration to fail

Wireless Remote Control

Remote Vision

Main Menu

Hand Compass
Setup

Pairing with the Wireless Port

Press the ON key to switch the RemoteVision on. If the unit is not paired with the Wireless Port the Main Menu will immediately appear as shown on the left. If the Main Menu appears in full (i.e. with Instruments as the top item) the unit is already paired and no further action is required.

Set the Wireless Port PIN code

On any FFD on the instrument system select BASE STN from the MISC menu on the upper line of the display.

On the lower line of the display SCROLL DOWN to CALBRATE, then CAL VAL 2, then press ENTER to display the PIN code for the Wireless Port.

You may set your own PIN code by pressing ENTER, then SCROLL UP/DOWN until you have selected your choice of PIN, press ENTER to fix your PIN number.

Setting the Wireless Port into Pairing Mode

After you have set the PIN code SCROLL DOWN to CAL VAL 1, press ENTER to display the PAIR setting which will be set to "0".

Press ENTER, then SCROLL UP so that the PAIR setting is "1", press ENTER. At this stage the Wireless Port becomes "visible" to the RemoteVision for a short period of approximately one minute.



Setup

Display
Sleep Mode
Key Lock
Wireless
Compass Setup
Status

Select

BaseStation

Name: Test
Address: 0002ee79b5a6

Set Pin Code

1234

Upper Limit 9999
Lower Limit 0
OK

Main Menu

Instruments
Pilot
Hand Compass
Instrument Control
Timer
Setup

Completing the Pairing

Whilst the Wireless Port is "visible" it is necessary to select "Pair Device" on the RemoteVision, this is found under Setup – Wireless – Pair Device.

After selecting Pair Device there will be a short delay while the RemoteVision searches for Wireless Ports in range. After this period the unit should report "Device Found", followed by the details of your Wireless Port. Select your Wireless Port from the list (there will usually only be one on the list) by pressing the NavStick. You will be prompted to enter your PIN code, enter the same PIN for the Wireless Port (above).

Numeric values are entered using the NavStick:

- Highlight the number field to edit using up/down on the NavStick
- Press Enter to enable editing
- Select the figure to edit using left/right on the NavStick
- Alter the figure selected using up/down on the NavStick
- When complete confirm by clicking the NavStick inwards

Once you have confirmed the PIN the unit should connect and display the full Main Menu after a short delay.

Ethernet Hubs

NEP (Navico Expansion Port)

NEP-1 000-10238-001



NEP-2 000-10029-001



LSS-1 000-10238-001

LSS-2 000-10800-001
(Has 3 ports built in)



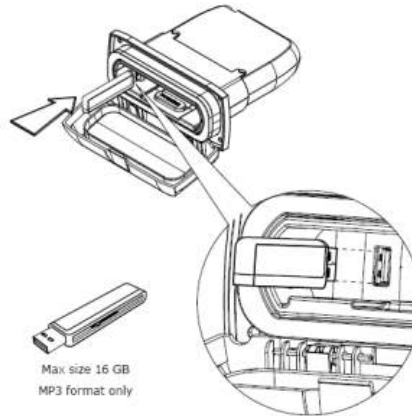
Zeus 8,12
(Has 3 ports built in)



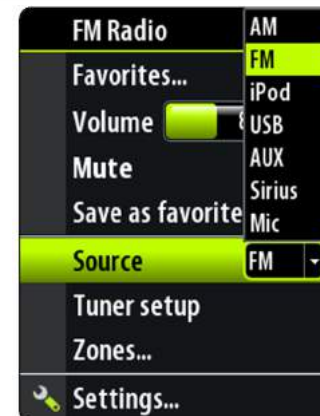
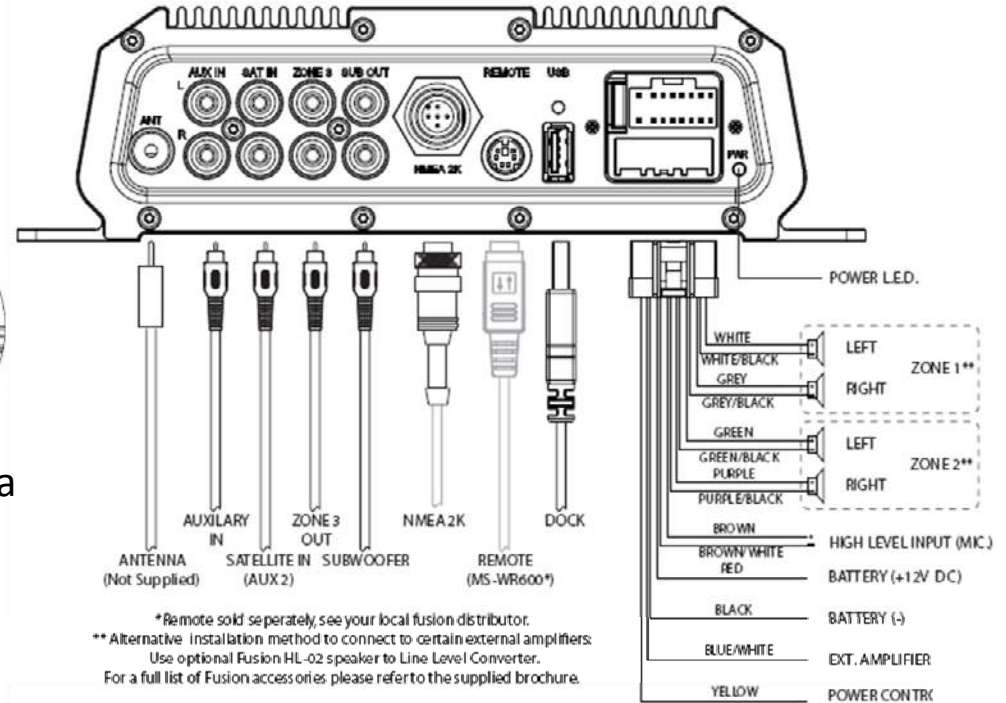
Ethernet Cables

Description	Length	P/N
Ethernet Cable Yellow	2 m	000-0127-51
Ethernet Cable Yellow	4.5 m	000-0127-29
Ethernet Cable Yellow	7.7 m	000-0127-30
Ethernet Cable Yellow	15.2 m	000-0127-37
Ethernet Adapter RJ45 Male- Yellow Female X-over		000-10437-001
Ethernet Adapter RJ45 Male- Yellow Female		000-10438-001
Ethernet Adapter RJ45Female-Yellow Male	2 m	000-0127-56
Ethernet Adapter RJ45Female-Yellow Male	0.3 m	000-0127-55
Ethernet RJ45 male-male	2 m	AA010080
Ethernet RJ45 male-male	10 m	AA010082
Ethernet RJ45 Field Termination		AA010085

Sonic Hub



Control of the Sonic Hub is via a Zeus, or Fusion remote. To enable Sonic Hub Press MENU:MENU>System>Advanced>Features and check Sonic Hub, Also in the System menu you can turn the Sonic Hub info bar on and off.



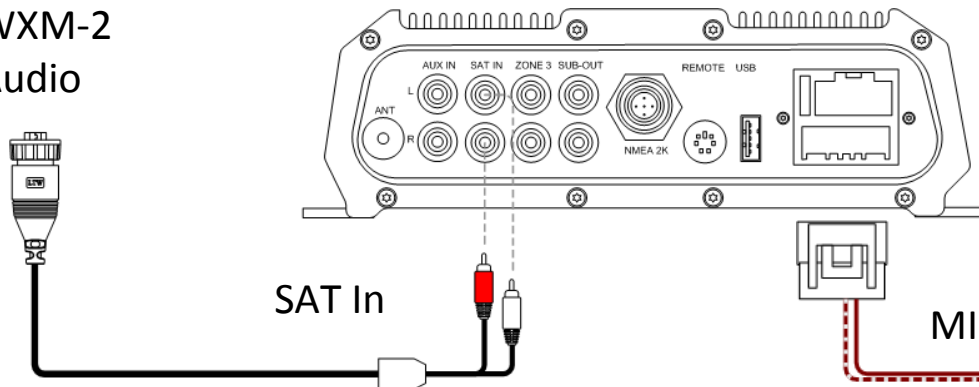
SonicHub



Sonic Hub can be used to relay Sirius Weather Audio (SAT) , and also VHF (MIC) if connected

Sonic Hub

WXM-2
Audio



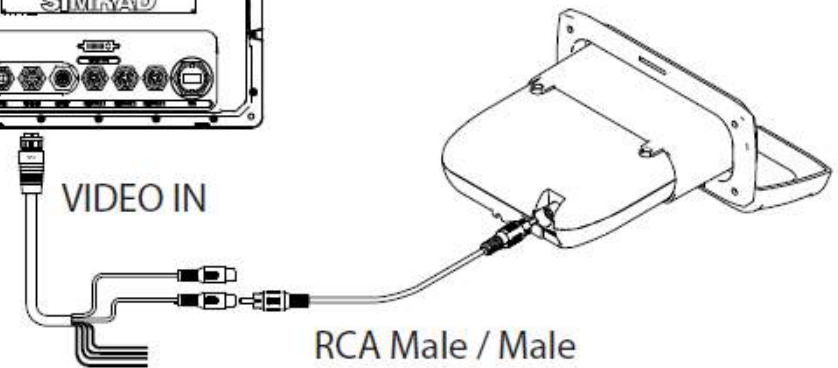
You can control the Sonic hub independent of having the NSO/NSE/NSS powered on using a Fusion MS-WR600. To do this Press the Power key, followed by the Mute key (hold for 2 Secs as remote is powering on) Then select the audio source

Connect Video Out

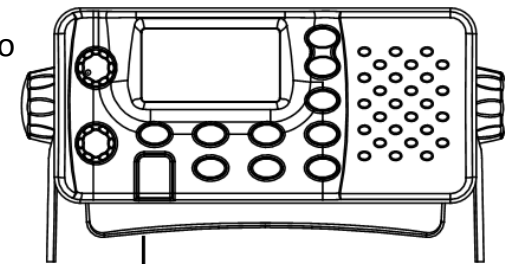


Connect video to the Video In / Data cable of the NSE port 1 or port 2.

An RCA Male / Male Video cable is required (not supplied).



RS10/25 Speaker out to Sonic Hub to allow the sonic hub to act as an external speaker



MIC In

SPKR OUT

RS10/25

PROGRAMMING

Programming Triton instruments is completed via an ST10 SWUP Programmer or by using a Zeus MFD , this can be done with the instrument in its existing system or independently.

SWUP Software Upgrade

File Edit View Tools

Software Upgrade Package: No packet loaded -click here>

Devices detected on SimNet Network:

Status	Addr	Device	Serial	Version
☐	1	Z12	001444	1.3.00.00
☐	0	Triton T41	001490	1.66.01.05
☐	1	Triton T41	004229	1.66.01.04
☐	245	ZG50	12E694	0.1.20.5
☐	3	GRC	003510	1.02.00
☐	49	Triton MHU	001076	1.2.00

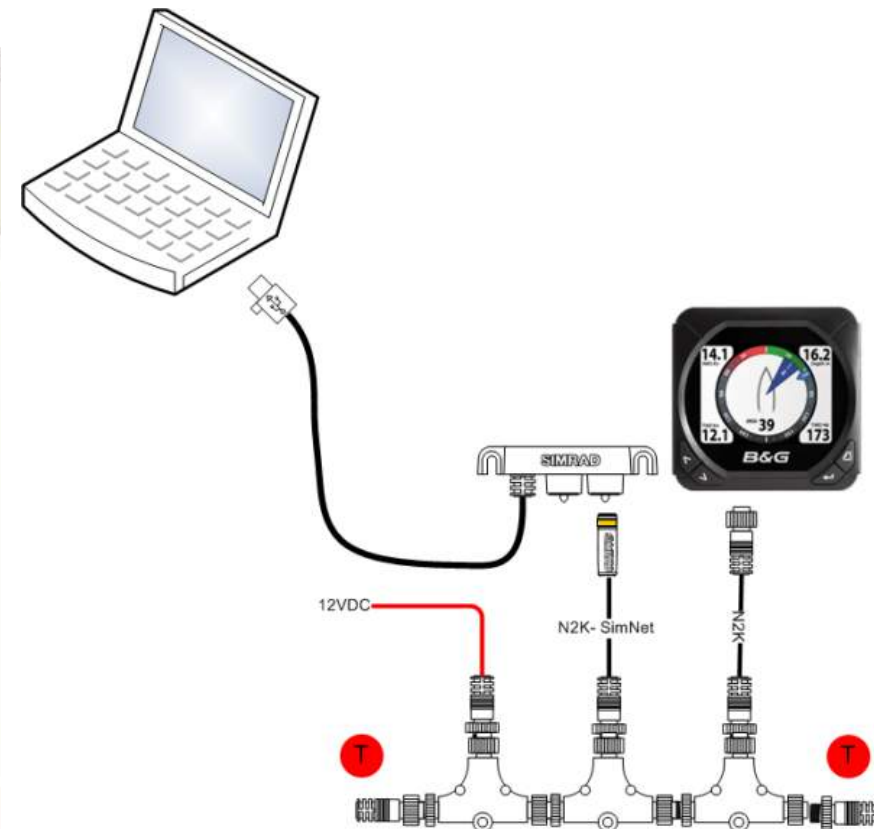
Callouts:

- Device's bus address (points to Addr)
- Selected Software upgrade (points to Software Upgrade Package)
- Device's Software Version (points to Version)
- Device's Name (points to Device)
- Last 4 digits of the unit's S/N (points to Serial)
- Status – whether active or in maintenance mode. A "!" indicates a device that cannot enter a maintenance mode (points to Status)

Buttons: Select all, Select none, Refresh (F5)

Wizards: Upgrade wizard..., Yard Commissioning wizard..., Service Report wizard...

Ready Connected to CAN



PROGRAMMING

Using ST10 SWUP Programmer

- According to our SWUP, the Triton software is old

	0	Triton T41	001490	1.66.01.04
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- To correct this, check the maintenance box beside the Triton

☒	0	Triton T41	001490	1.66.01.04
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- Now cursor up and click in <No Packet Loaded – click here> area

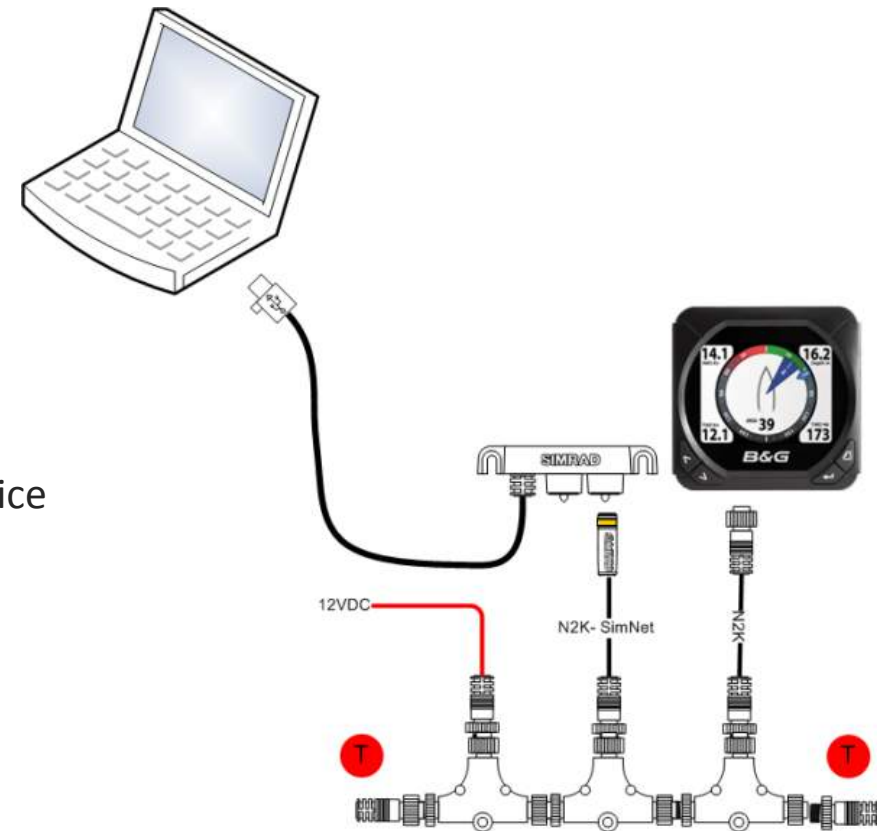
Software Upgrade Package: **<No packet loaded - click here>**

- When prompted, double click on the correct .swup file for the device



- The Packet window will now show the correct file to be loaded

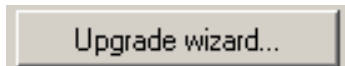
Software Upgrade Package: **C:\Documents and Settings\...\SOFTWARE\T41 1_66_01_05**
Supported Products: **Triton T41**



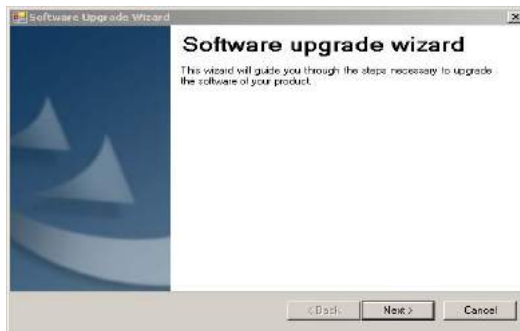
PROGRAMMING

Using ST10 SWUP Programmer

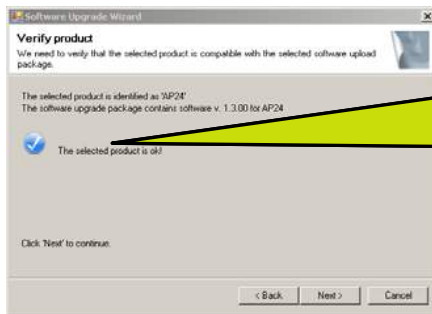
- When ready, press the Upgrade Wizard tab



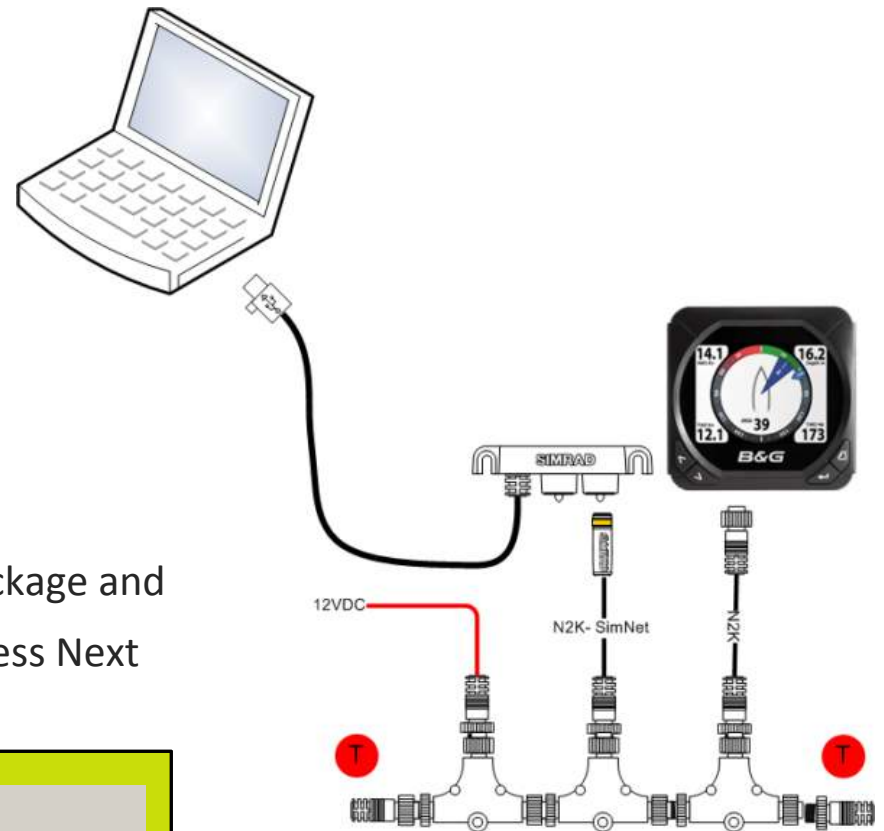
- When the Software upgrade wizard open, press Next



- The wizard will verify you've selected the correct upgrade package and if the selected device needs the upgrade - once verified, press Next



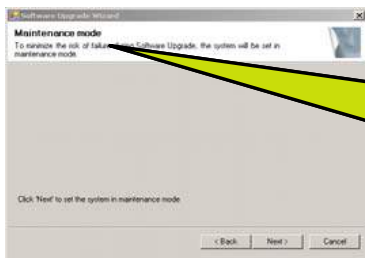
The selected product is ok!



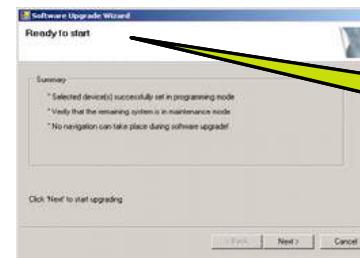
PROGRAMMING

Using ST10 SWUP Programmer

- The wizard will now put all SimNet devices into Maintenance Mode and restart the device – press Next through the next few prompts

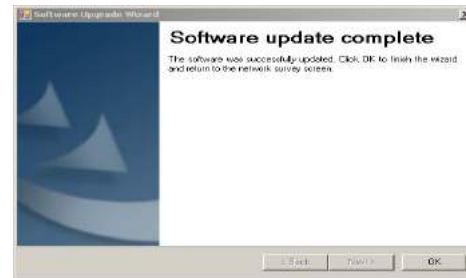
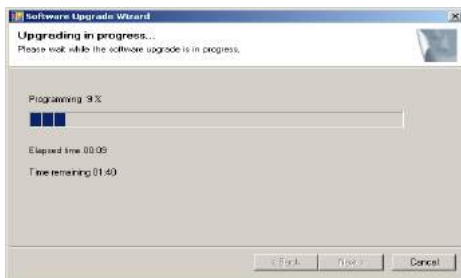


Maintenance mode
To minimize the risk of failure during maintenance mode.



Ready to start

- When ready, press Next – an upgrade status window will appear... once completed, press Next – an Upgrade Complete dialog will appear



- Pressing Next closes the dialog – the next s/w version should now appear in the SWUP main screen

☒		0	Triton T41	001490	1.66.01.05
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PROGRAMMING

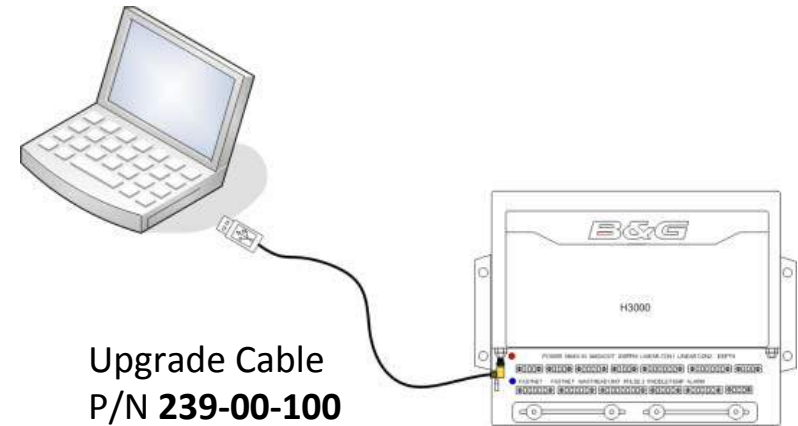
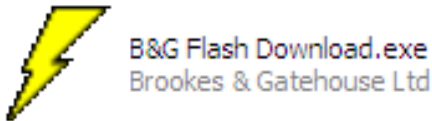
Programming the H3000 Processor

Connecting to the Processor Before you can connect to the main processor a Cal Val needs to be set. This Cal Val is set from any GFD on the system:

- Press the MENU button • Scroll down and highlight Setup
- Press the ENTER key • Scroll down and highlight Calibration
- Press the ENTER key • Scroll down and highlight Other Calibration
- Press the ENTER key • Scroll down through the menu and highlight the MISC menu
- Press the ENTER key • Scroll down through the menu and highlight Com Conf
- Press the ENTER key • Scroll down and highlight PROGLOCK
- With PROGLOCK highlighted, scroll RIGHT and highlight the number 0
- Once this number is highlighted, scroll up to 99
- Press the ENTER key

The Processor is now ready to be programmed

Start the B&G Flash Download Program



PROGRAMMING

Programming H3000

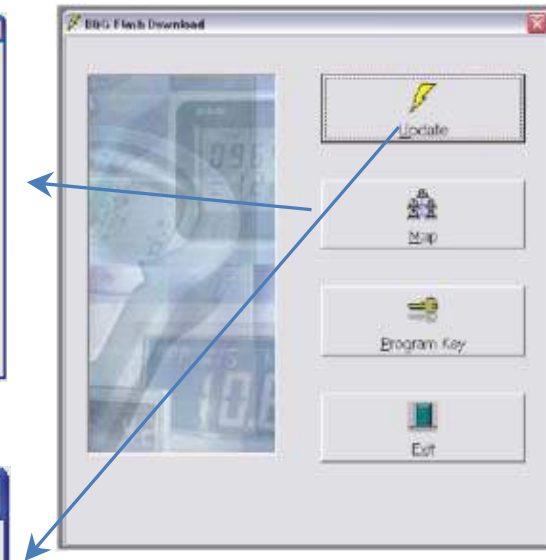
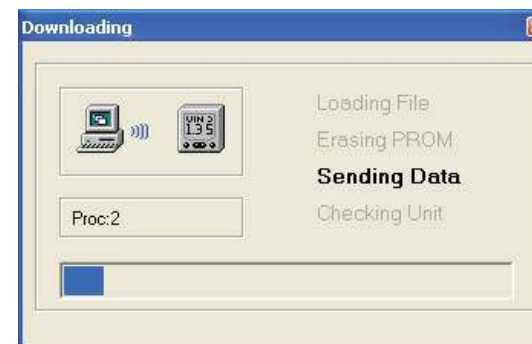
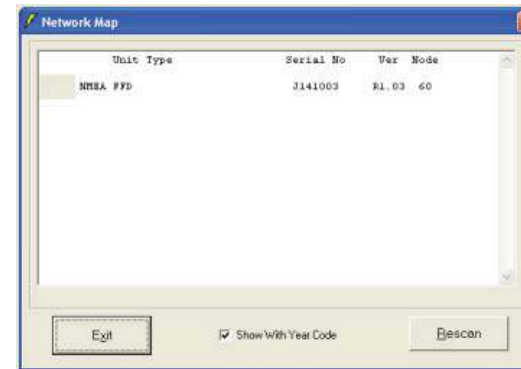
1) First click 'MAP' to check which software version is currently installed on the instrument.

The information box will show unit type, serial number, the current software version. Make sure that the software version is older than the version you wish to upgrade to. Click 'EXIT' to return to the front page.

2) Select **UPDATE** to start the upgrade.

3) A popup will appear asking you to 'Note the following serial number.' Write this number down. This number will be required if the unit fails to update or loses power midway through the update process. Click 'OK' to continue.

4) The unit will now be programmed with the updated software.



PROGRAMMING

Programming H3000

Click **OK** once complete.

Note: *If at any time during the programming process the unit fails to program cycle the power and re-start the procedure. If it fails a second time please refer to manual programming procedure below*

CAUTION: *It is possible to program an instrument with the wrong software when using manual mode. Make sure that the correct software is selected when updating instruments. We recommend that you attempt to update automatically before you update in manual mode.*

Cycle the power to the instrument.

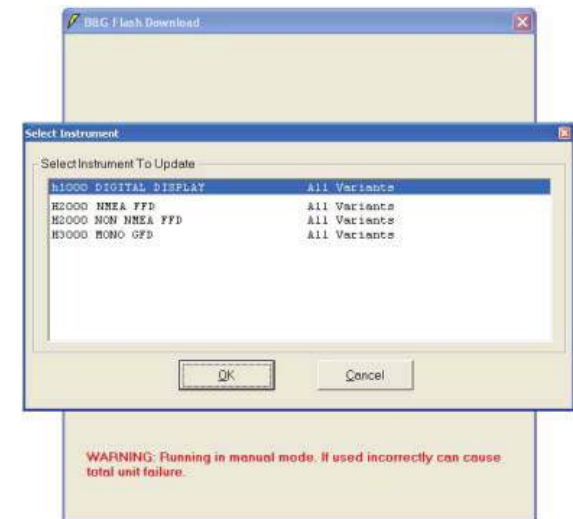
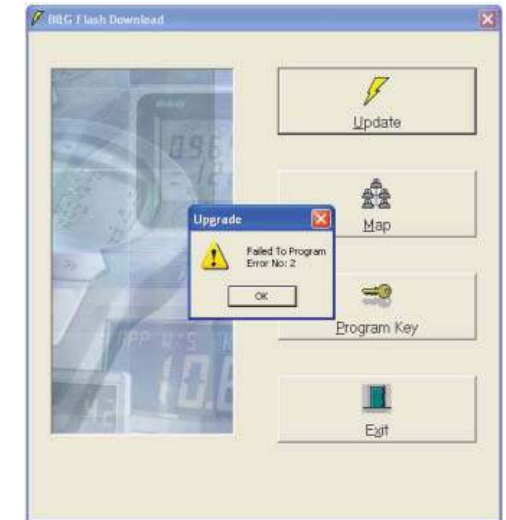
Restart the Update Utility.

Press the 'M' Key on your keyboard. A warning message will appear at the bottom of the window.

WARNING: Running in manual mode. If used incorrectly can cause total unit failure.

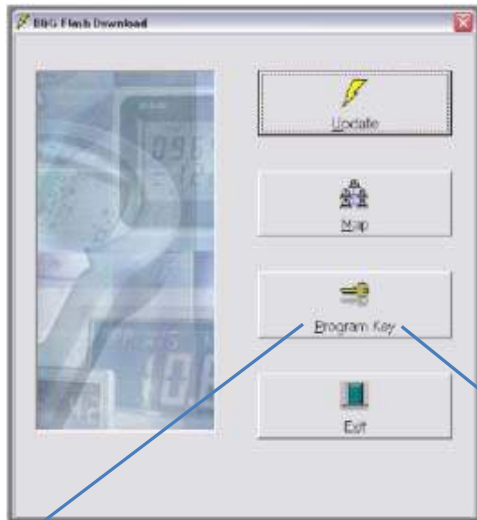
Click 'Update' It will then ask you which instrument you wish to program. Select the correct instrument and click 'OK'

The programmer will identify the unit and program accordingly. Follow steps 3 to 5 of the auto setup.

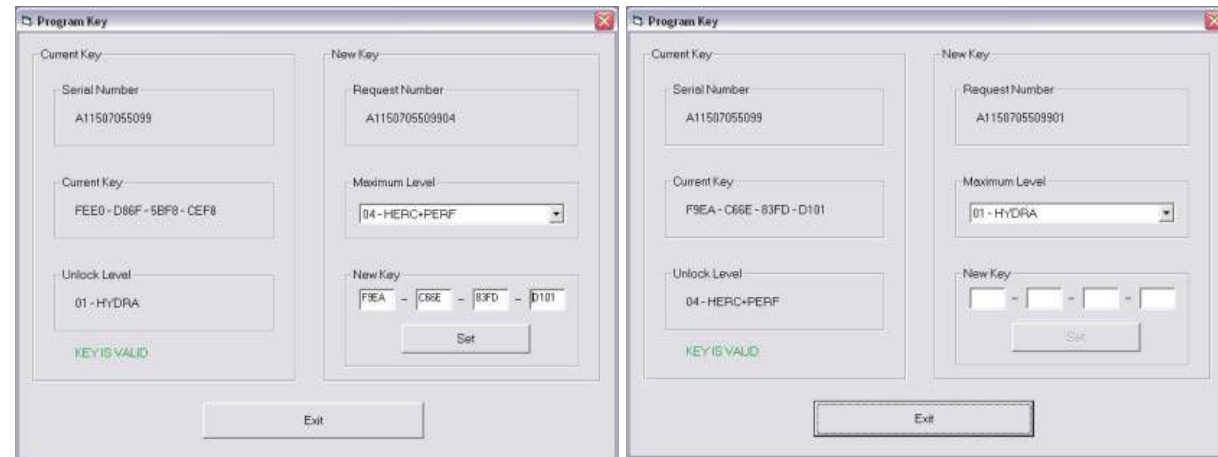
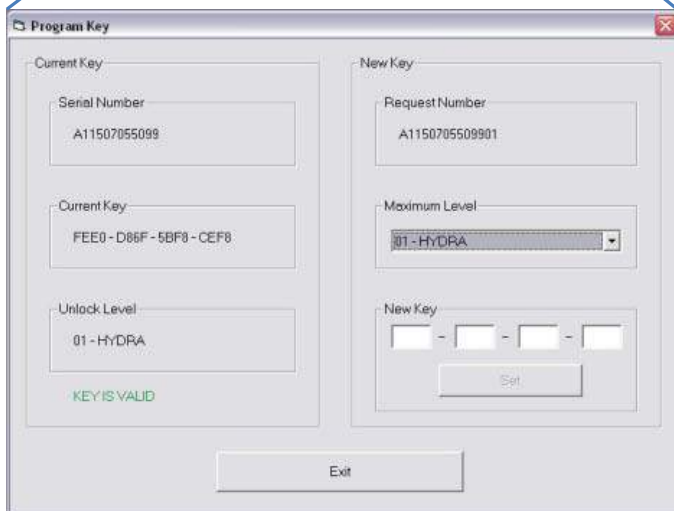


PROGRAMMING

Upgrading the H3000 Processor



Click on **Program Key**. This option will bring up the following screen from which you can change the software level.



To change the software level you need to enter the unlock key provided by B&G and also change the Maximum Level to suit the new key. To select the new level, click on the drop down box and select the required level. Once you have chosen the required level, enter the unlock key

Once the unlock key is entered and the required level is set, click **Set** and this will send the new code down to the processor and unlock the new functionality. If everything is successful in sending the new code, the screen will change and show the new unlock level.

Click **Exit** and you will return to the main screen. Click Exit again and this will close B&G Flash Download program down.

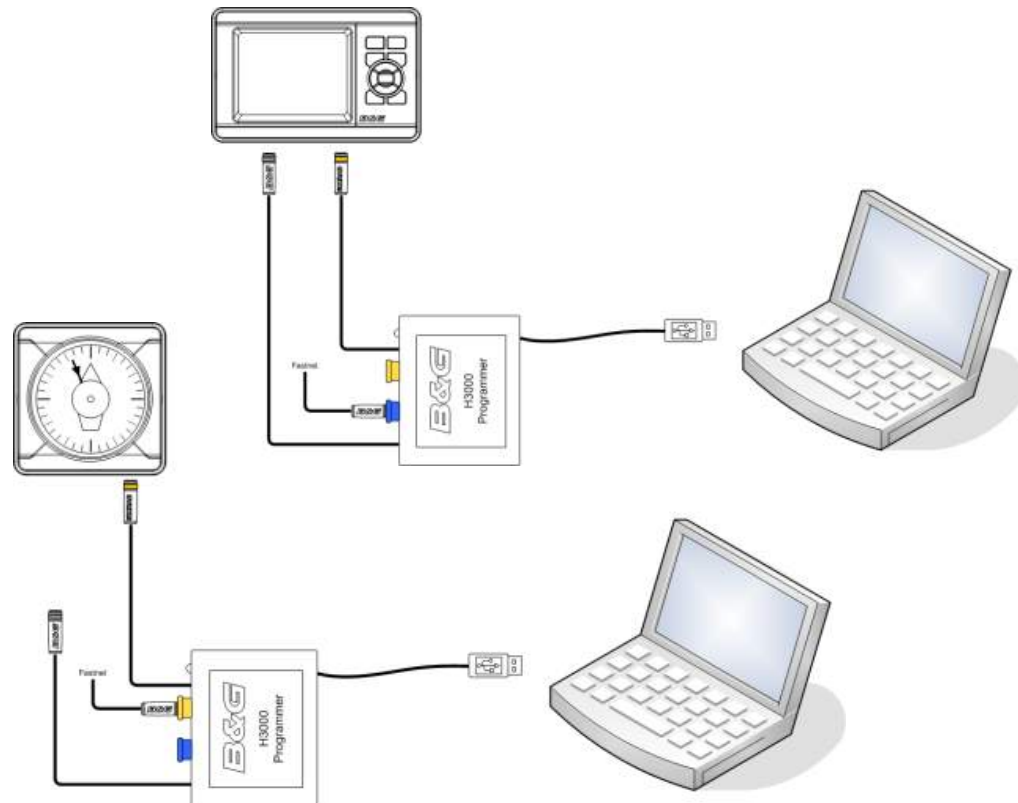
To allow the processor to load the new software and function correctly, the main processor needs to be reset. The procedure for resetting the main processor can be found in the H3000 User Manual.

PROGRAMMING

Programming the H3000 instruments

Programming these devices is the same as a Main Processor, except you do not need to set the Prog lock value.

A GFD or Analogue uses the H3000 Field Programming unit
P/N **BGH390001**

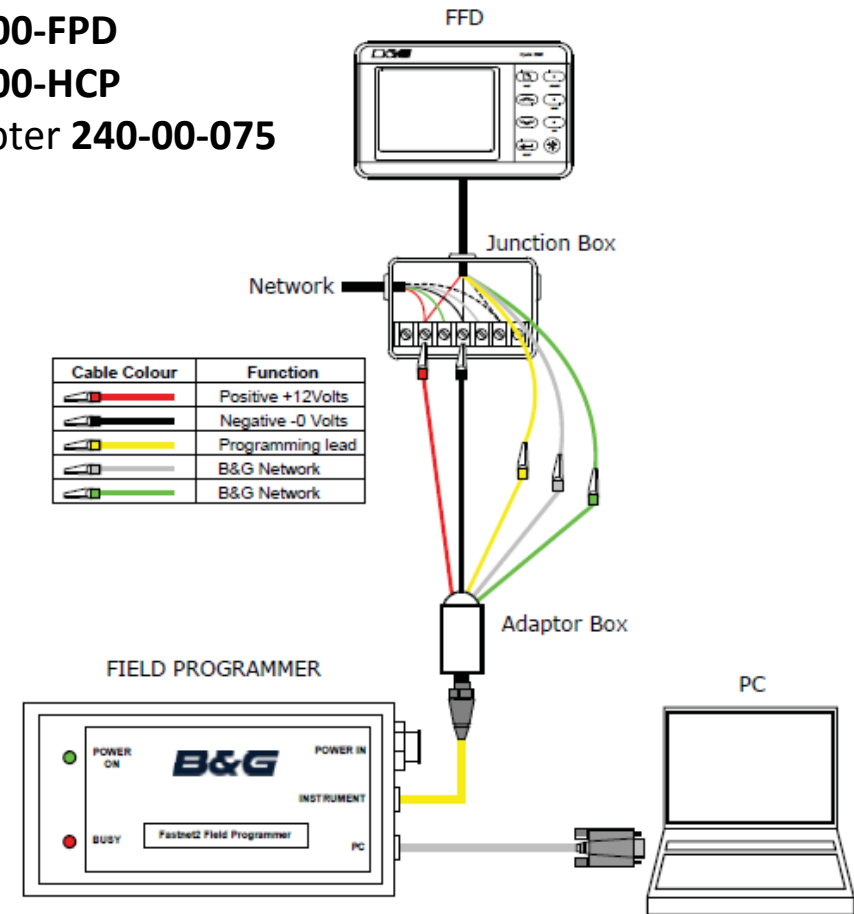


An FFD uses Fastnet 2 Field Programming unit

P/N **H1000-FPD**

P/N **H1000-HCP**

P/N Adapter **240-00-075**



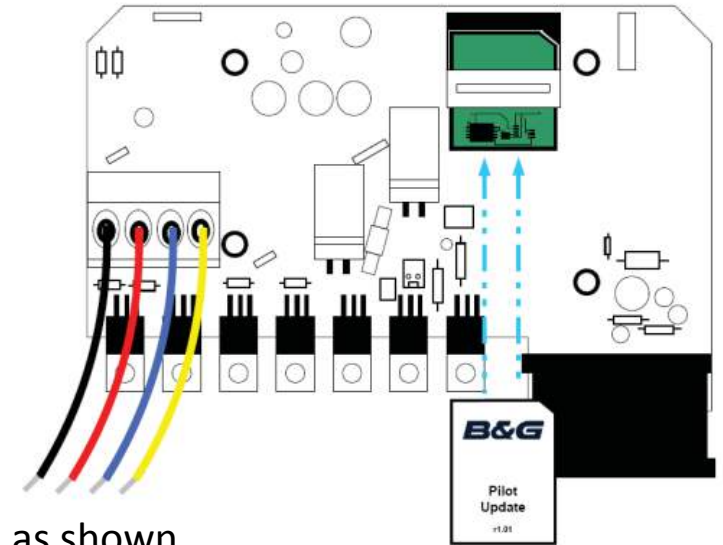
PROGRAMMING

Programming the H3000 ACP

To upgrade the ACP Pilot Processor follow the instructions below. You do not need to use the upgrade utility to update this equipment.

How to use the upgrade flash card

- 1) Switch off the power to your B&G system.
- 2) Unscrew the two screws in the outer case of the Pilot Processor. The processor lid is hinged. Open the lid and you will see four screws retaining an inner plate in the top half of the unit.
- 3) Unscrew the four screws holding this plate in place. This will expose the PCB as shown above. Slide the memory card out and slide the Flash Upgrade Card in.
- 4) Restart the processor. The programming process should take approx 1 minute. When the software has finished upgrading the Pilot display should flash Fault 114. This means the Processor has been upgraded successfully.
- 5) Turn off the power, remove the upgrade card and replace it with the original memory card. Replace the screws and return the processor to its original state. The new software will now work when you turn the power back on.
- 6) Carry out a system reset on node '0012' (Pilot ACP) unit. Upgrade process is now complete.
- 7) You will need to re commission the Pilot ACP again.



CZone



Systems In Operations

This Mode enables you to monitor all onboard parameters including tank levels displayed in Graphical, percentage and volume remaining formats.



Control, Monitor

The display is split in half with the left side showing controls, and the right showing Monitoring information, such as AC/DC volts/Amps, Hydraulic pressures and temp



Alarms

The alarms page shows visual and audible alarms that can be set for high or low levels

Advanced Settings:

MENU:MENU>SYSTEM>ADVANCED...>

FEATURES> CZONE

Features		
Structure	☒	Controls whether StructureScan features are shown. Turn on to allow Structure simulation.
Sirius weather	☒	Controls whether Sirius weather features are shown. Turn on to allow Sirius weather simulation.
CZone	☒	Controls whether CZone features are shown.
Additional Instruments	☒	Controls whether additional instrument dashes can be added.
SonicHub	☒	Controls whether SonicHub audio features are shown.

CZone		
CZone at startup	☒	Controls whether CZone is shown immediately at startup.
CZone dipswitch	00...	Controls CZone dipswitch for this display.
CZone backlight	☐	Controls whether CZone backlight integration is enabled for this display.

Replacing with a new un-configured module:

1. Remove the damaged module.
2. Copy its dipswitch setting to the replacement module.
3. Plug the module into the system.

The system will recognise the module by its dipswitch setting and configure it automatically. This will only take a few seconds before the module is operational.

Replacing with a configured module:

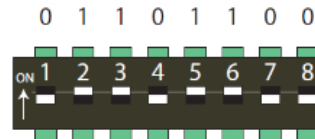
1. Remove the damaged module.
2. Plug the replacement module in.

The new module will begin operating instantaneously.

Replacing with a module that has been configured for another network, or the Same network but as a different dipswitch setting

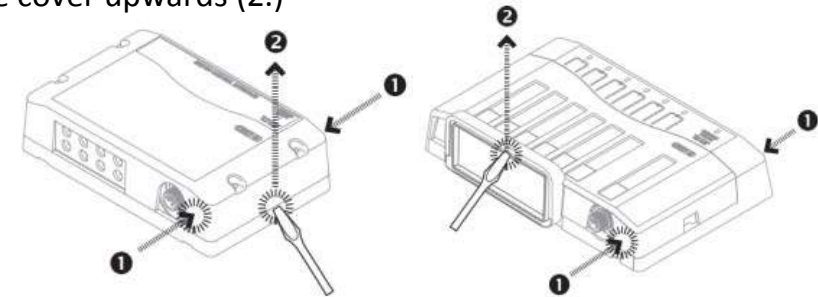
(i.e. replacing a damaged OI that supplies Critical circuits with one that supplies non-critical circuits)

1. Remove the damaged module.
2. Set the dipswitch settings of the replacement module to all off i.e. **00000000**.
3. Plug the network cable into the module. Wait 15 seconds, then unplug the module.
4. Set its dipswitch settings to match the dipswitch settings of the damaged module.
5. Plug it into the network.



Removing the Covers

The procedure for the module cases is almost identical, as each side is clipped down with pressure clips. Press firmly inwards near a short edge of the Case (1.) Assisting with a screwdriver if necessary, lever the cover upwards (2.)

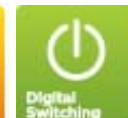


Global Fault Codes (All Interfaces)	
Flashes	Fault
2	There are conflicting Configuration Scripts present on the network.
3	There is at least one other module on the network with the same dipswitch setting as this module.
4	The controller is unable to read from the E-PROM.
5	No other CZONE modules are detected on the network.
Motor Control Interface (MOI)	
6	Output channel run current has fallen below 80% of configured run current when on.
7	Output channel run current has risen above configured fusing current.
8	The last switch to control this output channel is no longer detected on the network.
9	The last switch to set the mode of this output channel is no longer detected on the network.
10	A negative current has been detected on this output channel (Battery plug inverted/output connected to a generator).
11	The auto current sense calibration level for this channel is out of bounds.
SOLID GREEN	Output channel is on.
FLASHING GREEN	(2s On, 2s Off) Output Channel is off but under automatic control. This will come on automatically at some point in the future, as per channel configuration.

Signal Interface (SI)	
Flashes	Fault
6	A fault condition exists on an output channel controlled by this switch.
7	An output channel controlled by this switch is not currently detected on the network.
SOLID GREEN	Input signal is within the limits expected for the configured input type.
Meter Interface (MI)	
6	No errors detected.

Interface status lights flash red if faults are detected, except on SCI switches which have a single color System On Indicator. Flashes are 0.3s On, 0.3 s Off, with a 1.5 s delay after the end of the sequence.

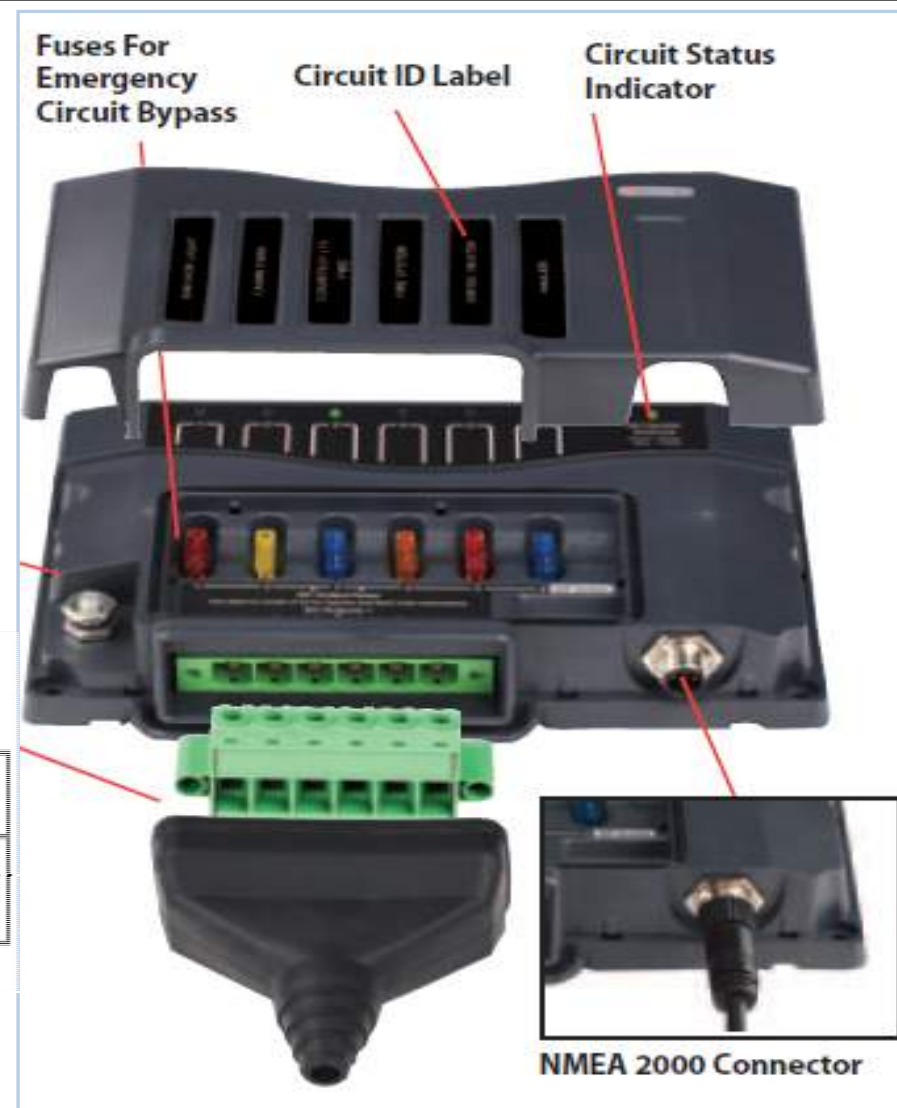
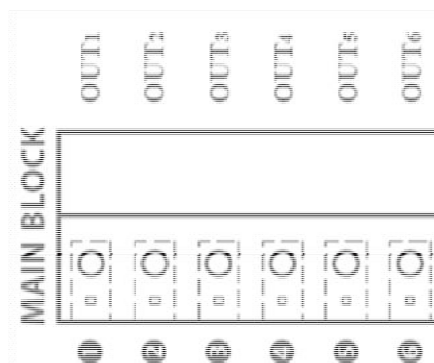
Output Interface (OI)



The CZone™ Output Interface provides decentralised power management and protection for dc circuits

Features

- 6 x 20 amp outputs (bridgeable for higher output)
- LED status indicator
- No crimps or special tools required
- Programmable software fuses from 200mA to 20A, motor start, slow blow etc.
- Manual circuit override
- Built in timer and dimmer functionality
- Micro C Network Connector
- IPX5 Water ingress protection

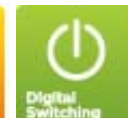


PART# DESCRIPTION

80-911-0009-00 Output interface w/connector, boot

80-911-0010-00 Output interface only

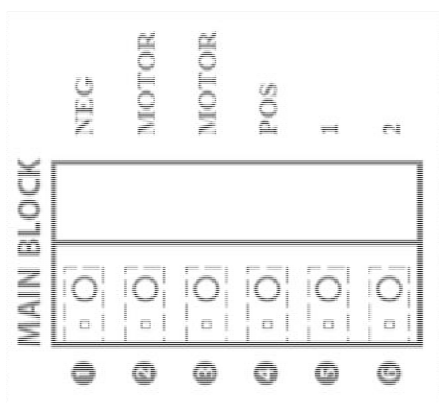
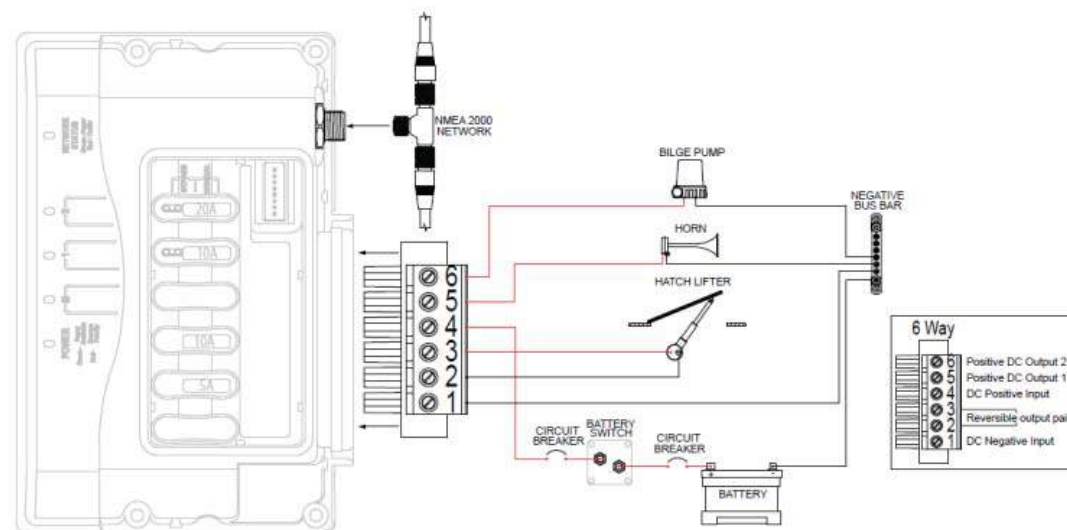
Motor Output Interface (MOI)



The Motor Output Interface (MOI) is designed for loads that require polarity reversal to change the direction of a motor. Two standard DC outputs are also available on each MOI.

Features

- 2x 20A Circuits per module (MOI)
- 1x 20A “H Bridge” output for controlling direction of DC motors through polarity change
- Each circuit channel is programmable to offer circuit protection with settable break currents ranging from 200 mA to 20 A.



Hatch lift control



PART# DESCRIPTION

80-911-0007-00 Motor Output Interface w/connector, boot

80-911-0008-00 Motor Output Interface only

Switch Control Interface (SCI)



The Signal Control Interface (SCI) converts signals from traditional mechanical switches to NMEA data to control loads

Features

- 8 switch channels per module (sixteen switch points)
- Single switch position can control multiple OI channels
- Multiple SCI switches can control single OI channel
- Output for backlighting of switch labels (dimmable)
- Outputs systems on and function/fault codes to systems on LED of switches (dimmable)
- Programmable switch types

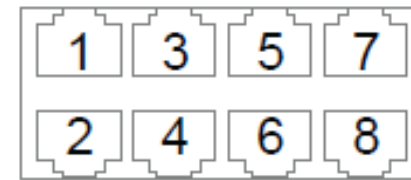
80-911-0018-00 .5 meter
80-911-0019-00 1 meter
80-911-0020-00 2 meter
80-911-0021-00 3 meter
80-911-0022-00 4 meter
80-911-0023-00 5 meter



Proprietary
cable assembly

PART# DESCRIPTION

80-911-0011-00 Switch Control Interface w/seal
80-911-0012-00 Switch Control Interface only



8-Way Connector Bank for SCI
Cable Assembly to Switches



Waterproof Cable Seal



NMEA 2000 Connector



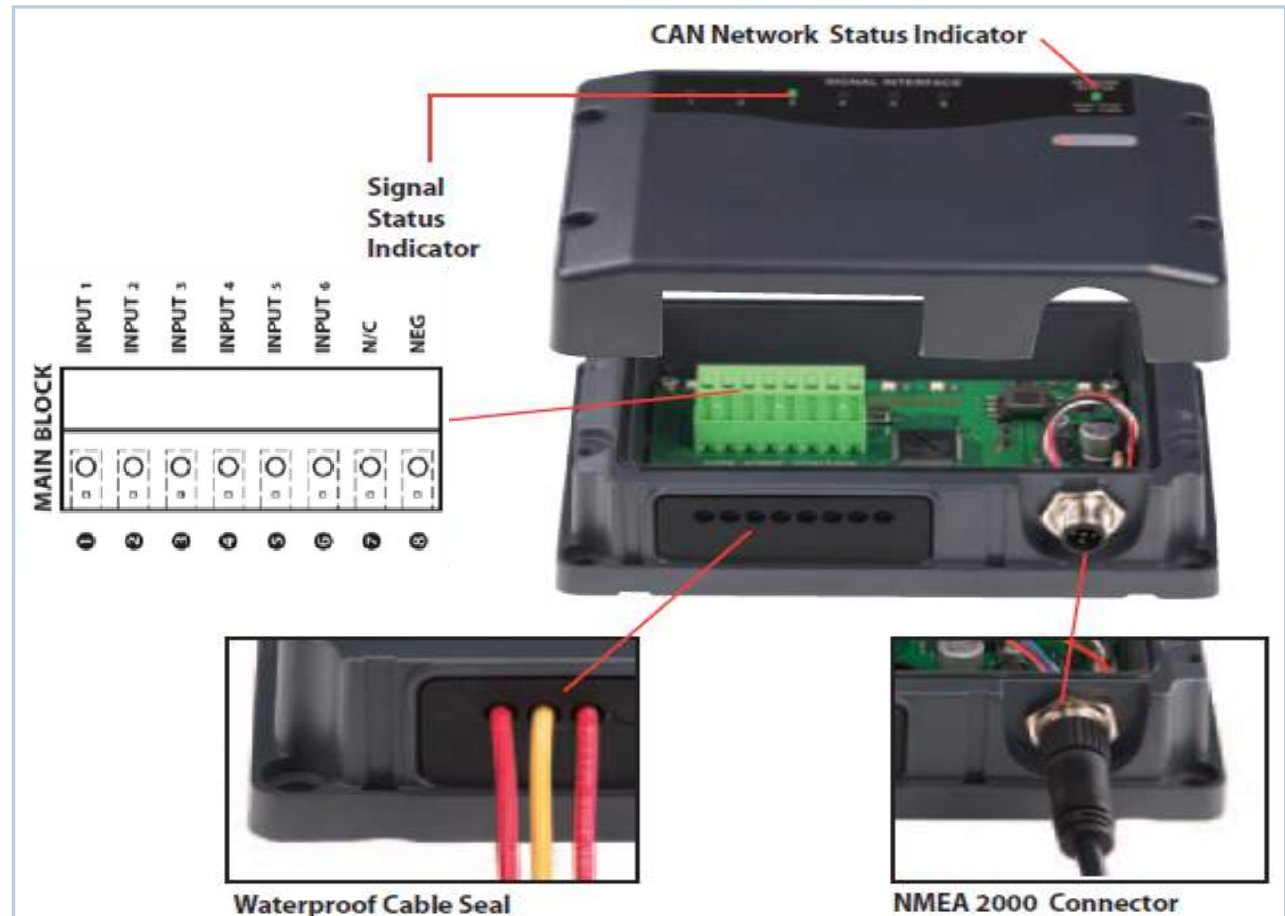
Signal Interface (SI)



Interface between sensors and switches, external to the system, and the CZone™ network

Features

- Inputs from traditional switch types being used to control outputs (Negative, Open circuit, 0-32 V DC)
- Inputs from switches to trigger alarm i.e. high water float switch (Negative, Open circuit, 0-32 V DC).
- LED status indicators for each input.
- Tank level senders (compatible with industry standard sender outputs, 0-5 V, 10-180 Ω , 240-33 Ω).
- General voltaic or resistive signals can be used for controlling outputs or to display a physical position ie show a hatch is partially open.

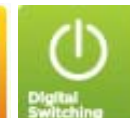


PART# DESCRIPTION

80-911-0013-00 Signal Interface w/seals, connector

80-911-0014-00 Signal Interface only

Meter Interface (MI)



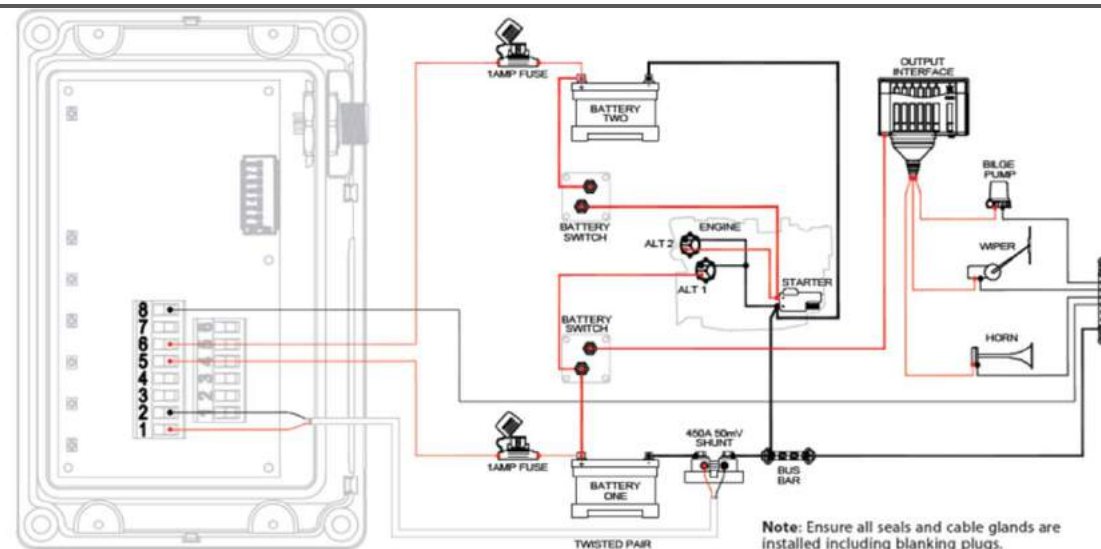
The Meter Interface (MI) takes multiple inputs from AC currents, voltage transformers, DC voltage and current inputs from shunts, processes these readings and broadcasts them to the CZONE network

Meter Interface

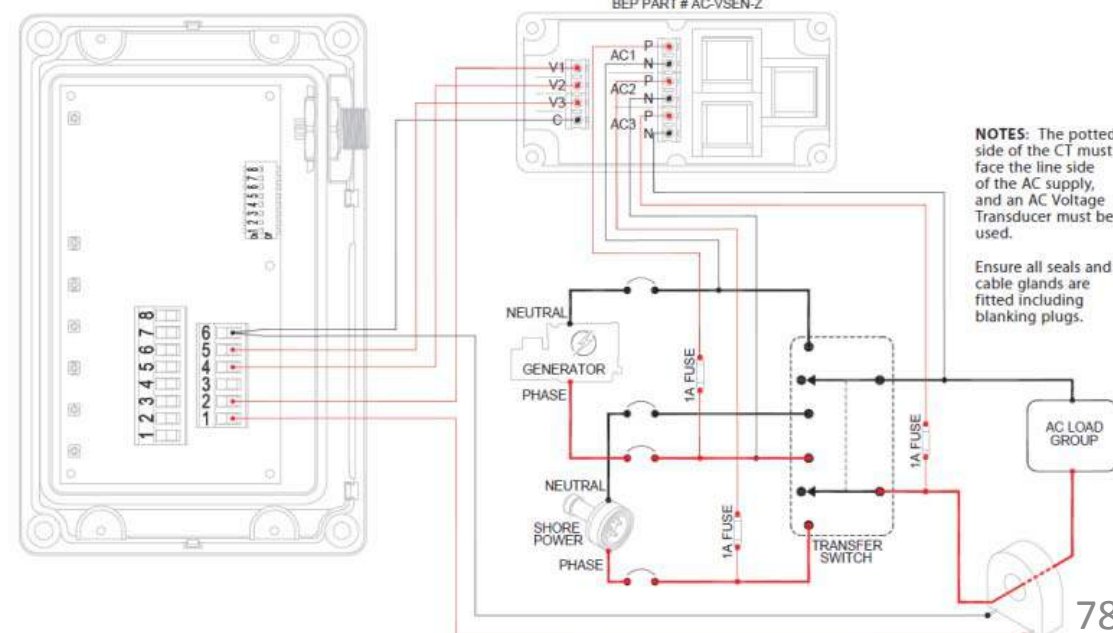


Battery

DC Sense



AC Sense



PART# DESCRIPTION

80-911-0005-00 Meter Interface, w/seal & plug

80-911-0006-00 Meter Interface only

Czone Support



Americas:

Actuant Electrical

Edward Saunders

Email: edward.saunders@actuant.com

Tel: +1-262-293-1932

APAC:

BEP Marine

Warren Wright

Email: warren.wright@bepmarine.com

Tel +64 9 448 0028 (DDI)

EMEA:

Mastervolt

Barry Oorbeek

Email: barry.oorbeek@mastervolt.com

Tel: +31-20-3422100

Instruments

Discontinued

H2000 Analogues



MWA



360 Wind Angle



BS 0-12Knts



BS 0-25Knts



BS 0-50Knts



Rudder Angle



Wind Speed



Cross Track Error



Depth m, Ft Fathoms



Compass

Instruments

Discontinued

H2000 Displays



20/20 Red



20/20 Green



40/40

UPGRADE OPTIONS

H3000



FFD Full Function Display
NMEA, Non-NMEA



Pilot Display



Instruments

Discontinued

H2000



Hydra
Processor

Speed, Depth, Wind, Heel, Trim, System and Meteorological data processed alongside more advanced functions such as Rudder Angle, Mast Rotation etc.

Advanced calibration control

Analogue display outputs (4)

Linear function inputs (4)

Audible alarm output

Battery voltage monitoring

Mast rotation correction - corrects



Performance
Processor

Performance Processor enables the h2000 system to communicate fully with a PC running a tactical software package such as B&G Deckman. Alongside this the processor has the ability to store a polar table of your boat's predicted performance making it a valuable tool for both racing and fast passage making.

Integrated polar table for performance optimisation
Fully functioned RS232 Interface for connection to PC Software

NMEA 0183 Input and Output

Secondary NMEA Input/Output option if RS232 not required



Hercules
Processor

Wind information calculated and displayed at twice the rate of Hydra. More "real time" information when making manoeuvres or accelerating.

Boat speed linearity correction - allows faster boats to correct for speed sensor errors caused by non-linearity in sensors and water-flow

Enhanced True Wind Speed calibration - enables the user to set the downwind angle at which correction is applied.

Dual boat speed input to allow active switching of sensor and calibration by wind angle or heel angle
NMEA 0183 input allows integration of external NMEA Depth sensor, Heading data and GPS data
NMEA Depth can be used as an additional sensor to give dual depth information, e.g. for aft Depth reading



Expansion
Processor

The Expansion Processor is for systems requiring additional Analogue Meter outputs or Linear inputs.

Four additional analogue meter outputs
Twelve additional linear inputs.

Instruments

Discontinued

H1000



H1000-DSP



H1000-Compass, MWA,TWS,AWA



H1000-PDU

H1000-3FD



UPGRADE OPTIONS

TRITON



H1000- WDI
H1000-SDI, SEN-DEPTH-A2
H1000-SDIE, SEN-DEPTH-P
H1000-UNI



H1000-CND



H1000-ACP

Instruments

Discontinued

Network



Network
Data



Network
Speed



Network
Quad



Network
Nav



Network
Depth



Network
Compass



Network
GPS Plus12



Network
Pilot



Network
Wind



Network
Tack



Network-3FD



LCD Elite

UPGRADE OPTIONS

TRITON





**SERVICE BEYOND
THE STANDARD.**

The B&G Advantage Service program offers the most comprehensive service and support available in the industry.

There are Advantage Service programs to support both the Dealer and Consumer during the initial sales inquiry, through the warranty period, and then extending significantly beyond the expiration of the warranty period

Ensuring we are the Partner of choice for our Dealers and the product of choice for our Consumers



5-YEAR ADVANTAGE

- ▶ 5 Year Service Period
- ▶ Guaranteed upgrade options to current technology



ONBOARD ADVANTAGE

- ▶ 2 Years OnBoard Support
- ▶ On systems >\$2500
- ▶ Global network of Certified Dealers to provide service and support



+1 ADVANTAGE

- ▶ Extra year of product warranty
- ▶ Extra year of onboard support
- ▶ Total of 3 years warranty with registration



24 SWAP ADVANTAGE

- ▶ Replacement product shipped within 1 business day during the first year of warranty
- ▶ Available for Zeus, Zeus Touch, WTP3



CERTIFIED DEALER

- ▶ Framework for Dealers to demonstrate technical proficiency
- ▶ Access to the vessel portal, providing extensive vessel details and servicing history
- ▶ Are designated on the website as Certified Dealers

When your customers choose a product from B&G, they are automatically protected by a standard service and support program, which includes the following features:

Two Years Product Warranty*

One Free year of extended product warranty protection, including OnBoard tech support when installed by a Certified Installer

Global service networks for the duration of the product warranty

Easily accessible self help sections, Frequently Asked Questions and email support on the B&G website (www.bandg.com)

Local customer care lines offering product information, technical assistance and warranty support by technical experts

Worldwide Technical support lines to help with Dealer warranty and product queries

In addition to these standard warranty features, B&G has now expanded its service offerings with the Advantage Service Program. This Program, which is free and available to all B&G customers, offers the most comprehensive service and support available today. The Advantage Service Program aims to provide the best possible experience with B&G products, even on occasions when support or replacement is required. B&G stands for high quality both in its products and support offerings.

Advantage Service



5-Year Advantage

The 5-Year Advantage offers comprehensive customer support for five years, including upgrade options to the latest technology products.

In the event of product failure, the 5-Year Advantage supports your customers for a full five years from the date of purchase.

The program features a unique Trade-Up option, where customers can upgrade to the latest technology products at a discount price.

In addition, the 5-Year Advantage allows Dealers to extend the discount to complementary B&G components when replacing a faulty product.

This includes items such as displays, radar, sounders and autopilots that comprise an overall system

Item: BGH250001 - H3000 CPU Hydra

Alternate Item:

Brand: B&G		EOL Date	
Product Category: Instruments		EOL Date	
Replacement Item			
Region	Item	Description	
Replacement - ROW	Y51-10155-001	H3000 CPU BGH250001	
		In Warranty Pricing	In Service Pricing
		RRP Pricing (ex-tax)	Available Inventory
			PLC
			YES
Trade-In Options			
	Item	Description	
Opt 1 ROW	BGH250002	H3000 CPU Hercules	
Opt 2 ROW	BGH250002	H3000 CPU Hercules Performance	
		In Warranty Pricing	In Service Trade-In Pricing
		RRP Pricing (ex-tax)	Inventory
			PLC
			NO
			YES

Disposition:

Service times

Key / Notes

* All Prices are ex Sales Tax.
* Inventory availability is indicative only. Navico Technical Support will confirm an estimated delivery date when the RMA is requested.
* Replacement or Trade-In options are covered by Navico's warranty terms and conditions for the remainder of the original product's warranty or six months, whichever is the greater.

	In warranty, free
	In service
	Trade in under warranty



Advantage Service



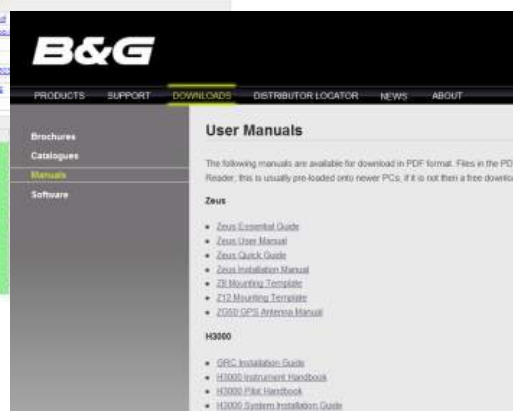
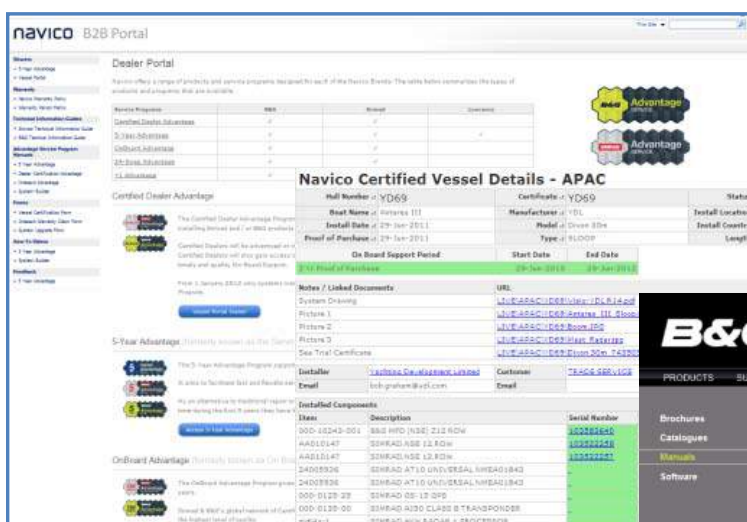
Certified dealer advantage

The Certified Dealer Advantage Program provides a framework for dealers to demonstrate technical proficiency regarding B&G products to become Certified Dealers.

Designated as a marine electronics authority, Certified Dealer accreditation builds customer confidence, trust, loyalty and referrals. Along with in-store signage, Certified Dealers receive worldwide recognition with preferential website listings on the B&G website.

Dealers also gain access to online tools and technical information via a new B2B Vessel Portal and Advantage Hub, which provide extensive vessel detail, and technical information on current and legacy products such as technical bulletins, product briefs, manuals, FAQs and troubleshooting training documents, videos and calendars.

All Certified Dealers are required to recertify annually.



Advantage Service



Onboard advantage

The OnBoard Advantage Program provides customers with the option to receive warranty service by a Certified Dealer onboard their vessel for the first 2 years. Systems under warranty with a value of over \$2,500 USD qualify when they have been installed or signed-off by a Certified Dealer, or installed by a Navico-authorized factory. Select limits apply to labor and travel related costs as detailed by the program guidelines.

Certified Installation Certificate

THIS DOCUMENT CERTIFIES:

Vessel type: HANSE - HANSE 445
Hull number: DE-HANN0042F111
Certificate: AP000017

Qualifies for **B&G Advantage**
On Board Support

DECLARATION OF VESSEL COMPLIANCE

DATE OF ISSUE:
28-Oct-2011

Chris Ottosen
LEIF OTTOSSEN, CEO

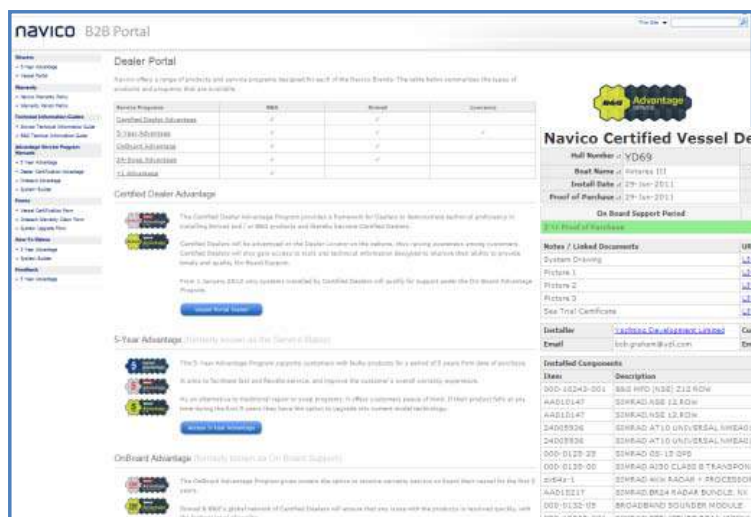
simrad-yachting.com

Hull Number	DE-HANN0042F111
Certificate Number	AP000017
Manufacturer	HANSE
Model	HANSE 445

Item Code	Description	Serial Number	Alternate S/N	Warranty Start	Warranty End
080-0135-02	SIMRAD NAV-200 SYSTEM		808906	28 Oct 2011	28 Oct 2013
080-0132-06	LOWRANCE STRUCTURE SCAN BLACK BOX MODEL			28 Oct 2011	28 Oct 2013
080-10145-801	LOWRANCE SERVER PACK,SERVER+HOOK+SPW3			28 Oct 2011	28 Oct 2013
080-10185-801	AC42 COMPUTER WITH SIMNET-1		803676		
20094067	SIMRAD AP44 CONTROL UNIT		804852	28 Oct 2011	28 Oct 2013
22080552	SIMRAD - DIST808 T110.5.7 SIMNET TDX			28 Oct 2011	28 Oct 2013
27110882	SIMRAD WIND VANE		812421	28 Oct 2011	28 Oct 2013
20095382	SIMRAD 830 GRAPHIC w/SIMNET CABLE		805385	28 Oct 2011	28 Oct 2013
20094420	SIMRAD 830 COMB w/SIMNET CABLE		805381	28 Oct 2011	28 Oct 2013
22095538	SIMRAD 830 WIND w/SIMNET CABLE		805385	28 Oct 2011	28 Oct 2013
080-10214-801	SIMRAD VHF RADIO RS25 DISC EU			28 Oct 2011	28 Oct 2013
080-10214-801	SIMRAD VHF HANDSET, 16000, 16000, 16000			28 Oct 2011	28 Oct 2013
24009336	SIMRAD AT10 UNIVERSAL NMEA0184			28 Oct 2011	28 Oct 2013
22014280	SIMRAD RF25 RFU		803583	28 Oct 2011	28 Oct 2013
080-10219-801	SIMRAD 8.0" TOUCHSCREEN NSEA ROW CARTO		808011	28 Oct 2011	28 Oct 2013
080-10219-801	SIMRAD 8.0" TOUCHSCREEN NSEA ROW CARTO		808009	28 Oct 2011	28 Oct 2013
22090187	SIMRAD PC40 FLIRGATE COMPASS		803140	28 Oct 2011	28 Oct 2013
080-0125-35	SIMRAD 88-15 GPS		913488	28 Oct 2011	28 Oct 2013

Navico Certified Vessel Details - APAC

Hull Number: YD69	Certificate: YD69	Status: VESSEL LEISURE CERTIFIED				
Boat Name: Vitesse III	Manufacturer: YDL	Install Location: YDL Yacht Imports/Exports, Auckland				
Install Date: 29-Jan-2011	Model: Dron SDR	Install Country: NEW ZEALAND				
Proof of Purchase: 29-Jan-2011	Type: SLOOP	Length: 4.30-5.9m / 99-19ft				
On Board Support Period:						
29-Jan-2011 to 29-Jan-2013						
Notes / Linked Documents:						
System Drawing:	http://www.navico.com/Products/PLR/PLR.pdf					
Pictors 1:	http://www.navico.com/Products/PLR/PLR.pdf					
Pictors 2:	http://www.navico.com/Products/PLR/PLR.pdf					
Pictors 3:	http://www.navico.com/Products/PLR/PLR.pdf					
See Trial Certificate:	http://www.navico.com/Products/PLR/PLR.pdf					
Installer: Yachting Services Limited	Customer: TRAC SERVICE					
Email: info@yachting.com	Email:					
Installed Components:						
Item	Description	Serial Number	NMEA S/N	Warranty End	Install Order	Notes
000-10243-001	88G HFD (N34) 212 ROW	800000000	N34 6251	28-Oct-13	1.00	
A4010147	88RAD, 150 12 ROW	800000000	N34 7101	28-Jun-13	1.00	
A4010147	88RAD, 150 12 ROW	800000000	N34 7080	28-Jun-13	1.00	
24009336	88RAD AT10 UNIVERSAL NMEA0184	800000000	N34 10068	28-Jun-13	1.00	
24009336	88RAD AT10 UNIVERSAL NMEA0184	800000000	N34 10078	28-Jun-13	1.00	
000-0128-28	88RAD 05-12 0P0	800000000	N34 120029	28-Jun-13	1.00	
000-0128-00	88RAD 030 CLASS B TRANSPOUNDER	800000000	N34 9080	28-Jun-13	1.00	
2004011	88RAD 88W RADAR 1 PROCESSOR	800000000		28-Jun-13	1.00	
A4010147	88RAD 88W RADAR BUNDLE, 1K	800000000		28-Jun-13	1.00	
000-0132-00	BROADBAND SOUNDER MODULE	800000000		28-Jun-13	1.00	
000-10238-001	88RAD STRUCTURE SCAN MODULE	800000000		28-Jun-13	1.00	
88330001	88G UNBALANCED RATE COMPASS	800000000	N34 278	30-Jun-13	1.00	



Formerly known as the B&G Gold Card Program

Minimum system value of USD \$2,500

Customers simply have their products installed by a Certified Dealer and register on-line with B&G within 90 days of installation

Vessel Certification

Type:	APAC
Model:	YD69
Unit:	L1318m Rd
Year Commenced:	2008
Capacity:	
Project/Contract Reference:	
File #:	PAC11100

INSTALLATION VELD **COMPLETED**

Customer Details	
Name:	
Address:	
Email:	
Phone:	
Fax:	

Installation Details	
Installed System:	
Accessories:	
Weatherstripping:	
Sealing:	
Insulation:	
Exterior Finishes:	
Interior Finishes:	

LOWRANCE **SARAS**

B&G

- PRODUCTS
- SUPPORT
- DOWNLOADS
- DISTRIBUTOR LOCATOR
- NEWS
- ABOUT

Brochures

Catalogues

Manuals

Software

User Manuals

The following manuals are available for download in PDF format. Files in the PDF Reader. This is usually pre-loaded onto newer PCs; if it is not then a free download.

Zeus

- Zeus Essential Guide
- Zeus User Manual
- Zeus Quick Guide
- Zeus Installation Manual
- ZE Mounting Template
- Z12 Mounting Template
- ZGS6 GPS Antenna Manual

H3000

- HRC Installation Guide
- H3000 Instrument Handbook
- H3000 Plot Handbook
- H3000 System Installation Guide

B&G

Certified Installation Certificate

THIS DOCUMENT CERTIFIES:

Vessel type: **HAIRIE, HAIRIE 400**
Hull number: **DE-HA3000142/111**
Certificate: **AP0000017**

Qualifies for: **B&G | Advantage
On Board Support**

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LEAF 01703300N.120

100% OF B&G

28-Oct-2011



simrad-yachting.com

Installations



Points to check

-Power Supplies

All network equipment require their own supplies, and the power control wire needs to be connected correctly, NMEA2000 should only have 1 supply point.

-Fuses

Correct values fitted for all equipment

-Power control

Zeus can control the yellow wire output but be aware this is the same as the supply voltage to the Zeus, the output is max 30mA

-System Voltage

Remember that some equipment is 12VDC only (h3000,NMEA2000, Structure scan, Sonic Hub)

-Commissioning

Ensure that ST10 report and Zeus Install record is completed and filed, it can save a lot of time if this record is available should the customer want to add equipment later, and can effect Onboard warranty if not completed.

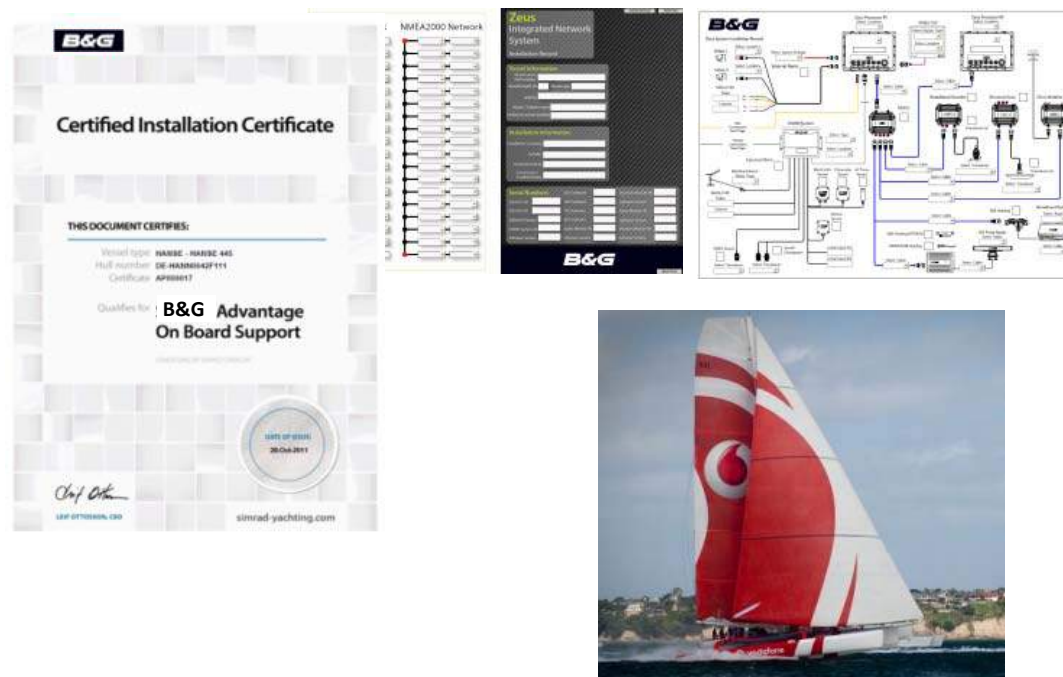
-Seatrial

Its useful to record any underway settings for future reference if equipment needs to be reset, or to compare against if performance decreases

-Owners Brief

Ensure that the customer is comfortable with the operation of the equipment, they know where equipment power switches and fuses are located. Ensure that all manuals are stored in the provided document wallet (Navigation Products)

Navico recommend saving user settings so that customer can return his equipment to defaults if they make changes that effect the performance of the equipment.



Onboard Advantage



For a Technician to attend a vessel for B&G Onboard support the following conditions must be confirmed by the customer-

1. Installation completed by a certified installer
2. Certificate of installation issued to the vessel by the installer
3. Value of equipment package must meet minimum requirement

It is the responsibility of the Technician to request an onboard authorisation number from Simrad support before attending the vessel and to complete the following-

1. Connect and produce a ST10 report of the system, both before and after work has been carried out on the system.
2. If there is any defect in the installation or concerns about the validity of warranty take photographs to indicate what was found.
3. Send copies of the ST10 report to Navico.service@navico.com
4. Submit an invoice within 30days of job completion with the onboard authorisation number as a reference



Formerly known as the B&G Gold Card Program

Under the +1 Advantage Program B&G customers can extend both their product warranty and ONBOARD ADVANTAGE coverage from 2 years to 3 years

Minimum system value of USD \$2,500

Customers simply have their products installed by a Certified Dealer and register on-line with B&G within 90 days of installation

Contacts

EMAIL:

Tech enquiries

Simrad simrad.emea@navico.com
B&G support@bandg.com
Lowrance lowrance.emea@navico.com

Warranty Enquiries EMEA

Navico.service@navico.com

Phone:

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B&G +44 1794 510042
Lowrance +44 1794 510134

Notes



Support.BG.apac@navico.com

Rev0211